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(54) **COMPOSITION COMPRISING ENGINEERED PLANT-DERIVED EXTRACELLULAR VESICLES AND USE THEREOF AS A VACCINE**

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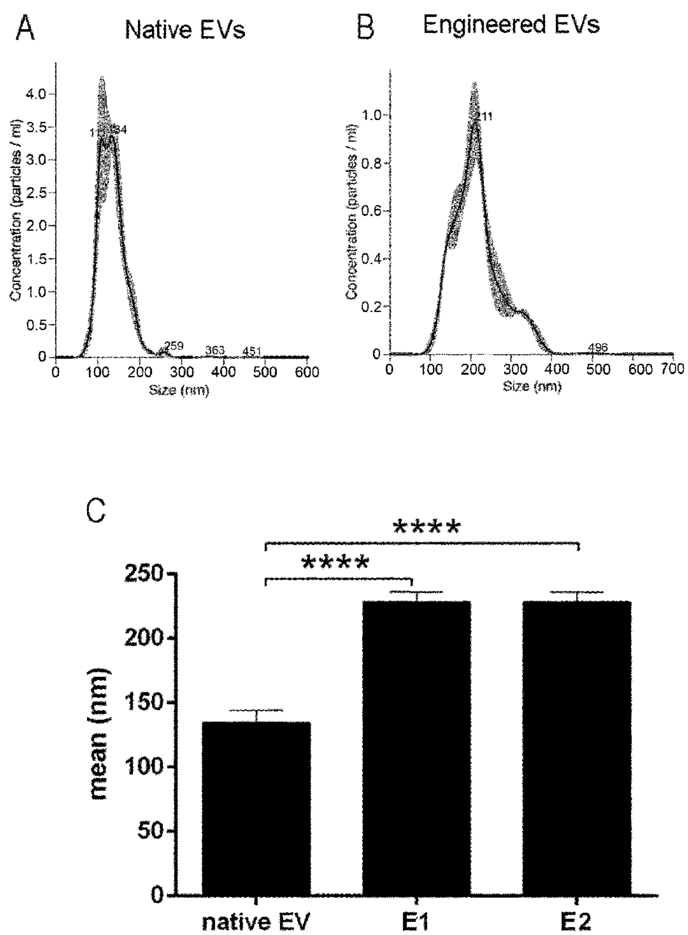
(57) **ABSTRACT**

A method for treatment or prophylaxis of a disease in a subject involving administering to the subject a vaccine composition including non-immunomodulating, engineered, plant-derived extracellular vesicles (EVs), is provided. The Evs are loaded with an exogenous nucleic acid molecule encoding a protein antigen. The disease is an infectious disease or cancer. A method for the preparation of the vaccine composition, which makes use of one or more polycationic substances and one or more sugar molecules is also provided.

Specification includes a Sequence Listing.



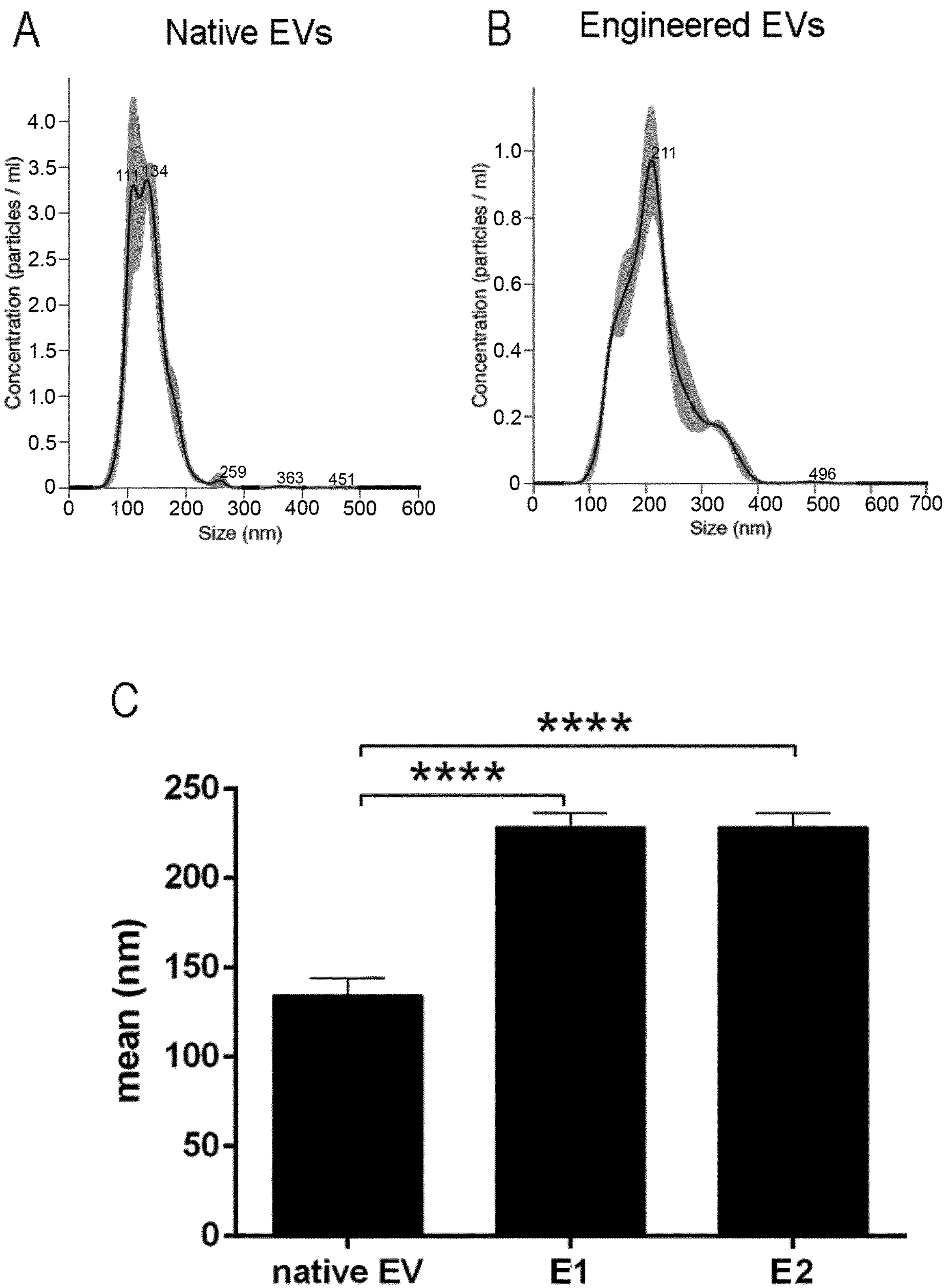

A
Native EVs
Concentration (particles / ml)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0
111
134
259
363
451
0
100
200
300
400
500
600
Size (nm)
B
Engineered EVs
Concentration (particles / ml)
1.0
0.8
0.6
0.4
0.2
0
211
496
0
100
200
300
400
500
600
700
Size (nm)
C
****
****
mean (nm)
250
200
150
100
50
0
native EV
E1
E2

FIG.1

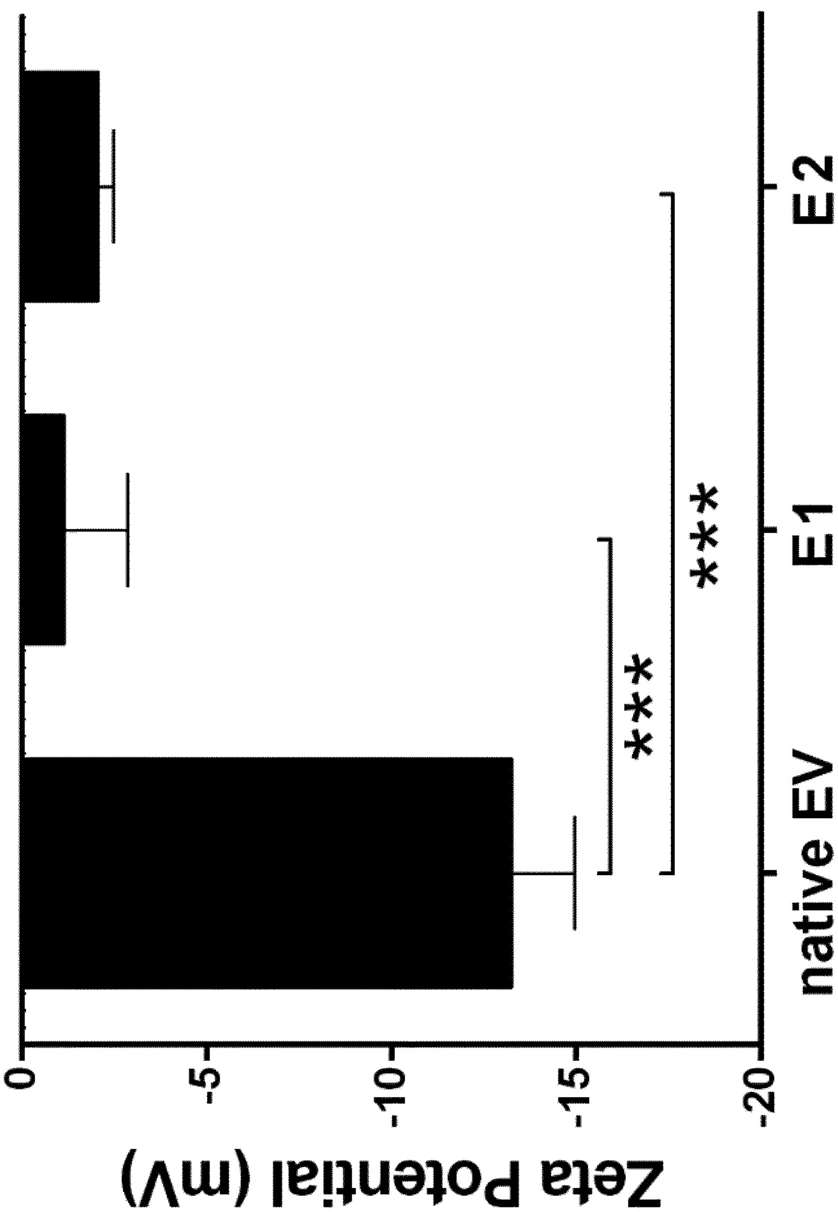


FIG.2

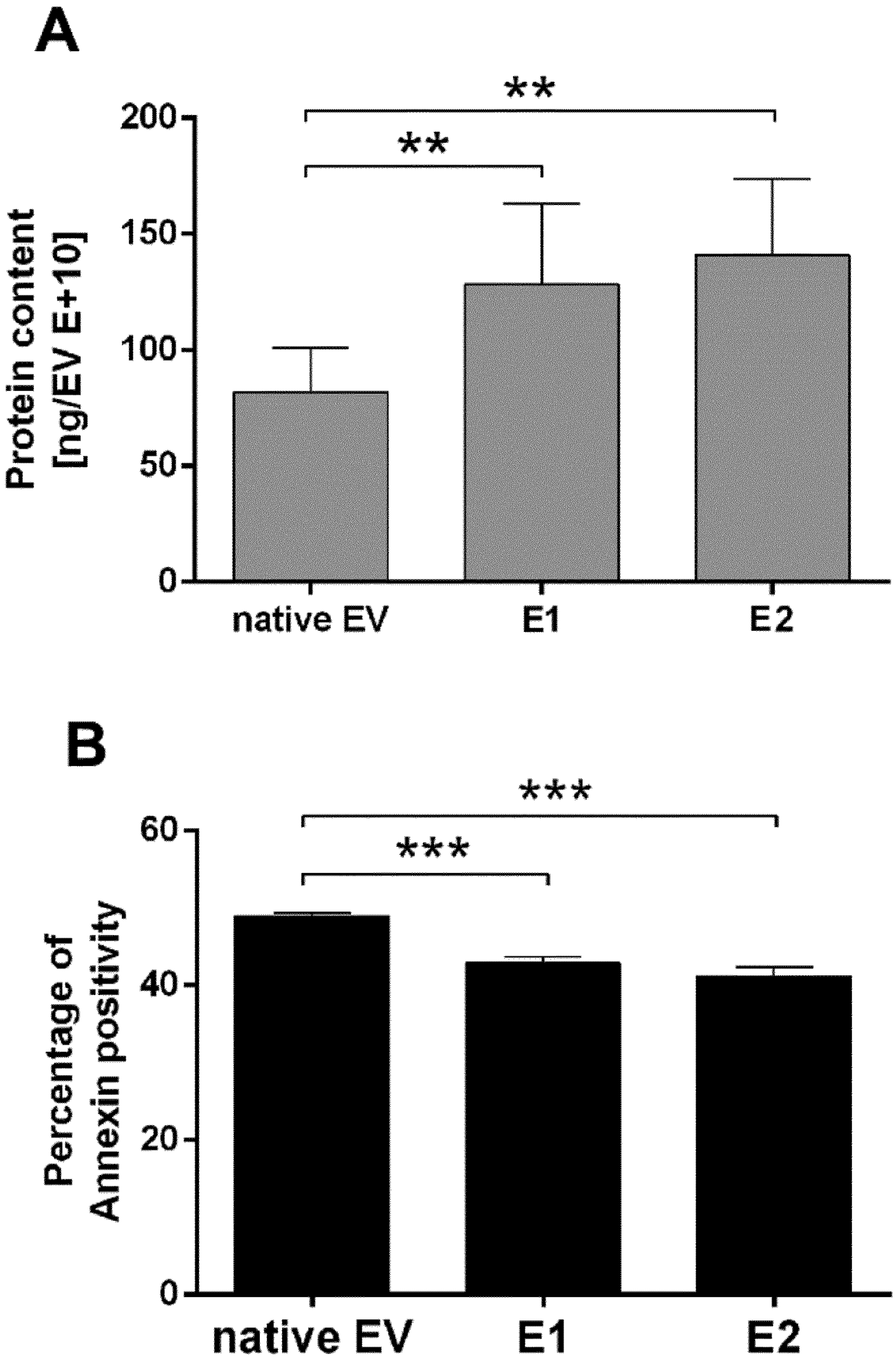

A
**
**
200
150
100
50
0
Protein content
[ng/EV E+10]
native EV
E1
E2
B
***
***
60
40
20
0
Percentage of
Annexin positivity
native EV
E1
E2

FIG.3

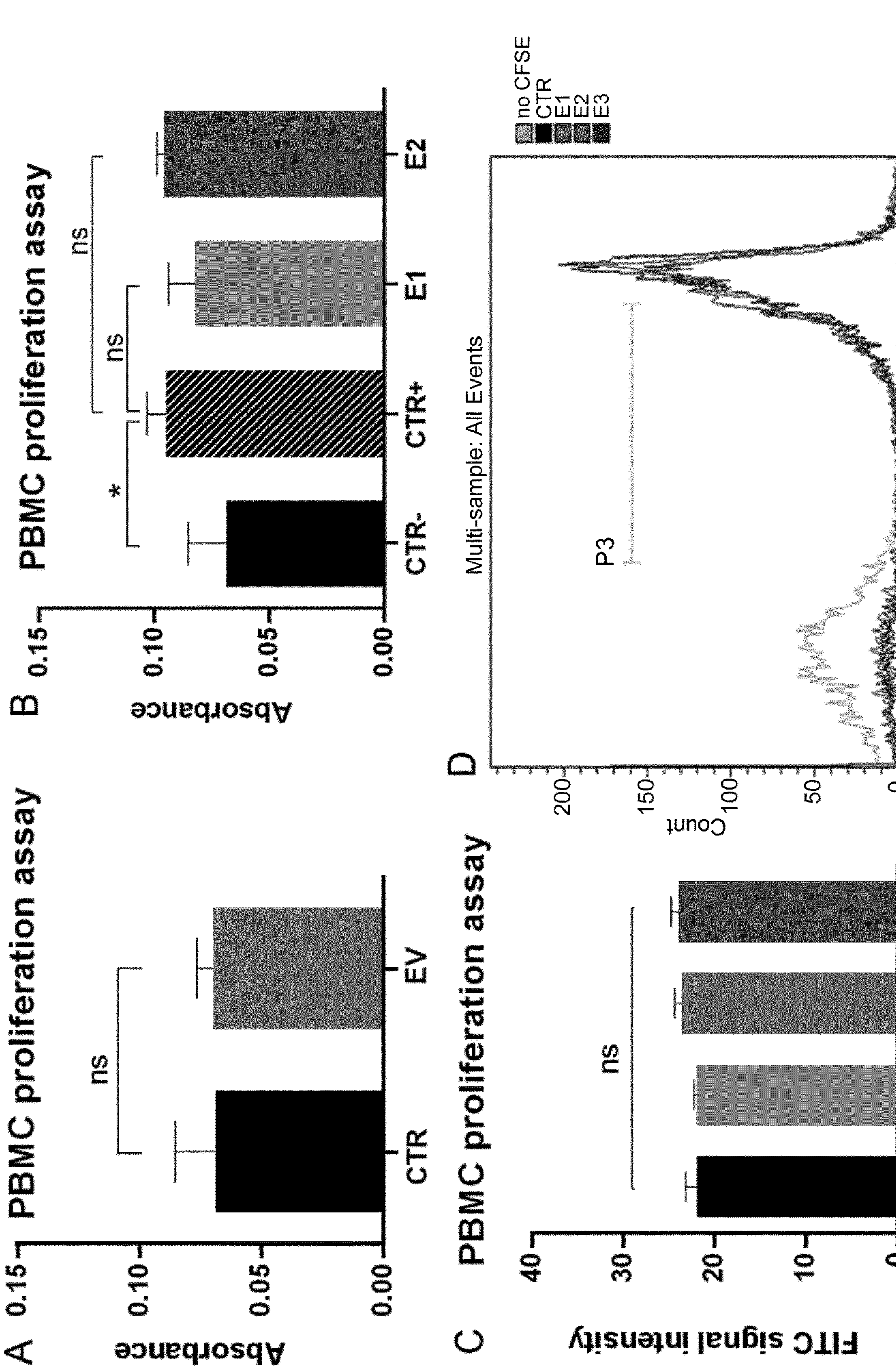

A
PBMC proliferation assay
Absorbance
0.15
0.10
0.05
0.00
ns
CTR
EV
B
PBMC proliferation assay
Absorbance
*
ns
ns
CTR-
CTR+
E1
E2
C
PBMC proliferation assay
FITC signal intensity
40
30
20
10
0
ns
CTR
E1
E2
E3
D
Multi-sample: All Events
no CFSE
CTR
E1
E2
E3
P3
Count
200
150
100
50
0
FITC-A

FIG.4

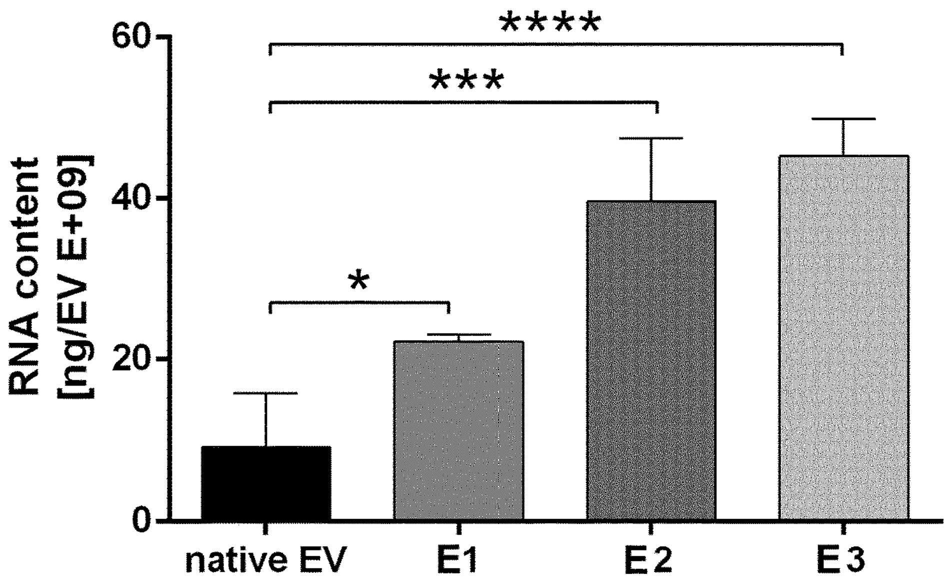

RNA content [ng/EV E+09]
60
40
20
0
****
***
*
native EV
E1
E2
E3

FIG.5

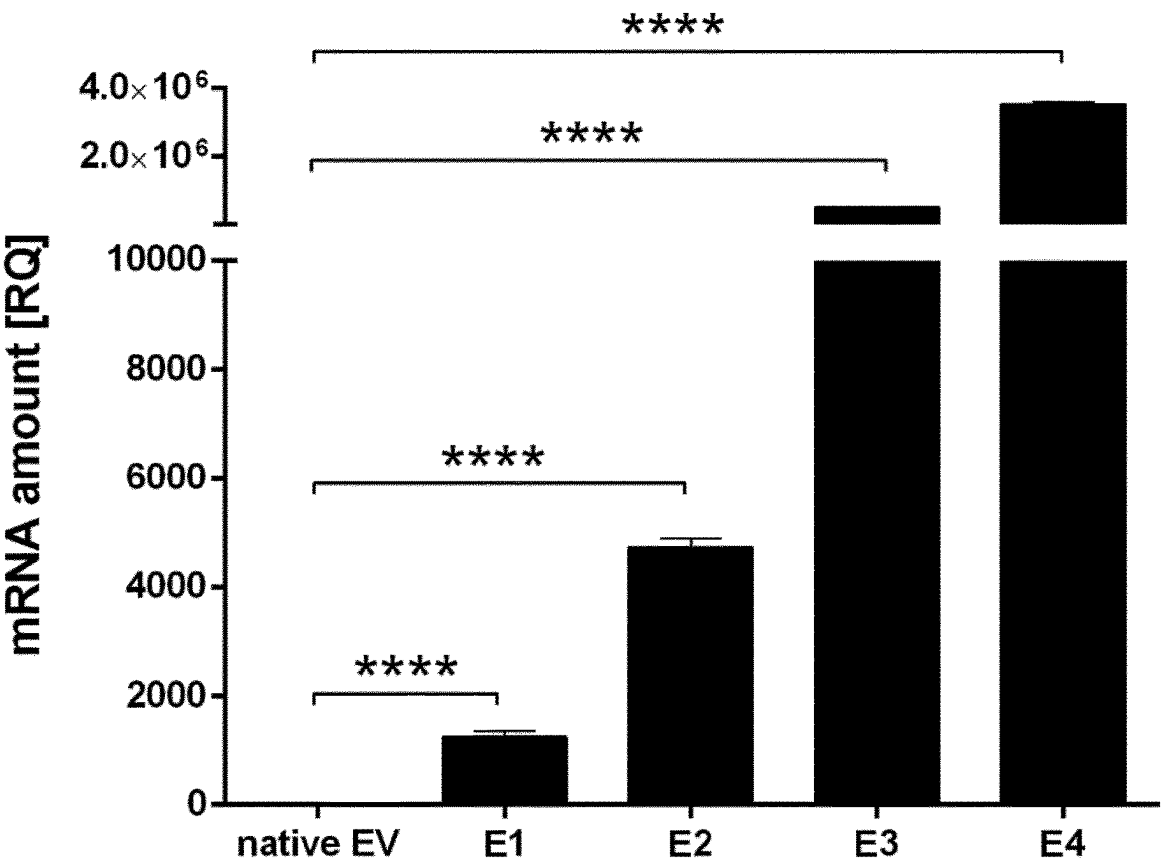

mRNA amount [RQ]
4.0×10⁶
2.0×10⁶
10000
8000
6000
4000
2000
0
****
****
****
****
native EV
E1
E2
E3
E4

FIG.6

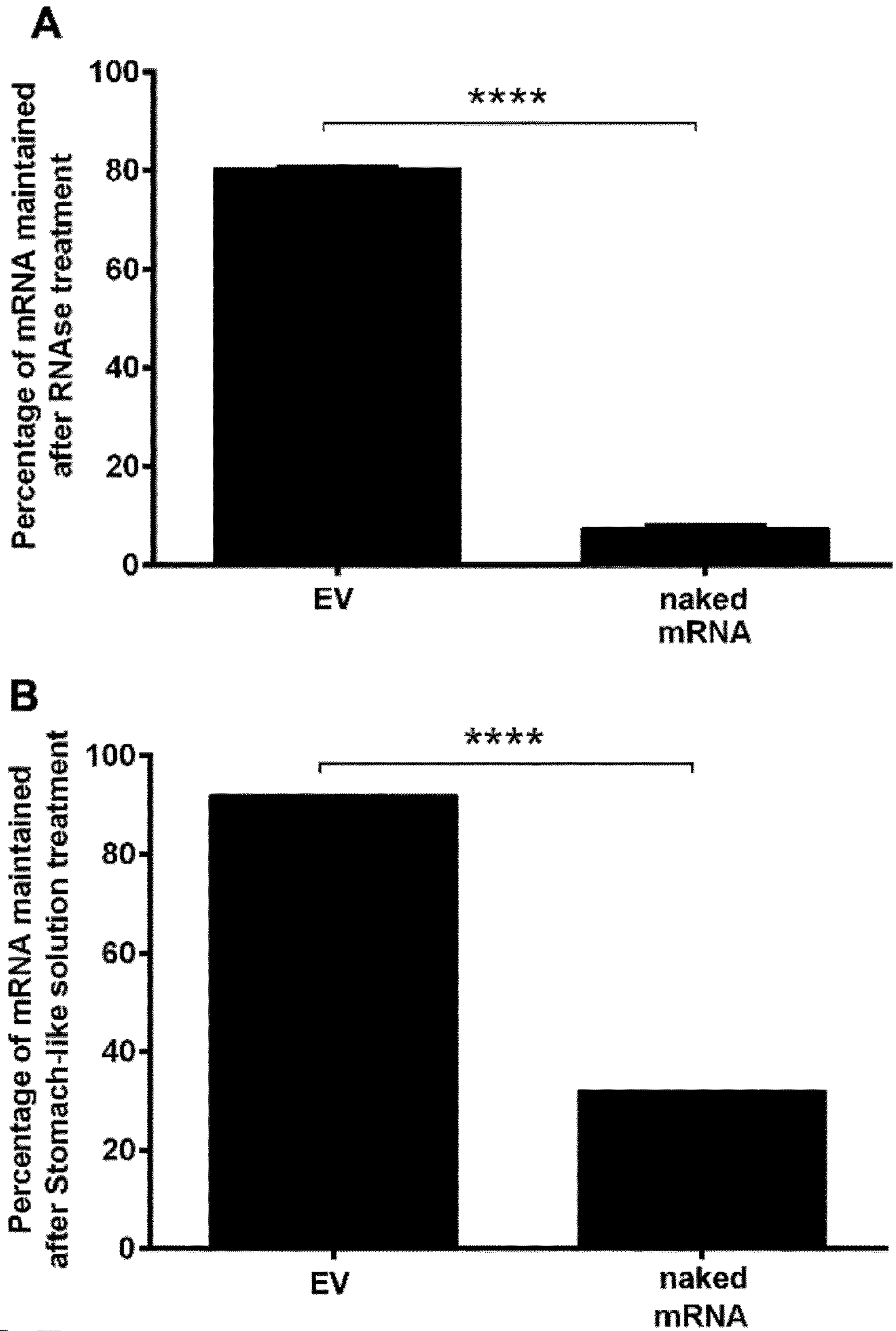

A
Percentage of mRNA maintained after RNAse treatment
100
80
60
40
20
0
****
EV
naked mRNA
B
Percentage of mRNA maintained after Stomach-like solution treatment
100
80
60
40
20
0
****
EV
naked mRNA

FIG.7


Percentage of mRNA maintained after storage
150
100
50
0
E1
E2

FIG.8

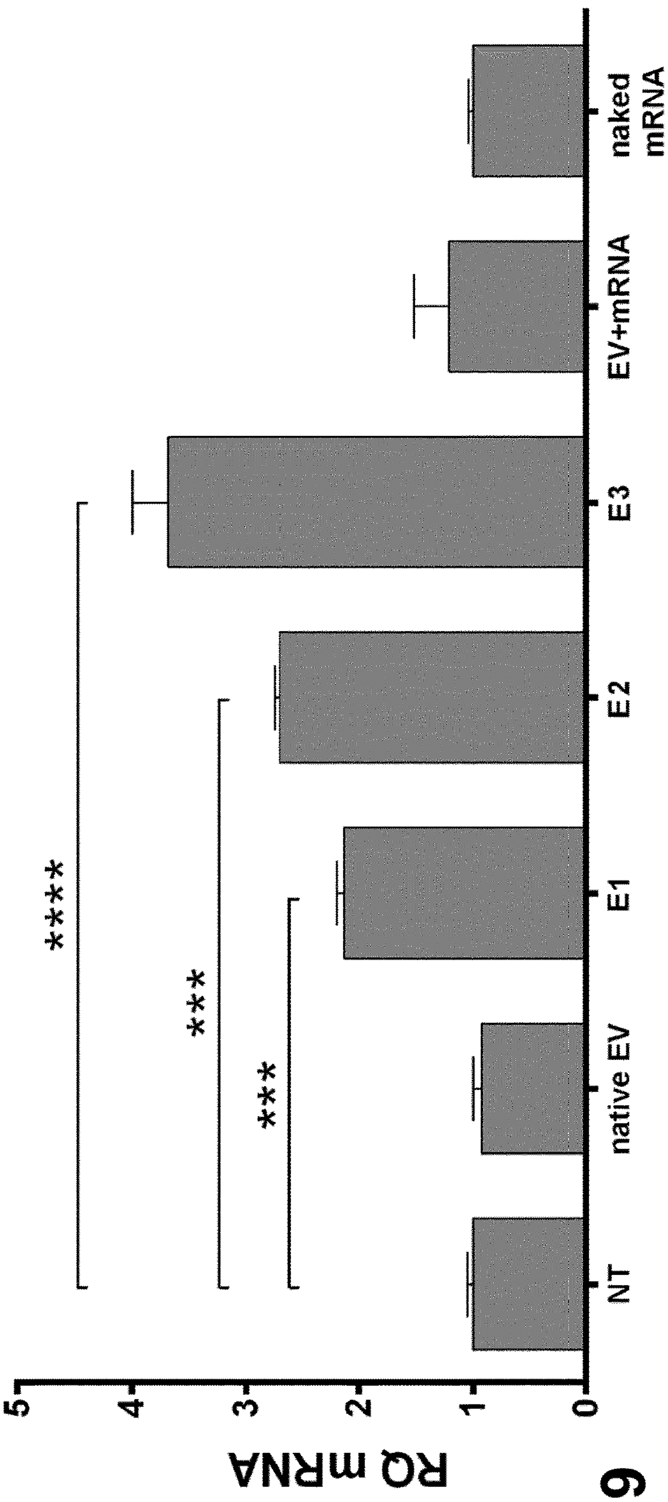

RQ mRNA
0
1
2
3
4
5
NT
native EV
E1
E2
E3
EV+mRNA
naked mRNA
***
***
****

FIG.9

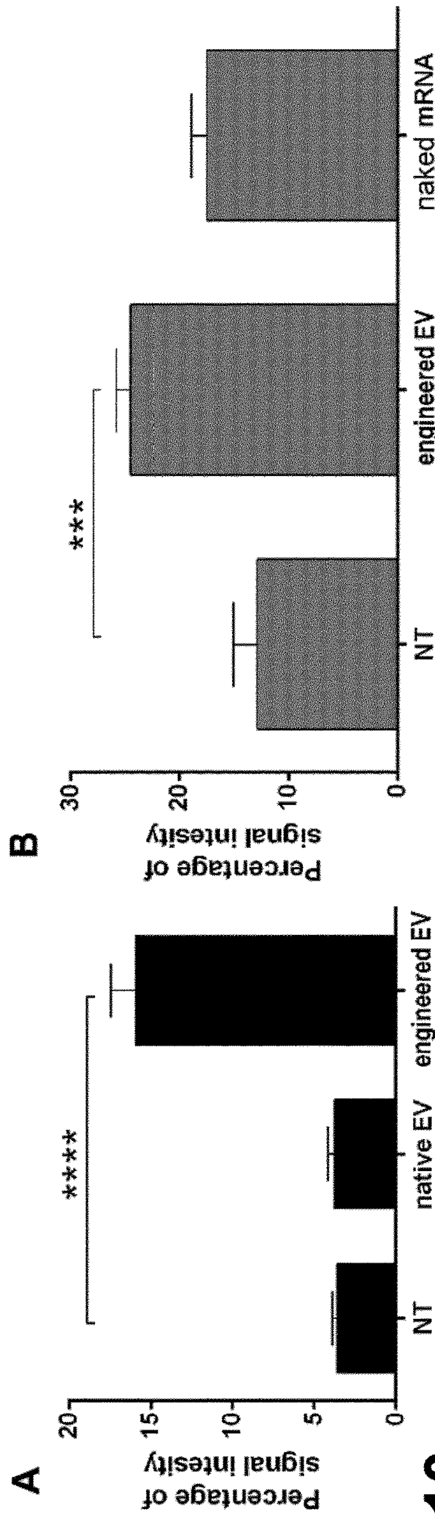

A
Percentage of signal intesity
0
5
10
15
20
NT
native EV
engineered EV
****
B
Percentage of signal intesity
0
10
20
30
NT
engineered EV
naked mRNA
***

FIG.10

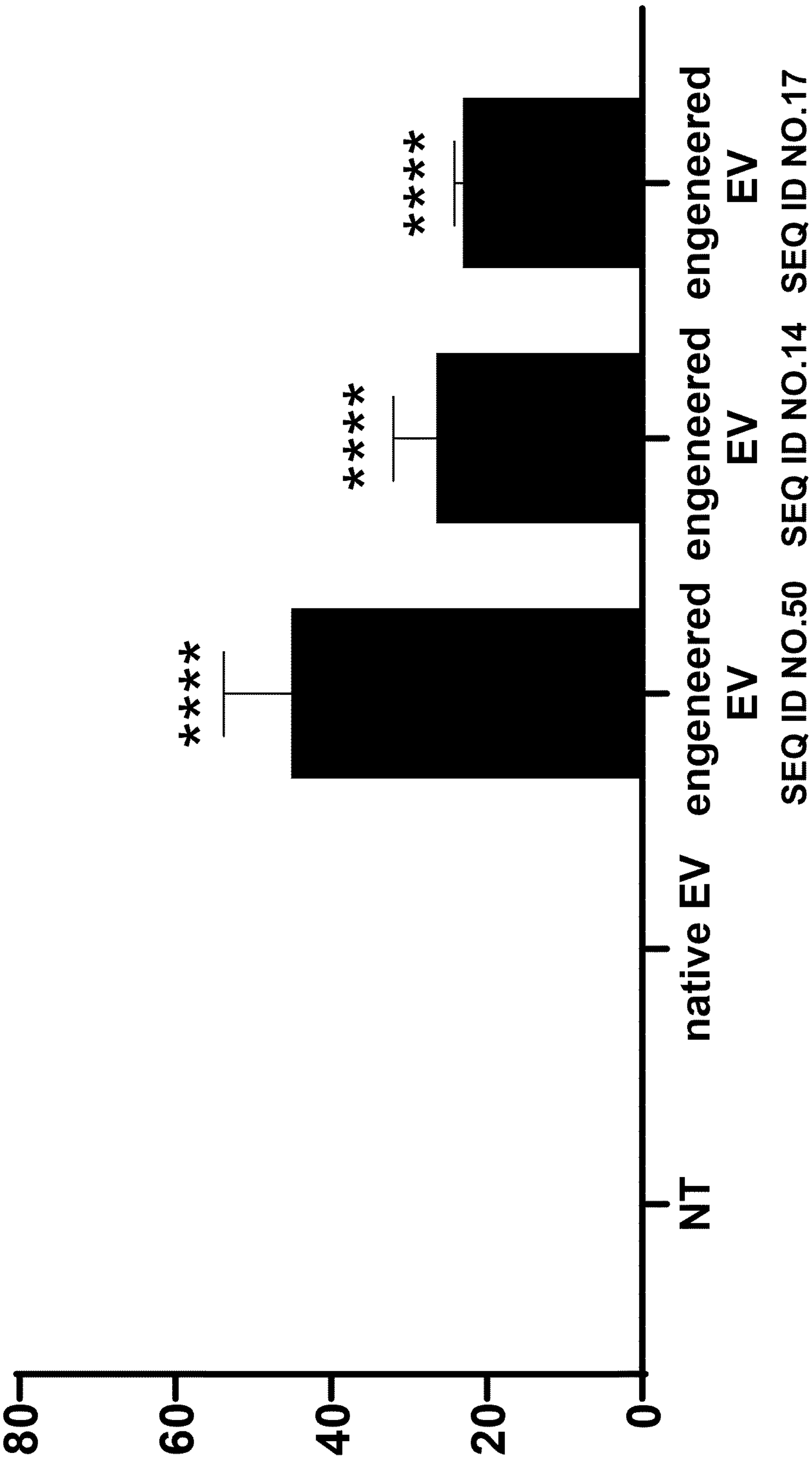
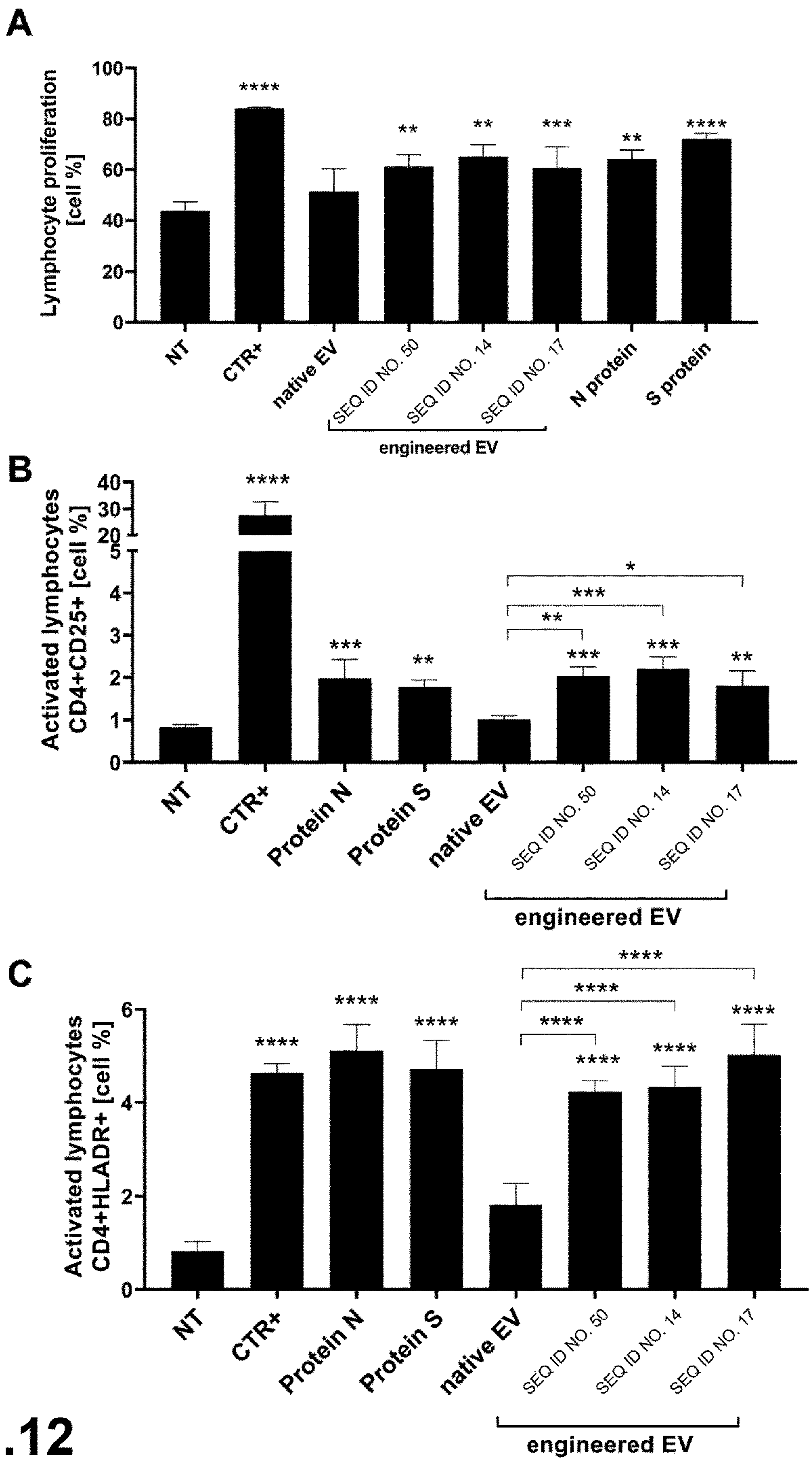

Protein expression
0
20
40
60
80
NT
native EV
engeneered EV SEQ ID NO.50
engeneered EV SEQ ID NO.14
engeneered EV SEQ ID NO.17
****
****
****

FIG.11



A
Lymphocyte proliferation [cell %]
100
80
60
40
20
0
****
**
**
***
**
****
NT
CTR+
native EV
SEQ ID NO. 50
SEQ ID NO. 14
SEQ ID NO. 17
N protein
S protein
engineered EV
B
Activated lymphocytes CD4+CD25+ [cell %]
40
30
20
5
4
3
2
1
0
****
***
**
*
***
**
***
***
**
NT
CTR+
Protein N
Protein S
native EV
SEQ ID NO. 50
SEQ ID NO. 14
SEQ ID NO. 17
engineered EV
C
Activated lymphocytes CD4+HLADR+ [cell %]
6
4
2
0
****
****
****
****
****
****
****
****
****
****
NT
CTR+
Protein N
Protein S
native EV
SEQ ID NO. 50
SEQ ID NO. 14
SEQ ID NO. 17
engineered EV

FIG.12

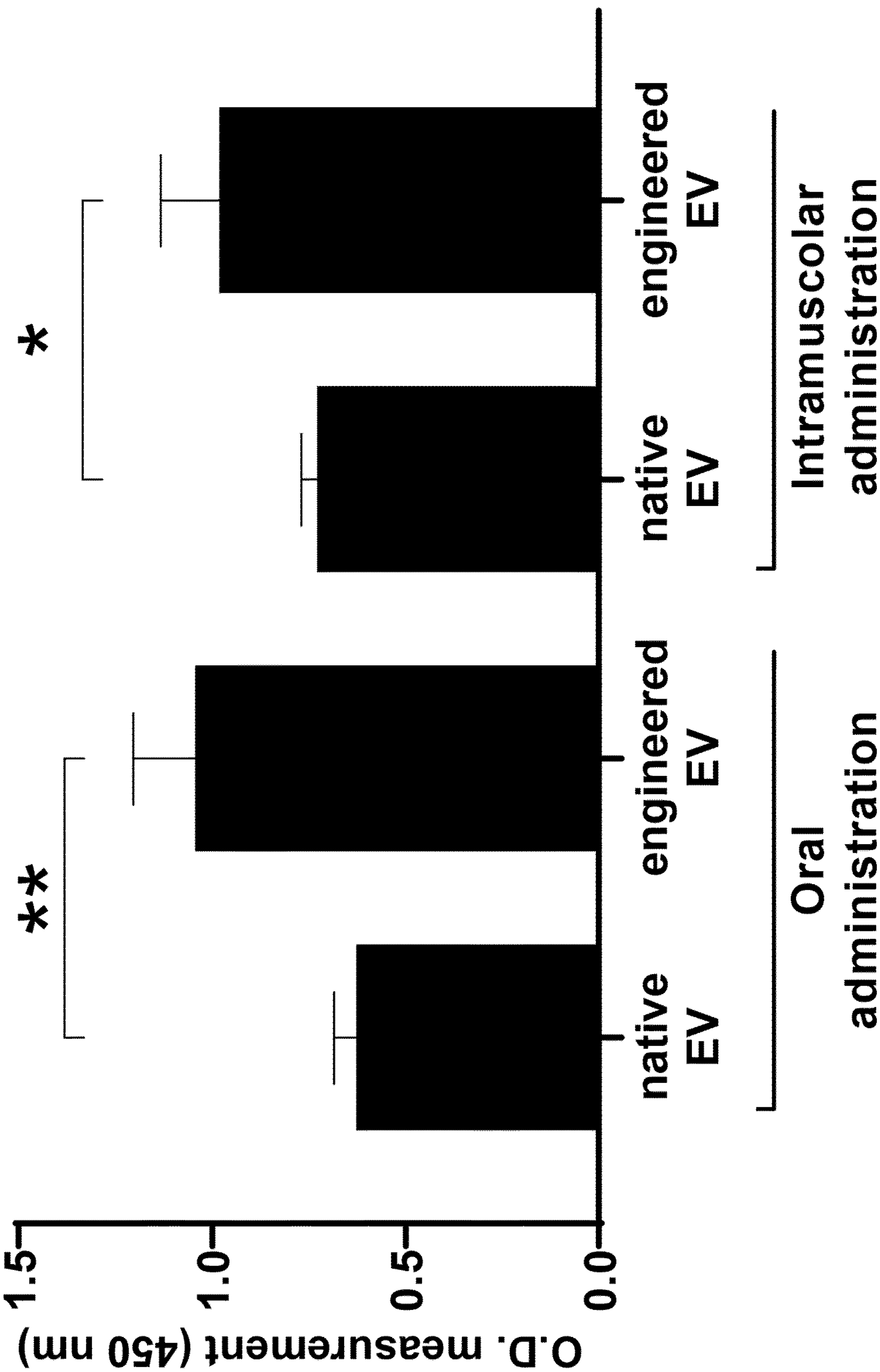

O.D. measurement (450 nm)
0.0
0.5
1.0
1.5
**
*
native EV
engineered EV
native EV
engineered EV
Oral administration
Intramuscolar administration

FIG.13

COMPOSITION COMPRISING ENGINEERED PLANT-DERIVED EXTRACELLULAR VESICLES AND USE THEREOF AS A VACCINE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a National Stage Application of International Patent Application No. PCT/EP2022/050590, having an International Filing Date of Jan. 13, 2022, which claims priority to Italian Application No. 102021000000569 filed Jan. 14, 2021, the entire contents of which are hereby incorporated by reference herein.

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING

[0002] The contents of the electronic sequence listing named "39447_283_sequence-listing.txt", Size: 192, 383 bytes created on Aug. 2, 2024 are herein incorporated by reference in their entirety.

TECHNICAL FIELD

[0003] The present invention relates to compositions comprising plant-derived extracellular vesicles for use as a vaccine and/or for prophylaxis applications. More specifically, the invention relates to a composition comprising non-immunomodulating, engineered, plant-derived extracellular vesicles (EVs) loaded with an exogenous nucleic acid molecule.

BACKGROUND

[0004] Vaccination is one of the most effective public health interventions to prevent and control infectious and non-infectious diseases. Different types of vaccines exist including live-attenuated vaccines, inactivated vaccines, subunit or recombinant or conjugate vaccines, toxoid vaccines, and those based on nucleic acids.

[0005] Nucleic acid-based vaccines comprise viral vectors, plasmid DNA, and mRNA. They have emerged as promising alternatives to conventional vaccine approaches because of their ability to induce broadly protective immune responses and their potential of being produced by rapid and flexible manufacturing processes. Among nucleic acid-based vaccine, RNA vaccines have several attributes that provide potential advantages over other vaccine types. In fact, RNA vaccines are characterized by the absence of eukaryotic contaminants. In contrast to DNA vaccines, RNA vaccines do not need to reach the nucleus to work and they are safer because plasmid DNA vaccines can integrate into the genome of the immunized host. In this context, mRNA molecules induce immune response against an encoded antigen.

[0006] The mechanism has been demonstrated for a variety of target genes including reporter genes, viral antigens, tumor antigens, and allergens. However, mRNA instability, high innate immunogenicity and inefficient in vivo delivery limit the clinical application of RNA vaccines. The main challenge faced by these vaccines is the intracellular delivery. Because of its sensitivity toward degradative enzymes, mRNA is highly unstable under physiological conditions in the body.

[0007] To date, in clinical applications mRNA molecules are vehicled by complexation with cationic polymers or using synthetic lipid nanoparticles (LNP) also called liposomes, cationic nanoparticles, EV-mimetic nanovesicles or polypeptide-based vesicles, to improve their efficacy. LNP allow mRNA protection from enzymes, promoting a higher stability, increasing RNA circulation lifetime and in vivo delivery. LNP particles are created by mixing mRNA molecules with different synthetic lipids or polymers. Nevertheless, LNP represent an inefficient delivery system.

[0008] In fact, LNP may accumulate in unintended tissues thereby limiting their effect on the target tissue of interest and LNP are characterized by a rapid clearance by the reticuloendothelial system or the mononuclear phagocyte system (Koppers-Lalic D., et al. "Virus-modified exosomes for targeted RNA delivery; a new approach in nanomedicine". Adv Drug Deliv Rev. 2013 March;65(3):348-56). In addition, LNP can induce a pro-inflammatory response and apoptosis in vivo.

[0009] Moreover, the cellular uptake of LNP is mediated by endocytosis, which could activate the cells' autophagic-lysosomal pathway. Accumulating evidence indicates that endocytosis of nanoparticles generates autophagosomes, and their subsequent fusion with lysosomes leads to the digestion of their content.

[0010] In recent years, attempts have been made to overcome limitations of LNP by using extracellular vesicles (EVs). In fact, EVs are naturally secreted by cells and are safer compared to synthetic nanomaterials, such as LNP. EVs can exploit their natural mechanism of action and overcome some of the limitations of assembled-particles, including immunogenicity, toxicity, administration of exogenous particles, limited cell uptake and chemical assemblage of particles. The routes of EV uptake differ from those of LNP and are not likely to elicit the autophagy-lysosomal pathway, as they release their content into the cytoplasm probably without undergoing lysosomal trapping. Additionally, because of their small size, EVs can escape from rapid phagocytosis, and steadily carry and deliver nucleic acids in circulation, passing through the vascular endothelium to the target cells.

[0011] Moreover, EVs show several advantages in comparison to LNP in terms of biocompatibility, low clearance throughout the circulation, low toxicity, low safety concerns and high specificity (Sancho-Albero M, et al (2020) "Use of exosomes as vectors to carry advanced therapies". RSC Adv 10, 23975-23987). The natural origin of EVs allows a high and inherent biocompatibility of their membrane and an efficient uptake in recipient cells. In addition, plant derived EVs can resist to stomach environment and reach the intestine after oral administration. EVs encapsulating nucleic acid molecules have been studied for multiple clinical applications, including RNA interference, RNA-based gene therapy for neurodegenerative disorders, cancer, cancer vaccine with miRNA and siRNA molecules.

[0012] US20200069594 discloses the use of plant-derived extracellular vesicles comprising a cationic polymer for delivering therapeutic agents, including coding and non-coding nucleic acid molecules.

[0013] It is well known in the art that EVs are able to protect and deliver nucleic acid molecules. For vaccine formulations, the beneficial activity of EVs is mainly based on their promoting effect on the innate and adaptive immune system cells, as shown for instance in the studies by Jesus S., et al, carried out on vaccine formulations for HBV (Jesus S., et al "Exosomes as adjuvants for the recombinant hepatitis B antigen: First report". Eur J Pharm Biopharm. 2018 December; 133:1-11).

[0014] WO2020191361 discloses the use of EVs as vaccine to induce a cellular immune response and to treat and/or prevent a range of medical disorders.

[0015] WO2020050808 describes the use of plant-derived exosomes as adjuvants in vaccine applications along with the immunomodulating properties of these vesicles in activating or suppressing the immune system cells.

[0016] Extracellular vesicles isolated from various plants have been shown to exert a modulating activity on the cells of the immune system by reducing inflammation in the intestine. Plant sources include curcumin (Ohno M., et al, "Nanoparticle curcumin ameliorates experimental colitis via modulation of gut microbiota and induction of regulatory T cells" PLOS One. (2017) October 6;12 (10): e0185999), ginger (Zhang M., et al, (2016) "Edible ginger-derived nanoparticles: A novel therapeutic approach for the prevention and treatment of inflammatory bowel disease and colitis-associated cancer", Biomaterials 101:321-40), orange (Berger E., et al, "Use of Nanovesicles from Orange Juice to Reverse Diet-Induced Gut Modifications in Diet-Induced Obese Mice" (2020) Mol Ther Methods Clin Dev. 18:880-892), grapes, grapefruit, and carrots (Ju S, et al. "Grape exosome-like nanoparticles induce intestinal stem cells and protect mice from DSS-induced colitis" (2013) Mol Ther. 21(7):1345-57). Moreover, EV from blueberry have shown to reduce gene expression of pro-inflammatory genes in endothelial cells stimulated with TNF-α and to protect endothelial cells from TNF-induced cytotoxicity and oxidative stress (De Robertis M, et al. "Blueberry-Derived Exosome-Like Nanoparticles Counter the Response to TNF-α-Induced Change on Gene Expression in EA.hy926 Cells" (2020) Biomolecules 10(5):742).

[0017] Notwithstanding the beneficial effects of EVs, the immunomodulating properties of these vesicles may represent a significant limitation in vaccine applications. In fact, the non-specific activation or inhibition of the immune system can be detrimental for vaccines. The use of EVs having immunosuppressive activity can significantly reduce vaccine efficiency by inhibiting immune cells response. Moreover, the development of an immune response to EVs may lead to accelerated vaccine clearance. On the contrary, immune system promotion by immune-stimulatory EVs can be harmful and led to detrimental activation and/or overreaction of a subjects' immune system. The immune responses to the vesicles can limit repeated application of vaccines. In addition, the innate immune sensing of vesicles can be associated with the inhibition of antigen expression and may negatively affect the immune response (Pardi N, et al "mRNA vaccines—a new era in vaccinology" (2018) Nat Rev Drug Discov. 17(4):261-279). Taken together, the above-illustrated evidences highlight significant disadvantages in using EVs in vaccine formulations based on the effects exhibited by these vesicles on the immune system.

SUMMARY OF THE INVENTION

[0018] In order to overcome the limitations and drawbacks of the prior art, the present invention provides a composition comprising non-immunomodulating, engineered, plant-derived extracellular vesicles (EVs), for use as a vaccine as well as a method for the preparation of said composition as defined in the appended independent claims. The dependent claims identify further advantageous features of the claimed composition and method. The subject-matter of the appended claims forms an integral part of the present description.

DETAILED DESCRIPTION OF THE INVENTION

[0019] The present invention relates to a composition comprising non-immunomodulating, engineered, plant-derived extracellular vesicles (EVs), wherein said extracellular vesicles (EVs) are delimited by a lipid bilayer membrane comprising an outer lipid layer and an inner lipid layer,

- **[0020]** wherein said EVs are internally loaded with an exogenous nucleic acid molecule encoding at least one protein antigen;
- **[0021]** wherein said EVs have a diameter ranging from 20 to 500 nm, preferably ranging from 200 to 300 nm;
- **[0022]** wherein the membrane potential across the lipid bilayer membrane of said EVs ranges from +5 to −5 mV;
- **[0023]** wherein ≤44% of the EVs in the composition comprise phosphatidylserine in the outer layer of the lipid bilayer membrane, for use as a vaccine.

[0024] As used herein, the term "extracellular vesicles (EVs)" refers to a heterogeneous population of particles released by virtually all living cells, which are delimited or encapsulated by a phospholipid bilayer and which carry lipids, proteins, nucleic acids and other molecules derived from the cell they are derived from. These vesicles mainly include microvesicles, released through the budding of the plasma membrane, and exosomes, derived from the endosomal compartment. Extracellular vesicles are referred to as "particles", "microparticles", "nanovesicles", "microvesicles" and "exosomes". The inherent cellular targeting properties of EVs that are dictated by their lipid composition and protein content as well as their intrinsic stability in circulation qualify these vesicles as vehicle for therapeutic agent delivery.

[0025] Within the present description the term "immunomodulation" refers to a process in which a function of the immune system is altered by enhancing (immunostimulation) or decreasing (immunosuppression) an immune response. Accordingly, the expression "non-immunomodulating EVs" as used herein refers to extracellular vesicles which do not exert any promoting nor immunosuppressive effect on the immune system.

[0026] As used herein, the term "engineered EVs" refers to extracellular vesicles which have been modified in vitro to express a heterologous component by loading a nucleic acid molecule exogenous to the vesicles' donor cells. It is therefore to be intended that an engineered EV is a non-naturally occurring vesicle.

[0027] The expression "internally loading" in the context of the present description means introducing a nucleic acid molecule in an extracellular vesicle, for example a plant-derived EV, by means, for example, of transfection, transformation or transduction.

[0028] The term "exogenous nucleic acid molecule" as used in the present description relates to a heterologous nucleic acid molecule which is not part of the natural cargo of the EVs of the invention as such. The expression "heterologous" refers to a nucleic acid molecule derived from an animal or another vegetal species than the extracellular vesicles according to the invention, or from different donor cells, different conditions, or from genetically modified donor cells.

[0029] The term "antigen protein" as used herein refers to a protein molecule capable of evoking an immune response.

[0030] In accordance with the present invention, the exogenous nucleic acid molecule loaded in the plant-derived EVs is preferably selected form the group consisting of: DNA, cDNA, messenger RNA (mRNA), pre-mRNA, long-chain RNA, coding RNA, single-stranded RNA, double stranded RNA, linear RNA, RNA oligonucleotide, self-replicating RNA (replicon RNA), retroviral RNA, a viral RNA (vRNA).

[0031] In a preferred embodiment of the invention, the exogenous nucleic acid molecule is a messenger RNA (mRNA) molecule. Within the context of the present invention, the exogenous mRNA molecule may comprise one or more modifications such as, for example, 5' cap structure, 5' UTR, open reading frame, 3' UTR and poly A tail.

[0032] According to the invention, the EVs in the composition may be loaded with a single nucleic acid molecule or with a combination of two or more nucleic acid molecules.

[0033] In one preferred embodiment of the invention, the content of the loaded exogenous nucleic acid molecules in the EVs is in the range of from 20 to 200 ng/109 EVs, preferably from 30 to 100 ng/109 EVs, more preferably from 40to 60 ng/109 EVs.

[0034] The loading of exogenous nucleic acid molecules into the EVs according to the present invention may be accomplished by a number of different techniques known in the art, including, for example, electroporation, sonication, lipofectamine mediation, microinjection, co-incubation, dialysis and freeze-thaw cycles.

[0035] The present invention makes use of extracellular vesicles which have a diameter in the range of from 20 to 500 nm, preferably from 100 to 400 nm, more preferably in the range of from 200 to 300 nm.

[0036] According to the invention, the value of the membrane potential across the lipid bilayer membrane of the EVs in the composition ranges from +5 to-5 mV, preferably from +2 to −4 mV, more preferably from 0 to −3.

[0037] In a further more preferred embodiment of the invention, the value of the membrane potential of the EVs is −2 mV.

[0038] In the composition according to the invention, an amount of EVs less than or equal to (≤) 44% of the total EVs in the composition comprise phosphatidylserine in the outer layer of the lipid bilayer membrane.

[0039] Preferably, the amount of EVs in the composition comprising phosphatidylserine in the outer layer of the lipid bilayer membrane is comprised within the range of from 25% to 44% of total EVs, more preferably from 35% to 44% of total EVs, even more preferably from 40% to 44% of total EVs.

[0040] The plant-derived EVs that are used in the present invention are preferably derived from one or more plants selected from the group consisting of: genus *Citrus*, including lemon and orange; genus *Actinidia*, including kiwifruit; genus *Cucurbita*, including courgette; genus *Brassica*, including cabbage and kale; genus *Punica*, including pomegranate; genus *Vaccinium*, including blueberry, and genus *Apium*, including celery.

[0041] The scope of the invention includes both compositions containing EVs derived from a single plant species and compositions containing EVs derived from a plurality of plant species. It is understood that plant-derived EVs can be used in their native form or with chemical modifications.

[0042] Preferably, the plant-derived EVs in the composition according to the invention are purified from fruit juice, part of plant or culture medium of plant cells. Plant cells and parts may be derived from leaf, fruit pulp, shoot or sprout.

[0043] Suitable purification techniques of EVs include, but are not limited to, ultracentrifugation, filtration and tangential flow filtration. The selection of the most suitable method to be used for the purification of plant-derived EVs falls within the knowledge and skills of the ordinary person of skill in the art.

[0044] In one embodiment of the invention, the total protein content of the EVs in the composition of the invention is in the range of from 100 to 200 ng/10^{10} EVs, more preferably from 120 to 160 ng/10^{10} EVs.

[0045] In another embodiment, the total RNA content of the EVs in the composition of the invention is in the range of from 20 to 200 ng/10^9 EVs, more preferably from 30 to 100 ng/10^9 EVs, even more preferably from 40 to 60 ng/10^9 EVs.

[0046] The expression "total protein content" encompasses both the endogenous protein cargo (internal and the membrane content of the EVs) and the loaded proteins in the EVs used in the present invention.

[0047] Within the context of the present description, the expression "total RNA content" encompasses both the endogenous RNA cargo and the loaded exogenous RNA in the EVs according to the invention.

[0048] As further illustrated in the Experimental Section below, the present inventors have surprisingly found that the engineered, plant-derived EVs having the structural and functional features as above defined do not exhibit any immunomodulatory activity, i.e. they are devoid of any ability to affect the cells of the immune system neither promoting nor reducing the activation and efficacy of these cells.

[0049] Differently from native plant-derived EVs, the non-naturally occurring EVs according to the invention are advantageously capable to deliver antigenic molecules to target cells without exerting per se any effect on the cells of the immune system. Therefore, the use of the EVs according to the invention enables to overcome the safety concerns in connection with EV-based vaccine formulations and to avoid detrimental activation or inhibition of the immune system, thereby enhancing the efficacy of vaccines.

[0050] Moreover, the EVs according to the invention are proved to efficiently load and vehicle nucleic acid agents to recipient cells and protect them from environment degradation. In particular, the high resistance to stomach environment allows the oral administration of the composition according to the invention.

[0051] After administration, the interaction of the loaded EVs with antigen presenting cells (APC), including macrophages and dendritic cells, allows the transfer of the nucleic acid molecules to the antigen-presenting cell. In the target cells, the nucleic acid molecules, comprising DNA and mRNA molecules, are expressed leading to protein antigen translation. Then, the antigen is presented on the surface of the APC inducing the specific activation of immune cells direct against the tumor cells or pathogen allowing an efficient immune protection.

[0052] Thanks to the advantageous features of the non-naturally occurring EVs as above illustrated, the composition of the invention is particularly suitable for use as a vaccine.

[0053] In the context of the present invention, the composition of the invention may be used as a vaccine for the treatment of an existing disease or prophylactically to prevent the occurrence of this disease.

[0054] Exemplary protein antigens encoded by the exogenous nucleic acid molecules encapsulated into the EVs of the invention include, but are not limited to, bacterial, viral, fungal, protozoan and tumor antigens, mammalian homologs thereof, and homologs from animals of veterinary or industrial interest thereof.

[0055] Accordingly, the composition of the present invention is particularly useful for the treatment or prophylaxis of infectious diseases or cancer diseases.

[0056] Exemplary cancer diseases include, but are not limited to, bladder cancer, cervical cancer, renal cell cancer, testicular cancer, colorectal cancer, lung cancer, head and neck cancer, ovarian, lymphoma, liver cancer, glioblastoma, melanoma, myeloma, leukemia, pancreatic cancer.

[0057] By way of example, but without limitation, the infectious disease may be, a viral disease, a bacterial disease, a fungal disease or a protozoan disease, such as, for example, COVID-19 disease, influenza, HPV infection, HIV infection, rhinovirus infection, hepatitis, flavivirus infections, encephalitis, meningitis, gastroenteritis, cholera, diphtheria, chlamydia, tuberculosis, typhoid, Sexually Transmitted Infections (STI), malaria, mycoses, toxoplasmosis.

[0058] In one embodiment of the invention, the at least one antigen encoded by the exogenous nucleic acid molecule loaded into the EVs is a tumor antigen selected from the group consisting of human kallikrein related peptidase 3, also called prostate specific antigen (PSA), human prostate stem cell antigen (PSCA), human prostate specific membrane antigen (PSMA), human metalloreductase (six transmembrane epithelial antigen of the prostate 1 (STEAP1), human Receptor tyrosine-protein kinase erbB-2, also called Tyrosine kinase-type cell surface receptor HER2, human cell surface associated mucin 1 protein (MUC1), also called Breast carcinoma-associated antigen DF3, human Tyrosinase-related protein 2 (TRP-2), human Serine/threonine-protein kinase B-raf, also called Proto-oncogene B-Raf, human Mast/stem cell growth factor receptor Kit, also called Proto-oncogene c-Kit, human GTPase NRas, also called Transforming protein N-Ras, human melanoma-associated antigen 1, human melanoma-associated antigen 1 protein, human NY-ESO-1 protein, and any combination thereof.

[0059] In another embodiment of the invention, the at least one protein antigen is a bacterial antigen from a bacterium selected from the group consisting of *Staphylococcus aureus, Mycobacterium tuberculosis, Chlamydia trachomatis, Streptococcus pyogenes, Streptococcus pneumoniae, Borrelia burgdorferi, Borrelia mayonii* (e.g., Lyme disease), *Klebsiella* sp., *Pseudomonas aeruginosa, Enterococcus* sp., *Proteus* sp. (e.g. vulgaris, mirabilis, penneri), *Neisseria gonorrhoeae, Enterobacter* sp., *Actinobacter* sp., coagulase-negative *Staphylococci* (CoNS), *Mycoplasma* sp., *Clostridium difficile, Bacillus anthracis, Vibrio cholerae, Clostridium botulinum, Clostridium tetani, Salmonella* sp., *Treponema pallidum, Plasmodium* sp., and any combination thereof.

[0060] In yet another embodiment of the invention, the at least one protein antigen is a fungal antigen from a fungus selected from the group consisting of *Blastomyces, Cryptococcus gattii, Cryptococcus neoformans, Fusarium, Aspergillus, Candida, Candida albicans, Candida auri, Cryptococcus, Histoplasma, Blastomyces, Coccidioides, Mucormycetes, Pneumocystis jirovecii, dermatophyte, Sporothrix*, and any combination thereof.

[0061] In a still another embodiment, the at least one protein antigen is a protozoan antigen from a protozoa selected from the group consisting of *Plasmodia* species (e.g., vivax and falciparum), *Giardia intestinalis, Hexamita salmonis, Histomonas meleagridis, Trichomonas foetus, Dientamoeba fragilis, Trichomonas vaginalis, Leishmania, Trypanosoma cruzi, Trypanosoma brucei rhodensiense, Trypanosoma brucei gambiense, Plasmodium parasite, Entamoeba histolytica, Naeglaria, Acanthomoeba, Peronosporomycetes, Phytophthora infestans, Giardia lamblia, Giardia duodenalis, Toxoplasma gondii, Balantidium Coli, Theileria parva, Theileria annulate, Phipicephalus appendiculatus, Prototheca moriformis*, and any combination thereof.

[0062] Preferably, the protozoan antigen is selected from the group consisting of dense granule protein 6 (GRA6), rhoptry protein 2A (ROP2A), rhoptry protein 18 (ROP18), surface antigen 1 (SAG1), surface antigen 2A (SAG2A), apical membrane antigen 1 (AMA1) of *Toxoplasma gondii*, and any combination thereof.

[0063] In a still further embodiment, the at least one protein antigen is a viral antigen from a virus selected from the group consisting of Human Papilloma Virus (HPV), Human Immunodeficiency virus HIV (e.g. HIV-1, HIV-2), Hepatitis A virus, Hepatitis B virus (HBV), Hepatitis C Virus, Hepatitis D Virus, Hepatitis E Virus, Herpes virus (Human Gamma herpes virus 4 (Epstein Barr virus), herpes simplex virus 2 (HSV2), human herpes virus 8, Influenza Virus (e.g. influenza A virus, influenza B virus), cytomegalovirus, Crimean-Congo hemorrhagic fever orthonairovirus, corona viruses, Human polyomavirus 2, BK virus, Severe acute respiratory syndrome coronavirus (SARS-CoV, SARS-Cov-2, COVID-19), Middle East respiratory syndrome-related coronavirus (MERS-CoV), noroviruses, filoviruses (Cueva, Marburg and Ebola virus), Chikungunya virus, Human alphaherpesvirus 3 (HHV-3) or varicella-zoster virus (VZV), Rubella virus, Merkel cell polyomavirus (MCV), bunyavirus (e.g., hanta virus), arena virus (e.g., lymphocytic choriomeningitis mammarenavirus (LCMV) and Lassa virus), flavivirus (Dengue virus, Zika virus, Japanese encephalitis, West Nile, Tick-borne encephalitis virus (TBEV) and Yellow fever), rhinovirus, Human parainfluenza viruses (HPIVs), enterovirus (e.g polio), Respiratory syncytial virus (RSV), Mumps virus, Coxsackievirus, Measles virus, astrovirus (e.g., gastroenteritis), rhabdoviridae (e.g., rabies), Adenovirus, Adeno-associated virus (AAV), oncogenic viruses, including human papillomavirus, hepatitis B virus, hepatitis C virus, Epstein-Barr virus, Kaposi's sarcoma-associated herpesvirus, human T-lymphotropic virus and Merkel cell polyomavirus, and any combination thereof.

[0064] According to a preferred embodiment of the invention, the viral antigen is selected from the group consisting of Spike proteins also called Surface Glycoprotein of Severe acute respiratory syndrome coronavirus 2 or SARS-COV-2 or COVID-19, N protein also called Nucleocapside phosphoprotein of Severe acute respiratory syndrome coronavirus 2 or SARS-COV-2 or COVID-19, M protein also called Membrane Glycoprotein of Severe acute respiratory syndrome coronavirus 2 or SARS-COV-2 or COVID-19, Hemagglutinin (HA) protein of influenza A virus H5N1, Hemagglutinin (HA) protein of influenza A virus H3N2, Hemagglutinin (HA) protein of influenza A virus H1N1, Hemagglutinin (HA) protein of influenza A virus H7N9, Hemagglutinin (HA) protein of influenza A virus H1N1, Hemagglutinin (HA) protein of influenza A virus H2N2, Hemagglutinin (HA) protein of influenza B virus, Neuraminidase (NA) protein of influenza A virus H5N1, Neuraminidase (NA) protein of influenza A virus H1N1, Neuraminidase (NA) protein of influenza A virus H3N2, Neuraminidase (NA) protein of influenza A virus H7N9, Neuraminidase (NA) protein of influenza A virus H9N2, Neuraminidase (NA) protein of influenza A virus H2N2, Neuraminidase (NA) protein of influenza A virus H1N1, Neuraminidase (NA) protein of influenza B virus, envelope protein of Human immunodeficiency virus (HIV1), envelope protein of Human immunodeficiency virus (HIV2), Major Capsid Protein L1 of Human Papilloma Virus (HPV), Minor Capsid Protein L2 of Human Papilloma Virus (HPV), glycoprotein of Rabies lyssavirus, glycoprotein of Human Cytomegalovirus, envelope glycoproteins E1E2 of Hepatitis C virus, Fusion protein (F) of Respiratory syncytial virus (RSV), spike glycoprotein of Zaire ebolavirus, Protein prM of Zika virus, Serine protease NS3 of Zika virus, Serine protease subunit NS2B of Zika virus, Envelope protein E of Zika virus, Capsid protein C of Zika virus, SARS-CoV-2 Spike(S) RBD protein, and any combination thereof.

[0065] In an exemplary embodiment, the encoded at least one protein antigen as above defined comprises, consists essentially or consists of an amino acid sequence selected from the group consisting of SEQ ID NOs.: 1-13, 15, 16, 18, and 20-49.

[0066] In another embodiment of the invention, the exogenous nucleic acid molecule loaded in the EVs is a mRNA molecule comprising or consisting of a nucleotide sequence selected from the group consisting of SEQ ID NOs. 14, 17, 19 and 50. More particularly, SEQ ID NOs. 14, 17, 19 and 50 correspond to mRNA sequences coding for SARS-COV-2 S protein, N protein, M protein and Spike(S) RBD protein, respectively.

[0067] According to the present invention, it is envisaged that the composition may comprise engineered, plant-derived EVs loaded with a single exogenous nucleic acid molecule or, alternatively, a combination of engineered, plant-derived EVs loaded with different exogenous nucleic acid molecules.

[0068] It is understood that the protein antigen within the scope of the invention may comprise one or more modifications in order to improve antigen immunogenicity and/or stability. Exemplary modifications include post-translational modifications.

[0069] The composition according to the invention may be used alone or in combination with other vaccines.

[0070] In one embodiment, the composition according to the invention further comprises one or more polycationic substances, said one or more polycationic substances being associated with the outer lipid layer of the lipid bilayer membrane of the EVs through electrostatic interactions.

[0071] Preferably, the one or more polycationic substances are selected from the group consisting of cationic proteins, including protamine, calcitonin peptides, plectasin, lactoferrin, protamine-like proteins, such as spermine or spermidine, nucleoline, histones, cell penetrating peptides (CPPs); cationic peptides, including histidine-rich peptides, arginine-rich peptides, lysine-rich peptides, cationic arginine-rich peptides (CARPs); polypeptides, including poly-arginine, poly-lysine, poly-histidine, histidine-rich peptides, arginine-rich peptides, lysine-rich peptides; polysaccharides, including chitosan, glycosaminoglycan such as polysulfated glycosaminoglycan (PSGAG), cationic dextrans; glycerol, polyethylene glycol (PEG).

[0072] A preferred polycationic substance is protamine.

[0073] Preferably, the content of the one or more polycationic substances in the composition is in the range of from 0.001 to 2 $\mu g/10^{10}$ EVs, more preferably from 0.05 to 1 $\mu g/10^{10}$ EVs, even more preferably from 0.1 to 0.4 $\mu g/10^{10}$ EVs.

[0074] According to the invention, the one or more polycationic substances may be used alone or in combination. It is understood that the polycationic substance can be used in its native form or with chemical modifications. Such components may be used individually or in combination.

[0075] In another embodiment according to the invention, the EVs in the composition of the invention are additionally loaded with one or more sugar molecules, said one or more sugar molecules being associated with the exogenous nucleic acid molecule loaded into the EVs through electrostatic interactions and hydrogen bonding.

[0076] Preferably, the one or more sugar molecules are selected from the group consisting of disaccharides, including trehalose, maltose, lactose, sucrose, cellobiose, chitobiose, kojibiose, nigerose, isomaltose, β,β-trehalose, α,β-trehalose, sophorose, laminaribiose, gentiobiose, trehalulose, turanose, maltulose, leucrose, iso-maltulose, gentiobiulose, mannobiose, melibiose, melibiulose, rutinose, rutinulose, xylobiose; sugar alcohols, including arabitol, erythritol, glycerol, HSHs, isomalt, lactitol, maltitol, mannitol, sorbitol, xylitol; polysaccharides, including starch, glycogen, galactogen, inulin, arabinoxylans, cellulose, chitin and pectin.

[0077] A preferred sugar molecule is trehalose. Trehalose is a non-reducing disaccharide sugar commonly used as a cytoprotectant to stabilize proteins and nucleic acids. Additionally, trehalose can resolve secondary structures of RNA.

[0078] Preferably, the content of the one or more sugar molecules in the EVs according to the invention is in the range of from 0.1 to 10 $mg/10^{10}$ EVs, more preferably from 0.5 to 5 $mg/10^{10}$ EVs, even more preferably from 1 to 2 $mg/10^{10}$ EVs.

[0079] In another embodiment, the content of the one or more sugar molecules in the EVs according to the invention is in the range of from 0.1 to 20 mg/μg of loaded exogenous nucleic acid, preferably from 1 to 10 mg/μg of loaded exogenous nucleic acid, more preferably from 2 to 6 mg/μg of loaded exogenous nucleic acid.

[0080] It is understood that the sugar molecules can be used in their native form or with chemical modifications. Such components may be used individually or in combination.

[0081] Before usage, the non-naturally occurring EVs in the composition of the invention may be lyophilized and resuspended with water. Alternatively, the non-naturally occurring EVs used in the composition of the invention may be freshly prepared or stored at 4° C., −20° C. or −80° C.

[0082] The composition according to the present invention may be formulated in several administrable forms, including powders, granules, tablets, capsules, suspensions, emulsions, syrups, aerosols, pills, sugar-coated tablets, capsules, liquids, gels, syrups, slurries, and suspensions.

[0083] The composition of the invention may optionally contain suitable excipients, preservatives, solvents or diluents according to conventional method.

[0084] Exemplary excipients include, but are not limited to, sugars, including sucrose, D-mannose, D-fructose, dextrose, anhydrous lactose, D-trehalose, D-sorbitol; proteins, including human serum albumin, hydrolyzed casein, MRC-5 cellular proteins, hydrolyzed gelatin, CRM197 carrier protein, proteins from plants, yeast, bacteria, eggs; essential and non-essential aminoacids such as asparagine, phenylalanine, arginine, histidine; sodium, including sodium chloride, sodium bicarbonate, sodium carbonate, sodium borate, sodium benzoate, sodium taurodeoxycholate, sodium deoxycholate, monobasic sodium phosphate, dibasic sodium phosphate, sodium metabisulphite; potassium, including potassium phosphate, polacrilin potassium, monobasic and dibasic potassium phosphate, potassium chloride; magnesium stearate, calcium chloride, calcium phosphate, calcium silicate, glutamate, cellulose, microcrystalline cellulose, cellulose acetate phthalate, aluminum, aluminum hydroxide, aluminum phosphate, amorphous aluminum hydroxyphosphate sulfate, potassium aluminum sulfate, citric acid, iron ammonium citrate, castor oil, neomycin, streptomycin, aminoglycoside, kanamycin, gentamicin, chlortetracycline, amphotericin B, plasdone C, alcohol, acetone, benzethonium chloride, formaldehyde, glycerin, ascorbic acid, trometamol, urea, glutaraldehyde, 2-phenoxyethanol, polysorbate 80 (Tween 80), polymyxin B, ammonium thiocyanate, tromethamine, host cell DNA benzonase, formalin, phosphate-buffered saline solution, polysorbate 20, deoxycholate, dibasic dodecahydrate, monobasic dehydrate, formalin, polymyxin B, beta-propiolactone, hydrocortisone, squalene, sorbitan trioleate, barium, cetyltrimethylammonium bromide (CTAB), octoxynol-10 (TRITON X-100), α-tocopheryl hydrogen succinate, cetyltrimethlyammonium bromide, and β-propiolactone, thimerosal, ethylenediaminetetraacetic acid (EDTA), phenol, beta-propiolactone, DMEM, HEPES, polydimethylsiloxane, vitamins, dioleoyl phosphatidylcholine (DOPC), 3-O-desacl-4'monophosphoryl lipid A (MPL), lipids, cholesterol, panthenol, gums, including guar gum, boric acid and borates, including sodium tetraborate, glycerol, allantoin, triethanolamine, alginate, pluronic, poloxamer, including P188, P331; PEG, including PEG8000; glycols, including ethylene glycol, propylene glycol, and glycerol; citicoline (cyti-dine-5-diphosphocholine; CDP-choline), cholesterol.

[0085] Illustrative, non-limiting examples of preservatives suitable for use in the composition of the invention include parabens, including ethyl paraben, methyl paraben, propyl paraben, formaldehyde donors including DMDM hydantoin, imidazolidinyl urea, and glutaraldehyde, phenol derivatives, benzoic acid, benzyl alcohol.

[0086] Suitable solvents or diluents to be used in the invention may be selected from purified water, ethanol and benzyl alcohol.

[0087] According to the invention, it is contemplated that an adjuvant can be added to the composition for use as a vaccine.

[0088] Illustrative, non-limiting examples of adjuvants suitable for use in the immunogenic composition of the invention are mineral compositions, including aluminum salts such as aluminium hydroxide, aluminium potassium phosphate, AS04, and others, calcium salts, hydroxides (e.g. oxyhydroxides), phosphates (e.g. hydroxyphoshpates, orthophosphates), sulphates; emulsions, including oil-in-water and water-in-oil emulsions, such as Freund's adjuvant, complete Freund's adjuvant, incomplete Freund's adjuvant, MF59, AF03, AS03, AS02, glucopyranoside lipid adjuvant (GLA-SE), glucopyranosyl lipid adjuvant (GLA);

[0089] bacterial or microbial derivatives, including non-toxic derivatives of enterobacterial lipopolysaccharide (LPS), monophosphoryl lipid A (MPL), 3-O-deacylated MPL (3dMPL), lipid A, lipid A from *Escherichia coli* such as OM-174. OM-174; immunostimulatory oligonucleotides, including nucleotide sequences containing a CpG motif, bacterial double stranded RNA, oligonucleotides containing palindromic or poly(dG) sequences, ADP-ribosylaling toxins and detoxified derivatives, RC529; cyclic GMP-AMP adjuvant, STING agonists, CAF01, immunostimulating complexes (ISCOMs), ISCOMATRIX, AS01; polyoxyethylene ether and polyoxyethylene ester formulations, polymeric particles, such as poly(lactide-co-glycolide) (PLG) microparticles, polyphosphazene (PCPP), saponin formulations, such as saponin derived from Smilax ornata (sarsaprilla), Gypsophilla paniculata (brides veil), and Saponaria officinalis (soap root), purified formulations, such as QS7, QS17, QS18, QS21, QH-A, QH-B and QH-C; human immunomodulators, including cytokines, such as interleukins (e.g. IL-1, IL-2, IL-4, IL-5, IL-6, IL-7, IL-12, etc.), interferons (e.g. interferon-γ), macrophage colony stimulating factor, and tumor necrosis factor; bioadhesives and mucoadhesives, including esterified hyaluronic acid microspheres, or mucoadhesives such as cross-linked derivatives of poly (acrylic acid), polyvinyl alcohol, polyvinyl pyrollidone, polysaccharides and carboxymethylcellulose, chitosan and derivatives thereof; t muramyl peptides, including N-acetyl-muramyl-Lthreonyl-D-isoglutamine (thr-MDP), N-acetyl-normuramyl-L-alanyl-D-isoglutamine (nor-MDP), and N-acetylmuramyl-Lalanyl-D-isoglutaminyl-L-alanine-2-(1'-2'-dipalmitoyl-sn-glycero-3-hydroxyphosphoryloxy)-ethyl-amine MTP-PE); imidazoquinolone compounds, including Imiquamod and its homologues; virosomes and Virus Like Particles (VLPs).

[0090] The composition of the invention may be administered via various routes, including oral, intranasal, parenteral, including subcutaneous, intraperitoneal, intravenous, intradermal, intramuscular, intrasplenic, and intranodal.

[0091] Preferably, the pharmaceutical composition of the present invention is in a form suitable for oral, intranasal or parenteral administration.

[0092] The administration dose, the number and frequency of applications are determined according to various factors, such as the disease to treat or prevent and the patient's characteristics, and can be determined by a person of ordinary skill in the art by using his/her normal knowledge.

[0093] In addition, the composition according to the invention may be lyophilized and is stable without the need of a cold-chain storage.

[0094] Within the scope of the present invention is also a method for the preparation of a composition having the features as above-defined.

[0095] According to the invention, the method comprises the steps of:

[0096] (i) contacting and mixing a suspension of plant-derived extracellular vesicles (EVs) with one or more polycationic substances to obtain a first mixture;

[0097] (ii) contacting and mixing a preparation of nucleic acid molecules with one or more sugar molecules to obtain a second mixture, said nucleic acid molecules encoding at least one protein antigen;

[0098] (iii) admixing said first mixture and said second mixture to obtain a third mixture; and

[0099] (iv) adding to said third mixture a pre-determined volume of water, wherein the ratio of said pre-determined volume of water to the volume of the third mixture is comprised within 5:1 to 15:1, preferably within 8:1 to 12:1.

[0100] A preferred ratio of the pre-determined volume of water to the volume of the third mixture is 10:1.

[0101] Optionally, the method according to the invention may further comprise concentrating the composition obtained in step (iv). Concentration techniques are well known and include, for example, filtration, ultracentrifugation, tangential flow filtration, chromatography and precipitation. The skilled person will be aware of techniques for concentrating a composition, and any such suitable method may be used.

[0102] In one embodiment of the method of the invention, in step (i) mixing further comprises the step of incubating the first mixture for a time ranging from 30 minutes to 2 hours, preferably for 1 hour, at a temperature ranging from 30 to 40° C., preferably at 37° C.

[0103] In another embodiment of the method of the invention, in step (ii) mixing further comprises the step of incubating the second mixture for a time ranging from 5 to 30 minutes, preferably for 10 minutes, at a temperature ranging from 0 to 25° C., preferably at 20° C.

[0104] In yet another embodiment of the method of the invention, in step (iii) admixing further comprises the step of incubating the third mixture for a time ranging from 1 to 5 hours, preferably 3 hours, at a temperature ranging from 30 to 40° C., preferably at 37° C.

[0105] In a further embodiment of the method of the invention, step (iv) further comprises an incubation step performed for a time ranging from 5 to 24 hours, preferably 12 hours, at a temperature ranging from 0 to 10° C., at 4° C.

[0106] Suitable polycationic substances and sugar molecules for use in the method according to the invention are as above described with reference to the composition.

[0107] Without wishing to be bound by any theory, the inventors believe that the polycationic substance may alter the charge of the lipid bilayer membrane of the plant-derived EVs and allow the adsorption of the nucleic acid molecules on the outer surface of such membrane. Further, the inventors believe that the sugar may play a protecting role of nucleic acid molecules in order to allow an efficient introduction of these molecules into the plant-derived EVs.

[0108] Preferably, the concentration of plant-derived EVs in the first mixture is comprised within the range of from 5×10^{10} to 10^{12} EVs/ml on the total volume of said first mixture, more preferably from 1×10^{11} to 5×10^{11} EVs/ml on the total volume of said first mixture.

[0109] The first mixture according to the method of the invention may further comprise a salt, preferably NaCl, more preferably NaCl at a concentration of 0.9% (w/v) on the total volume of said first mixture.

[0110] In one embodiment of the method of the invention, the one or more polycationic substances are present in the first mixture at a concentration comprised within the range of from 0.1 to 2 μg/ml on the total volume of said first mixture, preferably from 0.1 to 1 μg/ml on the total volume of said first mixture, more preferably from 0.4 to 0.6 μg/ml on the total volume of said first mixture.

[0111] In another embodiment of the invention, the nucleic acid molecule is present in the second mixture at a concentration comprised within the range of from 0.1 to 10 μg/ml on the total volume of said second mixture, preferably from 0.1 to 1 μg/ml on the total volume of said second mixture, more preferably from 0.1 to 0.5 μg/ml on the total volume of said second mixture.

[0112] In a still another embodiment according to the invention, the one or more sugar molecules are present in the second mixture at a concentration comprised within the range of from 1 to 20% (w/v) on the total volume of said second mixture, preferably from 1 to 10% (w/v) on the total volume of said second mixture, more preferably from 1 to 5% (w/v) on the total volume of said second mixture.

[0113] According to the method of the invention, the mixing of the suspension comprising plant-derived EVs with the polycationic substance in step (i) and/or the mixing of the preparation of nucleic acid molecules with one or more sugar molecules in step (ii) may be performed by vortexing, preferably for a period of time of at least 30 seconds.

[0114] According to the invention, it is envisaged that the method may comprise further manipulations to improve the loading of nucleic acid molecules into plant-derived EVs including, but not limited to, electroporation, sonication, transfection, incubation, cell extrusion, saponin-mediated permeabilization, and freeze-thawing.

[0115] Another aspect of the present invention is a composition comprising non-immunomodulating, engineered, plant-derived extracellular vesicles (EVs), obtainable by a method as above defined, for use as a vaccine.

EXAMPLES

[0116] The following experimental section is provided purely by way of illustration and is not intended to limit the scope of the invention as defined in the appended claims. In the following experimental section, reference is made to the appended drawings, wherein:

[0117] FIG. **1** shows the characterization of engineered, plant-derived EVs of the invention in experimental example 1 compared to native plant-derived EVs. Representative image of Nanosight analysis of native EVs (A) and engineered EVs of the invention (B) from kiwifruit showing a significant difference in size. Statistical analysis of the mean diameter of n=3 preparations of native plant-derived EVs and engineered, plant-derived EVs of the invention analyzed by Nanosight (C). E1=engineered EVs from cabbage EV, E2=engineered EVs from blueberry EV. p: **** <0.001.

[0118] FIG. **2** shows the values of the membrane potential across the lipid bilayer membrane (Z potential) measured in EVs in experimental example 1. The membrane potential was measured as mVolt (mV) in native EVs (native EV) and engineered EVs of the invention from courgette (E1) and blueberry (E2). The statistical significance was calculated comparing the membrane potential measured for engineered plant-derived EVs with the values determined for native plant-derived EVs. p: *** <0.005. N=3 experiments were performed for each data set. Data are shown as mean±standard deviation (SD).

[0119] FIG. **3**A shows the protein content of native plant-derived EVs and engineered plant-derived EVs of the invention in experimental example 1 expressed as nanograms (ng) of protein in 10^{10} EVs. Measurements were performed on native plant-derived EVs (native EV) and engineered, plant-derived EVs from pomegranate (E1) and kiwifruit (E2). The statistical significance was calculated comparing the protein content of engineered plant-derived EVs with the values measured in native plant-derived EVs. p: ** <0.01. N=3 experiments were performed for each data set. FIG. **3**B shows the percentage of EVs in the composition of the invention in experimental example 1 containing phosphatidylserine in the outer layer of the lipid bilayer membrane. The presence of phosphatidylserine in the outer layer of vesicles membrane was analyzed in compositions comprising native plant-derived EVs (native EV) and compositions comprising engineered, plant-derived EVs from cabbage (E1) and blueberry (E2). In each sample, the phosphatidylserine content was measured using cytofluorimetric assay (FACS) as staining to Annexin V and expressed as percentage of fluorescent signal. The statistical significance was calculated comparing the percentage of engineered plant-derived EVs containing phosphatidylserine with native plant-derived EVs. p: ** <0.01. N=3 experiments were performed for each data set. Data are shown as mean±standard deviation (SD).

[0120] FIG. **4** shows the results of the immunomodulatory assay on engineered plant-derived EVs of the invention in experimental example 1. PBMC cells were incubated with engineered plant-derived EVs from pomegranate (dose of 50,000 particles/cell) for 48 hours and cellular proliferation was measured by BrdU incorporation. (A) The histogram shows the absorbance (mean±SD) for untreated PBMC (CTR) and PBMC stimulated with the EVs of the invention. Absorbance is directly proportional to cell proliferation. The proliferation rate of PBMC stimulated with the EVs of the invention is unchanged and not statistically significant compared to control (CTR). Then, lymphocytes were activated with LPS (dose of 100 ng/ml) and stimulated with the EVs of the invention (dose of 50,000 particles/cell) for 48 hours and proliferation was measured by BrdU incorporation. (B) The histogram shows the absorbance (mean±SD) of non-stimulated PBMC (CTR−), PBMC treated with LPS (CTR+), PBMC treated with LPS and the EVs of the invention from pomegranate (E1), and PBMC treated with LPS and the EVs of the invention from kiwifruit (E2). Absorbance is directly proportional to cell proliferation. LPS significantly activates PBMC proliferation compared to untreated cells, while the proliferation rate of PBMC stimulated with LPS and the EVs of the invention is unchanged and not statistically significant compared to PBMC treated with LPS. PBMC proliferation was also measured by using the fluorescent dye CFSE. PBMC were stimulated with the EVs of the invention (dose of 50,000 vesicles/cell) for 24 hours, then proliferation was analyzed by flow cytometry (C, D). The histogram (C) shows fluorescent FITC intensity (mean±SD) for untreated PBMC (CTR) and PBMC stimulated with the EVs of the invention from cabbage (E1), celery (E2), and courgettes (E3). The proliferation rate of PBMC stimulated with the different samples of EVs of the invention is unchanged and not statistically significant compared to CTR. In fact, the histograms of flow cytometry analysis of CTR, E1, E2, and E3 were completely overlapping (D). p: * <0.05; ns>0.05.

[0121] FIG. **5** shows the total RNA content of EVs in experimental example 1. The total RNA content was measured in native plant-derived EVs and engineered plant-derived EVs from celery (E1), pomegranate (E2) and kiwifruit (E3). The total RNA content was measured with absolute quantification of the RNA contained in each sample after the RNA extraction and was expressed as the amount of RNA (ng) normalized for the number of vesicles (ng/10^9 vesicles). The statistical significance was calculated comparing the total RNA content value of each sample of EVs of the invention with native plant-derived EVs. p: * <0.05, *** <0.005, **** <0.001. N=3 experiments were performed for each data set. Data are shown as mean±standard deviation (SD).

[0122] FIG. **6** shows the quantification of exogenous nucleic acid molecules loaded in the engineered, plant-derived EVs of the invention in experimental example 2. For the assay, mRNA molecules coding for the nucleocapsid (N) protein of SARS-CoV-2 were used and loaded nucleic acid molecules were measured by qRT-PCR in native plant-derived EVs (native EV) and engineered, plant-derived EVs from kiwifruit (E1 and E3) and celery (E2 and E4). Two different doses of mRNA were used: 0.1 μg/ml for samples E1 and E2, and 1 μg/ml for samples E3 and E4. The amount of loaded mRNA was expressed as RQ value accordingly to method described. The statistical significance was calculated comparing the amount of loaded nucleic acid in each sample of EVs of the invention with native plant-derived EVs. p: **** <0.001. N=3 experiments were performed for each data set. Data are shown as mean±standard deviation (SD).

[0123] FIG. **7** shows the resistance of the nucleic acid molecules loaded in engineered, plant-derived EVs of the invention to degrading environments in experimental example 3. For the experiments, mRNA molecules were used and measured by qRT-PCR assay. Graphs indicate the percentage of mRNA molecules still present after the degrading assay in comparison to the starting material and show that a total of 100% of mRNA is preserved in the EVs of the invention. (A) The resistance to enzyme degradation was measured after treatment with RNAse, whereas (B) the resistance to gastrointestinal environment was evaluated after the treatment with a stomach-like solution. In all experiments, naked mRNA was used as control. The experiments were performed on engineered, plant-derived EVs from pomegranate (7A) and kiwifruit (7B). The statistical significance was calculated comparing the percentage of the nucleic acid preserved in the EVs of the invention with naked mRNA. p: **** <0.001. N=3 experiments were performed for each data set. Data are shown as mean±standard deviation (SD).

[0124] FIG. **8** shows the resistance of nucleic acid molecules loaded in engineered, plant-derived EVs of the invention to storage in experimental example 4. For these experiments, EVs derived from celery (E1) and kiwifruit (E2) were loaded with mRNA molecules coding for the nucleocapsid (N) protein of SARS-CoV-2. The amount of preserved mRNA into the EVs after lyophilization and storage at +4° C. for 7 days was measured by qRT-PCR assay and expressed as percentage relative to the starting amount. N=3 experiments were performed for each data set. Data are shown as mean±standard deviation (SD).

[0125] FIG. **9** shows the transfer of the nucleic acid molecules loaded into the engineered plant-derived EVs of the invention to recipient cells in experimental example 5. EVs of the invention loaded with mRNA molecules coding for the nucleocapsid (N) protein of SARS-CoV-2were incubated with macrophages. After 24 hours, the amount of mRNA was measured in recipient cells using molecular analysis (qRT-PCR), normalized to GAPDH as housekeeping and expressed as RQ value as described in the method section. The RQ values were normalized to the control (untreated cells, NT) and a RQ value of 1 means that the mRNA is not detectable in the sample. Macrophages were treated with native plant-derived EVs (native EV), engineered plant-derived EVs (E1, E2, E3), plant-derived EVs incubated with mRNA (EV+mRNA) without nucleic acid loading, naked mRNA. Recipient cells were treated with a dose of 50,000 particles/cell. The experiment was performed with EVs from cabbage (E1), pomegranate (E2) and kiwifruit (E3). The statistical significance was calculated comparing the RQ value of the mRNA for each sample with untreated cells as control (NT). p: *** <0.005, **** <0.001. N=3 experiments were performed for each data set. Data are shown as mean±standard deviation (SD).

[0126] FIG. **10** shows the functionality of nucleic acid molecules carried by engineered plant-derived EVs of the invention in recipient cells in experimental example 5. For these experiments, the EVs of the invention loaded with mRNA molecules coding for the green fluorescent protein (GFP) were incubated with (A) endothelial cells and (B) macrophages as recipient cells. After 24 hours of co-incubation, the expression of the protein encoded by the exogenous mRNA in recipient cells was detected as fluorescent signal using cytofluorimetric analysis (FACS). Recipient cells were treated with native plant-derived EVs (native EV), engineered plant-derived EVs of the invention or naked mRNA at a dose of 50,000 particle/cell. The experiments were performed with plant-derived EVs from courgette (FIG. **10**A) and kale (FIG. **10**B). The statistical significance was calculated comparing the percentage of signal intensity for each sample with untreated cells as control (NT). p: *** <0.005, **** <0.001. N=3 experiments were performed for each data set. Data are shown as mean±standard deviation (SD).

[0127] FIG. **11** shows the protein expression in target recipient cells treated with nucleic acid molecules carried by engineered plant-derived EVs of the invention in experimental example 5. For these experiments, the EVs of the invention loaded with mRNA molecules coding for the SARS-CoV-2 Spike Glycoprotein (S1) RBD protein (SEQ ID NO. 50), SARS-CoV-2 Spike Glycoprotein full protein (SEQ ID NO. 14), or SARS-CoV-2 Nucleocapsid Protein (SEQ ID NO. 17) were incubated with endothelial cells as recipient cells. After 24 hours of co-incubation, the expression of the protein encoded by the exogenous mRNA in recipient cells was detected using fluorescent-labelled secondary antibody and the fluorescent signal was measured with cytofluorimetric analysis (FACS). Recipient cells were treated with native plant-derived EVs (native EV) or engineered plant-derived EVs of the invention at a dose of 1.2×10^{10} particles. The experiments were performed with orange-derived EVs. The statistical significance was calculated comparing the percentage of signal intensity for each sample with untreated cells as control (NT). p: **** <0.001. N=3 experiments were performed for each data set. Data are shown as mean±standard deviation (SD).

[0128] FIG. **12** shows that plant-derived EVs of the invention engineered with nucleic acids, and not native EVs, are able to activate lymphocytes after incorporation into macrophages in experimental example 6. For these experiments, the EVs of the invention loaded with mRNA molecules coding for the SARS-CoV-2 Spike Glycoprotein (S1) RBD protein (SEQ ID NO. 50), SARS-CoV-2 Spike Glycoprotein full protein (SEQ ID NO. 14), or SARS-CoV-2 Nucleocapsid Protein (SEQ ID NO. 17) were incubated with APC cells (macrophages) as recipient cells. After EV incorporation in APCs, PBMC cells were added and the treatment with engineered EVs was repeated every five days for two times. At the end, lymphocytes were analyzed by cytofluorimetric analysis (FACS). Lymphocytes, identified by the expression of CD4+, were evaluated for their activation. Lymphocyte activation was measured as increase of lymphocyte proliferation (A) and increase of the expression of lymphocyte activation markers CD25+ (B) and HLADR+ (C) of lymphocytes CD4+. Cells were treated with native plant-derived EVs (native EV) or engineered plant-derived EVs of the invention at a dose of 1.2×10^{10} particles. The experiments were performed with orange-derived EVs. The statistical significance was calculated comparing the percentage of signal intensity for each sample with untreated cells as control (NT) or cells treated with native EVs. Positive controls were represented by treatment with beads human T-Activator CD3/CD28 (CTR+) and purified proteins (SARS-CoV-2 Spike Glycoprotein RBD protein (S protein), or SARS-CoV-2 Nucleocapsid protein (N protein)). p: * <0.05, ** <0.01, *** <0.005, **** <0.001. N=3 experiments were performed for each data set. Data are shown as mean±standard deviation (SD).

[0129] FIG. **13** shows that plant-derived EVs of the invention engineered with nucleic acids, and not native EVs, are able to induce specific immune response in mice in experimental example 7. The graph shows the absorbance measurement of IgA immunoglobulins specific for SARS-CoV-2 Spike Glycoprotein (S1) RBD protein induced by vaccination. For this experiment, mice were immunized at day 0 and day 21 and serum was analyzed at day 35 following sacrifice. Mice were treated with native EVs (native EV) or EV engineered with mRNA molecules coding for the SARS-CoV-2 Spike Glycoprotein RBD (S1). The treatment was administered via intramuscular or oral routes. The humoral immune response as IgA induction was assessed by ELISA using ELISA plates coated with the SARS-CoV-2 Spike Glycoprotein RBD (S1). The statistical significance was calculated comparing the signal intensity of native and engineered EVs for each administration route. p: * <0.05, ** <0.01 N=3 animals for each data set. Data are shown as mean±standard deviation (SD).

MATERIALS AND METHODS

Extracellular Vesicles Isolation

[0130] Extracellular vesicles were isolated from fresh fruit juice (kiwifruit, pomegranate, blueberry, orange, lemon) or fresh plant extract (courgette, cabbage, kale, celery). The juice or extract was sequentially filtered using decreasing order of pores to remove fibers. EV were then purified with differential ultracentrifugation or tangential flow filtration.

For differential ultracentrifugation, the juice was centrifuged at 1,500 g for 30 minutes to remove debris and other contaminants. Then, EV were purified by ultracentrifugation at 10,000 g followed by ultracentrifugation at 100,000 g for 1 hour at 4° C. (Beck-man Coulter Optima L-90K). The final pellet was resuspended with phosphate buffered saline added with 1% DMSO and filtered with 0.22 micrometer filters to sterilize.

[0131] Extracellular vesicles were used or stored at −80° C. for long time. For tangential flow filtration, at first the juice was clarified by filtration with depth filter sheet discs Supracap 50 (Pall) to exclude fibers and debris. Then, the filtered juice was purified by concentration and diafiltration using a tangential flow filtration cassette TFF Omega (Pall Cadence). Finally, the retentate from tangential flow filtration was sterilized by filtration with a 0.2 nm filter.

Nanoparticle Tracking Analysis (NTA)

[0132] Nanoparticle tracking analysis (NTA) was used to define the EV dimension and profile using the NanoSight LM10 system (Malvern), equipped with a 405 nm laser and with the NTA 3.1 analytic software. The Brownian movements of EV present in the sample subjected to a laser light source were recorded by a camera and converted into size and concentration parameters by NTA through the Stokes-Einstein equation. Camera levels were for all the acquisition at 16 and for each sample, three videos of 30 s duration were recorded. Briefly, purified EVs were diluted 1:2000 in 1 ml vesicle-free saline solution (Fresenius Kabi). NTA post-acquisition settings were optimized and maintained constant among all samples, and each video was then analyzed to measure EV mean, mode and concentration.

Production of the EVs of the Invention

[0133] The engineered plant-derived EVs of the invention were produced by sequential steps as described as follows. Briefly, plant-derived EVs were mixed with a cationic peptide and the reaction was carried out at 37° C. for 1 hour. A preparation of nucleic acid molecules was mixed with a sugar and the reaction was carried out at 20° C. for 10 minutes. Then, the two solutions were mixed and the reaction was carried out at 37° C. for 3 hours. Then, water was added to the reaction and samples were put at 4° C. for 12 hours. In order to purify the engineered plant-derived EVs from remaining free nucleic acid molecules, samples were washed by ultracentrifugation at 100,000 g for 2 hours at 4° C. (Beckman Coulter Optima L-90K, Fullerton, CA, USA) and samples were resuspended in saline solution.

Measurement of EVs Membrane Potential

[0134] The analysis was performed by using the Zeta-sizer nanoinstrument (Malvern Instruments, Malvern, UK). All samples were analyzed at 25° C. in filtered (cutoff=200 nm) saline solution. Zeta-potential (slipping plane) was generated at x distance from the vesicle indicating the degree of electrostatic repulsion between adjacent, similarly charged vesicles in a dispersion.

Protein Extraction and Quantification

[0135] Proteins were extracted from EVs samples using RIPA buffer (150 nM NaCl, 20 nM Tris-HCl, 0.1% sodium dodecyl sulfate, 1% deoxycholate, 1% Triton X-100, pH 7.8) supplemented with a cocktail of protease and phosphatase inhibitors (Sigma-Aldrich, St. Louis, Missouri, USA). The protein content was quantified by BCA Protein Assay Kit (Thermo Fisher Scientific, Waltham, Massachusetts, USA) following manufacturer's protocol. Briefly, 10 μl of sample were dispensed into wells of a 96-well plate and total protein concentrations were determined using a linear standard curve established with bovine serum albumin (BSA).

Phosphatidylserine Analysis

[0136] For phosphatidylserine analysis, EVs samples were stained with Annexin V FITC and FITC isotype (Miltenyi Biotec, Germany) for 30 minutes and diluted with saline solution before acquisition. Samples were characterized by cytofluorimetric analysis using the CytoFLEX flow cytometer (Beckman Coulter) with CytExpert software and the percentage of signal positivity was measured for each sample using FITC isotype as background.

RNA Extraction and Quantification

[0137] Total RNA was isolated from EVs and cells using the miRNeasy Mini Kit (Qiagen, Hilden, Germany) according to the manufacturer's protocol and resuspended in water. RNA concentration of samples was quantified using spectrophotometer (mySPEC, VWR, Radnor, PA, USA).

mRNA Detection Using qRT-PCR

[0138] From RNA samples, cDNA was obtained using High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems). Five nanograms of cDNA were added to SYBR GREEN PCR Master Mix (Applied Biosystems) and run on a 96-well QuantStudio12K Flex Real-Time PCR (qRT-PCR) system (Thermo Fisher Scientific, Waltham, MA, USA). GAPDH was used as a housekeeping gene in cell samples. Fold change (Rq) in mRNA expression among all samples was calculated as 2-ΔΔCt respect control samples.

Cell Cultures

[0139] Human microvascular endothelial cells (HMEC) were obtained by immortalization with simian virus 40 of primary human dermal microvascular endothelial cells. HMEC were cultured in Endothelial Basal Medium supplemented with bullet kit (EBM, Lonza, Basel, Switzerland) and 1 ml Mycozap CL (Lonza). Macrophage MV-4-11 cell line (ATCC® CRL9591™) was obtained by ATCC and cultured in Iscove's Modified Dulbecco's Medium supplemented with 10% of fetal bovine serum (ATCC, USA). Peripheral blood mononuclear cells (PBMC) were isolated as follow: whole blood from healthy volunteer donors was diluted 1:1 with PBS, then 30 ml were gently layered above 15 ml of Histopaque (Sigma-Aldrich) in a 50 ml centrifuge tube. The tube was centrifuged for 30 minutes at 400 g. The white and cloudy layer containing PBMC was collected in a 50 ml centrifuge tube, diluted with 40 ml of PBS and centrifuged 5 minutes at 300 g for washing, for two times. The pelleted cells were counted, and the percentage viability estimated using Trypan blue staining. Cells were cultured in RPMI with 10% fetal bovine serum in 24 well plates.

Nucleic Acid Incorporation in Recipient Cells

[0140] In order to evaluate the uptake into cells of eGFP mRNA loaded into engineered plant-derived EVs of the invention, these vesicles were incubated with HMEC cells and macrophages. A total of 50,000 recipient cells/well were plated in 24-well plates and stimulated with 50,000 vesicles/recipient cell. After 24 hours, cells were extensively washed, detached with trypsin and the fluorescence of translated GFP protein was measured by FACS using the CytoFLEX flow cytometer with CytExpert software (Beckman Coulter Optima L-90K, Fullerton, CA, USA).

In Vitro Nucleic Acid Degradation Assay

[0141] In order to test the resistance to enzyme degradation of nucleic acid molecules loaded into the EVs of the invention, the inventors carried out a RNAse assay. Briefly, samples were treated with RNase A (ThermoFisher Scientific), using a concentration of 0.4 mg/mL, for 30 min at 37° C. The RNase inhibitor (Thermo Fisher Scientific) was used to stop the reaction as described by the manufacturer's protocol, and samples were washed by ultracentrifugation at 100,000 g for 2 h at 4° C. using a 10 mL polycarbonate tube (SW 90 Ti rotor, Beckman Coulter Optima L-90 K ultracentrifuge). Eventually, samples of EV pellets were resuspended in saline buffer solution and molecular analysis was performed.

[0142] In order to test resistance to stomach digestion of nucleic acid molecules loaded into the EVs of the invention, the inventors carried out a stomach digestion assay. Briefly, a stomach-like solution was prepared containing 18.5% w/v HCl (pH 2.0), 24 mg/mL of bile extract, pepsin solution (80 mg/mL in 0.1 N of HCl, pH 2.0; Sigma) and 4 mg/mL of pancreatin (Sigma) in 0.1 N of NaHCO3. An amount of 1 μl of each EVs sample in a water solution was incubated with slow rotation at 37° C. for 60 min with 1.34 μL of stomach-like solution. The pH value of the stomach-like solution was adjusted to 6.5 with 1 N $NaHCO_3$ and was referred to as an intestinal solution. Then, EV samples were incubated for additional 60 min in the intestinal solution. The stability of the nucleic acid molecules loaded into the EVs of the invention was evaluated by molecular analysis as above described. For all resistance experiments, naked RNA was used as control.

Sample Lyophilization

[0143] Samples were lyophilized using the instrument Heto lyolab 3000 (Thermo Fisher Scientific) for 3 hours and kept for 7 days at 4° C. After storage time point, EV samples were analyzed for their content of nucleic acid using molecular analysis. The content thus measured was compared to the starting amount, before lyophilization and storage.

Immune Cells Activation Assay

[0144] To assess PBMC proliferation by flow cytometric analysis, PBMC were stained with CFSE dye from CellTrace Cell Proliferation Kits (Invitrogen, ThermoFisher Scientific) according to manufacturer's instruction. PBMCs were then plated in a 48 well plate at the density of 50,000 cells/well. In order to evaluate whether the engineered, plant-derived EVs of the invention may affect PBMC proliferation, PBMCs were stimulated with these vesicles at the dose of 50,000 particles/cell. Unstimulated PBMCs were used as control. After 24 hours incubation, PBMCs were collected and fluorescence was measured by the CytoFLEX flow cytometer equipped with CytoExpert software (Beckman Coulter). CFSE dye is detected as FITC fluorescence.

[0145] To analyse PBMC proliferation by Bromo Deoxyuridine (BrdU) incorporation assay, PBMCs were plated in a 96 well plate at the density of 20,000 cells/well and 10 μl of BrdU labeling solution (BrdU colorimetric assay, Roche) were added to each well. In order to evaluate whether the engineered, plant-derived EVs of the invention may affect PBMC proliferation, PBMCs were stimulated with these vesicles at the dose of 50,000 particles/cell. Un-stimulated PBMCs were used as control. Moreover, to assess whether the EVs of the invention may reduce proliferation of activated PBMC, PBMCs were stimulated with these vesicles at the dose of 50,000 particles/cell and LPS (from *E.coli*, Sigma-Aldrich) at the concentration of 100 ng/ml. Unstimulated PBMCs were used as negative control. The effects of the stimuli were analyzed after 48 hours of incubation. The assay was carried out according to manufacturer's instructions. Absorbance was measured by an ELISA reader at 420 nm with the reference wavelength at 490 nm. The mean absorbance for each condition was calculated. Absorbance is directly proportional to proliferation rate.

Detection of Protein Expression

[0146] To assess the protein expression in target cells, endothelial cells were stimulated with 1.2×10^{10} EVs. The assayed cell samples included untreated cells (NT), cells treated with plant-derived native EVs and cells treated with plant-derived EVs of the invention engineered with mRNA molecules coding for the SARS-CoV-2 Spike Glycoprotein (S1) RBD protein (SEQ ID NO. 50), SARS-CoV-2 Spike Glycoprotein full protein (SEQ ID NO. 14), or SARS-CoV-2 Nucleocapsid Protein (SEQ ID NO. 17). After 24 hours, cells were extensively washed, detached with trypsin and fixed and permeabilized following manufacturer's instruction (Inside Stain Kit, Miltenyi Biotec). Then, cells were stained for 30 minutes at room temperature with specific antibody to detect protein expression (antibody against SARS-CoV-2 Spike Glycoprotein and Nucleocapsid protein, Invitrogen). Following a washing, fluorescent secondary antibodies were added for 1 hour at room temperature (Alexa Fluor Plus 594 or 488, Invitrogen, ThermoFisher Scientific). After a wash, cells were resuspended in appropriate buffer and acquired by FACS using the CytoFLEX flow cytometer with CytExpert software (Beckman Coulter Optima L-90K, Fullerton, CA, USA).

Lymphocyte Activation Assay

[0147] To assess the ability of engineered EVs of the invention to induce lymphocyte activation following incorporation into APCs, macrophages were plated 20.000 cells/well in a 24 well plate and stimulated with 1.2×10^{10} EVs. The assayed cell samples included untreated cells (NT), cells treated with plant-derived native EVs and cells treated with plant-derived EVs of the invention engineered with mRNA molecules coding for the SARS-CoV-2 Spike Glycoprotein (S1) RBD protein (SEQ ID NO. 50), SARS-CoV-2 Spike Glycoprotein full protein (SEQ ID NO. 14), or SARS-CoV-2 Nucleocapsid Protein (SEQ ID NO. 17). After incorporation, PBMC were added at the concentration of 200,000 cells/well and the treatment with EVs was repeated after 5 days. After 10 days from the initial treatment, cells were harvested and stained with fluorescent antibody for CD4, CD25 and

HLA DR using appropriate isotype (Miltenyi Biotec) for 30 minutes at room temperature. Following a wash, cells were acquired.

[0148] For proliferation analysis, the employed PBMCs were previously stained with CSFE dye from CellTrace Cell Proliferation Kits (Invitrogen, ThermoFisher Scientific) according to manufacturer's instruction.

[0149] Finally, samples were analyzed by FACS using the CytoFLEX flow cytometer with CytExpert software (Beckman Coulter Optima L-90K, Fullerton, CA, USA).

Mice Vaccination

[0150] Female BALB/cAnNCrl mice, 6-10 weeks old, received 2 immunizations at day 0 and day 21 with a dose of engineered plant-derived EVs of the invention equivalent to 30 μg of mRNA and were sacrificed at day 35. Mice were treated with plant-derived native EVs or plant-derived EVs of the invention engineered with mRNA molecules coding for the SARS-CoV-2 Spike Glycoprotein RBD (S1) using oral (using gavage) and intramuscular (right leg) routes. After the sacrifice, blood was collected to isolated sera for antibody detection.

Antibody Measurement

[0151] SARS-CoV-2 specific IgA antibody titers of sera were determined by ELISA. Briefly, MaxiSorp ELISA plates (Nunc) were coated with 1 μg/ml of SARS-CoV-2 Spike protein (Thermofisher Scientific) in 100 μl of 50 mM sodium carbonate/bicarbonate pH 9.6 buffer per well, overnight at 4° C. Coated plates were washed 3 times with 200 μl of 1× PBS and saturated with 200 μl 3% BSA in 1× PBS per well. Plates were washed three times with 1× PBS, and incubated in 3% BSA and with 100-fold mouse sera dilution for 2 h. This was followed by 3 washes with 200 μl of 1× PBS per well and incubation with 100 μl per well of secondary donkey anti-mouse IgA HRP conjugated antibody diluted 1:10,000. Following the incubation with the secondary antibody, plates were washed 5 times with 200 μl of 1× PBS per well and developed with 100 μl of TMB per well (Thermofisher Scientific) for 30 minutes. The reaction was stopped by adding 100 μl of stop solution (Thermofisher Scientific) per well. The 450 nm absorbance was read using plate reader.

Statistical Analysis

[0152] Data analysis was carried out with the software Graph Pad 8, demo version. Results are expressed as mean±standard deviation (SD). One-way analysis of variance (ANOVA) was used to substantiate statistical differences between groups, while Student's t-test was used for comparison between two samples. We used $p<0.05$ as a minimal level of significance.

RESULTS/EXAMPLES

Example 1

[0153] To investigate the feasibility of the method of the present invention, the inventors engineered different plant-derived EVs and characterized them according to a number of features traditionally used to characterize EVs in terms of physical properties, such as particle size and surface charge (Théry C., et al, (2018) "Minimal information for studies of extracellular vesicles 2018 (MISEV2018): a position statement of the International Society for Extracellular Vesicles and update of the MISEV2014 guidelines"; Journal of Extracellular Vesicles, 7:1, 1535750, DOI: 10.1080/20013078.2018.1535750). FIG. **1** shows the size of engineered, plant-derived EVs of the invention from different plants, including kiwifruit, cabbage, and blueberry. Similar results were also obtained with EVs from other plant sources such as pomegranate, kale, celery, courgette, orange, and lemon. Particle size is a fundamental parameter of EVs. Data obtained showed that the EVs of the invention have a higher mean diameter compared to native plant-derived EVs (FIG. **1**A, B, C). The measurement of EV size was carried out by the inventors with Nanoparticle tracking Analysis (NTA) technique, the most used method to measure EV size which utilizes the properties of both light scattering and Brownian motion in order to obtain the nanoparticle size distribution of samples in liquid suspension. In particular, NTA works by tracking particle motion via light scattering to assess the mean squared displacement of particles moving under Brownian motion, in a sample chamber illuminated by a laser beam. The tracking of particles enables a diffusion constant to be calculated, which is used in the Stokes-Einstein equation to calculate hydrodynamic diameters. The Stokes-Einstein equation also takes into account the temperature and viscosity of the suspension. The results of the analysis performed by the inventors demonstrate that the EVs of the invention have a higher size compared to native plant-derived EVs as the native vesicles have a diameter ranging from 100 to 150 nm, with a mean diameter of 134±6 nm, whereas the diameters of the EVs of the invention were in the range between 200 and 250 nm, with a mean diameter of 220 nm. The size distribution demonstrated that EVs of the invention have a diameter ranging from 20 to 500 nm, preferably ranging from 200 to 300 nm.

[0154] In order to characterize the membrane properties of engineered, plant-derived EVs of the invention, vesicle membrane potential was measured. In fact, Zeta potential is a popular method to measure the surface potential of EVs and it is used as an indicator of surface charge and colloidal stability. The surface charge of EVs depends on the nature of molecules expressed at their surfaces and it affects EV interaction in dispersed systems such as human body, defining their activity in biological processes. For example, the surface charge is known to influence different biological processes associated with particles, such as cellular uptake and cytotoxicity. Zeta potential is a measure of the magnitude of the electrostatic or charge repulsion/attraction between particles and it can be measured from the electrophoretic mobility in a suspension determined by applying an electric field and measuring the resulting velocity of the particles (Electrophoretic light scattering) (Midekessa G, et al. Zeta Potential of Extracellular Vesicles: Toward Understanding the Attributes that Determine Colloidal Stability. ACS Omega. 2020 Jun. 30;5(27):16701-16710. doi: 10.1021/acsomega.0c01582. PMID: 32685837; PMCID: PMC7364712.) As shown in FIG. **2**, plant-derived EVs are known to have a negative surface charge and the membrane potential (Z potential) was ranging between −10 and −15 mVolt, with a mean value of −13 mVolt. Instead, engineered plant-derived EV of the invention exploited a different membrane with a membrane potential comprises between 0 and −3 mVolt, with a mean value of −2 mVolt. The above described experiment was performed on engineered, plant-derived EVs from courgette (E1) and blueberry (E2), but similar results were obtained with EVs derived from other plant sources such as kiwifruit, cabbage, kale, pomegranate, lemon, orange and celery.

[0155] To further characterize the EVs of the invention, the protein content of these vesicles and native plant-derived EVs was measured (FIG. **3**A). The data thus obtained demonstrated that the EVs of the invention have a higher protein content ranging from 120 to 160 ng/10^{10} EVs, whereas native plant-derived EVs have a protein content ranging from 50 to 100 ng/10^{10} EVs. The inventors carried out the experiments on engineered plant-derived EVs from pomegranate (E1) and kiwifruit (E2), but similar results were obtained with EVs derived from other plant sources such as courgette, cabbage, kale, blueberry, lemon, orange and celery.

[0156] Moreover, the present inventors carried out dedicated experiments to better characterize the membrane of the EVs of the invention. As is known in the art, in native EVs phosphatidylserine (PS) is predominantly located along the outer surface of the plasma membrane. Without wishing to be bound by any theory, the inventors believe that, upon membrane rearrangement due to the osmotic stress, PS loses its asymmetric distribution in the phospholipid bilayer and translocates to the inner side of the membrane in the EVs of the invention. The detection of phosphatidylserine in the extracellular membrane of vesicles was achieved by means of fluorescently labeled Annexin V. In fact, Annexin V is known to specifically bind to PS on vesicle membrane. The quantity of fluorescent signal of Annexin V reflects the PS content on the outer surface of the EV membrane (Montoro-García S, et al. "An innovative flow cytometric approach for small-size platelet microparticles: influence of calcium". Thromb Haemost. 2012 August;108(2):373-83). The results obtained by the present inventors showed that a percentage ≤44% of EVs in the composition of the invention have phosphatidylserine in the outer layer of the membrane (FIG. **3**B), such percentage ranging from 40 to 44%, with a mean value of 43%. Differently, in a composition comprising native plant-derived EVs, the percentage of these vesicles having phosphatidylserine in the outer layer of the membrane ranges from 55 to 48%, with a mean value of 49%.

[0157] The experiments above described were performed on engineered, plant-derived EVs from cabbage (E1) and blueberry (E2), but similar results were also obtained with EVs from other plant sources such as kiwifruit, lemon, orange, courgetti, kale, pomegranate and celery.

[0158] As a further assessment, the present inventors conducted dedicated experiments with the aim of evaluating the immunomodulatory activity of the engineered, plant-derived EVs of the invention. Briefly, PBMC, i.e. a mixed population of lymphocytes, monocytes and other immune cells from the human blood, were stimulated with the EVs of the invention and cell proliferation rate was measured.

[0159] As shown in FIG. **4**A, the proliferation rate of PBMC stimulated for 48 hours with the EVs of the invention from pomegranate is the same as untreated PBMC, suggesting that these vesicles do not promote PBMC proliferation and do not exert an immunostimulatory effect. Further, to verify whether the EVs of the invention have an immunosuppressive effect, the present inventors treated PBMC with LPS, which is known to induce inflammatory responses and promote lymphocyte proliferation, and then stimulated the cells with the engineered, plant-derived EVs of the invention. As shown in FIG. **4**B, the proliferation rate of PBMC activated by LPS was not affected by the EVs of the invention (from pomegranate (E1) and kiwifruit (E2)).

[0160] These results confirm that the EVs according to the invention do not have immunostimulating nor immunosuppressive effects.

[0161] Furthermore, PBMC were also stained with a fluorescent dye (CFSE) which allows the detection of PBMC proliferation by flow cytometry. As shown in FIGS. **4**C and **4**D, proliferation rate of PBMC stimulated with engineered, plant-derived EVs from cabbage (E1), celery (E2), and courgettes (E3) is the same as unstimulated PBMC. Similar results were obtained with engineered EVs from other plants, such as blueberry, lemon, orange, and kale.

[0162] Overall, the above results demonstrate that engineered, plant-derived EVs of the invention are not able to stimulate nor suppress immune cell activation and proliferation, and confirm the non-immunomodulating properties of these vesicles regardless of vesicles plant source.

[0163] Finally, the total RNA content of EVs of the invention and native plant-derived EVs was measured (FIG. **5**). The data obtained demonstrated that the EVs of the invention have a higher RNA content compared to native vesicles, ranging from 30 to 100 ng/10^9 EVs, with a mean value of 50 ng/10^9 EVs. Native plant-derived EVs have an RNA content ranging from 5 to 15 ng/10^9 EVs, with a mean value of 10 ng/10^9 EVs. The experiments were performed on engineered EVs from celery (E1), pomegranate (E2) and kiwifruit (E3), but similar results were also obtained with EVs from other plant sources such as courgette, cabbage, kale, lemon, orange and blueberry.

[0164] Taken together, all these data demonstrate that engineered, plant-derived EVs of the invention differ significantly from native plant-derived EVs. In particular, upon exogenous nucleic acid loading, unique alterations surprisingly occur in the plant-derived EVs structure and function compared to the native vesicles, thereby resulting in higher mean diameter, higher surface charge, lower phosphatidylserine content and loss of immunomodulatory effect on immune system cells.

Example 2

[0165] With the aim of demonstrating the suitability of the EVs according to the invention as vehicles for nucleic acid delivery, engineered EVs were produced by internally loading mRNA molecules, and the amount of loaded mRNA was measured by qRT-PCR analysis (FIG. **6**). The results obtained showed that the EVs according to the invention can be loaded with increasing doses of nucleic acid molecules. In fact, engineered EVs were produced from kiwifruit (E1 and E3) and celery (E2 and E4) using two different doses of mRNA: 0.1 μg/ml for E1 and E2, and 1 μg/ml for E3 and E4. The nucleic acid dose increase was detectable as increase of the amount of mRNA in vesicles (E3 and E4 versus E1 and E2, respectively). Similar results were also obtained with EVs from other plant sources such as courgette, cabbage, kale, lemon, orange, pomegranate and blueberry. Taken together, these data demonstrated that EVs of the invention encapsulate the loaded nucleic acid molecules and their amount can be increased.

Example 3

[0166] The present inventors conducted dedicated experiments to assess the capacity by the EVs of the invention to preserve loaded nucleic acid molecules from degradation (FIG. **7**). In particular, these studies showed that engineered, plant-derived EVs of the invention were able to protect loaded nucleic acid molecules from the treatment with degrading enzyme (RNAse). Briefly, following EVs treatment with RNAse, qRT-PCR analysis revealed that about 80% of the loaded mRNA was still present in the vesicles of the invention, whereas naked mRNA used as control was almost completely degraded (FIG. **7**A)

[0167] Further experiments showed that the EVs according to the invention are also able to protect loaded nucleic acid molecules since, upon vesicles treatment with a stomach-like solution mimicking the gastrointestinal environment, about 90% of the loaded mRNA was still present in the EVs whereas the naked mRNA was almost completely degraded (FIG. **7**B). The above experiments were performed on engineered, plant-derived EVs from pomegranate (6A) and kiwifruit (6B), but similar results were also obtained with EVs from other plant sources such as courgette, cabbage, kale, blueberry, lemon, orange and celery.

[0168] Taken together, these data demonstrate that the EVs according to the invention protect the encapsulated exogenous nucleic acid from degrading conditions. Moreover, the protection from gastrointestinal environment supports the oral administration of the composition for use according to the invention.

Example 4

[0169] Engineered, plant-derived EVs according to the invention can be efficiently lyophilized and stored. In particular, following EVs lyophilization and storage at +4° C. for 7 days, the content of loaded mRNA in the vesicles did not decrease compared to the starting condition (FIG. **8**). The experiments were performed on engineered, plant-derived EVs from celery (E1) and kiwifruit (E2), but similar results were also obtained with EVs from other plant sources such as courgette, cabbage, kale, blueberry, lemon, orange and pomegranate.

[0170] These data demonstrated that the EVs according to the invention can be easily lyophilized and stored efficiently at +4° C. or at room temperature, without the need to use very low temperatures typical of storage of nucleic acid agents, such as −80° C.

Example 5

[0171] The inventors have further demonstrated that engineered, plant-derived EVs of the invention are suitable to be used for delivering loaded nucleic acids to recipient cells (FIG. **9**). In these experiments, macrophages were used as exemplary recipient cells and the transfer of mRNA molecules in these cells was measured by qRT-PCR analysis. In particular, the above experiments demonstrated that EVs of the invention derived from different types of plants (E1, E2, E3) were able to transfer the mRNA molecules to macrophages relative to untreated cells (not treated, NT), whereas no mRNA transfer was detected in native plant-derived EVs (native EV), plant-derived EVs co-incubated with the mRNA without nucleic acid loading (EV+mRNA) and naked mRNA. The experiments were performed on engineered, plant-derived EVs from cabbage (E1), pomegranate (E2) and kiwifruit (E3), but similar results were also obtained with EVs from other plant sources such as courgette, celery, kale, lemon, orange and blueberry.

[0172] In order to demonstrate that the nucleic acid delivered to recipient cells maintains its functional activity, EVs of the invention were assayed containing mRNA molecules coding for the GFP protein. After incorporation into recipient cells, the mRNA, if functional, is translated into the GFP protein and the fluorescence of the functional protein is detectable in cells.

[0173] The experiments carried out by the inventors showed that the mRNA carried by the EVs of the invention was functional and was detectable as fluorescent signal in endothelial cells (FIG. **10**A) and macrophages (FIG. **10**B), whereas there was no transfer of functional mRNA using native plant-derived (native EV, in FIG. **10**A) or naked mRNA (naked mRNA, in FIG. **10**B). The experiments were performed on engineered, plant-derived EVs from courgette (FIG. **10**A) and kale (FIG. **10**B), but similar results were obtained with EVs from other plant sources such as celery, cabbage, kiwifruit, blueberry, lemon, orange and pomegranate.

[0174] Furthermore, the inventors demonstrated that engineered, plant-derived EVs of the invention are able to transfer functional mRNAs which are translated into protein antigens into recipient cells and expressed as correctly folded antigen (FIG. **11**). In the course of the experiments, EVs of the invention were assayed containing mRNA molecules coding for different sequences of viral protein antigens: SEQ ID NO. 50, SARS-CoV-2 Spike Glycoprotein (S1) RBD protein; SEQ ID No. 14, SARS-CoV-2 Spike Glycoprotein full protein; SEQ ID NO. 17, SARS-CoV-2 Nucleocapsid Protein. After incorporation into endothelial cells, the mRNA, if functional, is translated into the protein which can be detected by specific antibody. FIG. **11** shows that engineered, plant-derived EVs of the invention are able to transfer the mRNA into target cells which translate it into the specific protein antigen, whereas no protein antigens were detectable in untreated cells (NT) or cells treated with native plant-derived, not engineered, EVs.

[0175] Taken together, these data demonstrated that the EVs according to the invention can efficiently deliver exogenous nucleic acid molecules to different types of recipient cells, including antigen presenting cells (APC) as macrophages, preserving at the same time the nucleic acid function and its ability to be translated into protein. Thus, the correctly folded expressed protein can function as antigen in promoting the immunization by the APC. Moreover, the experiments performed as above described showed that the EVs of the invention are suitable to be used with nucleic acid molecules encoding different proteins such as viral antigens (the nucleocapsid (N) protein and Spike Glycoprotein of SARS-CoV-2) or other proteins such as GFP.

Example 6

[0176] The inventors have further demonstrated that engineered, plant-derived EVs of the invention can deliver nucleic acids to APC cells which express and present the antigen and then stimulate specific immune activation (FIG. **12**). In these experiments, macrophages were used as exemplary APC recipient cells and were stimulated with engineered EVs of the present invention before the incubation with PBMC, i.e. a mixed population of lymphocytes, monocytes and other immune cells from the human blood. For the experiments, EVs of the invention were assayed containing mRNA molecules coding for different sequences of viral protein antigens used as example: sequence 1, SARS-CoV-2

Spike Glycoprotein (S1) RBD protein; sequence 2, SARS-CoV-2 Spike Glycoprotein full protein; sequence 3, SARS-CoV-2Nucleocapsid Protein. The activation of lymphocytes was measured after ten days using FACS analysis as increase of lymphocyte (cell CD4+) proliferation and expression of activation markers CD25 and HLA DR. In order to detect lymphocytes proliferation, PBMC were also stained with a fluorescent dye (CFSE) which allows the detection of PBMC proliferation by flow cytometry. As shown in FIG. **12**A, proliferation rate of lymphocyte CD4+ stimulated with engineered, plant-derived EVs is increased in comparison to negative controls, untreated cells (NT) and cells treated with native EVs. As expected, positive controls stimulated lymphocyte proliferation: human T-Activator CD3/CD28 (CTR+) and purified proteins (SARS-CoV-2 Spike Glycoprotein RBD protein (S protein), or SARS-CoV-2Nucleocapsid protein (N protein)). The increased proliferation rate of lymphocytes demonstrates their activation.

[0177] Furthermore, the stimulation of APC with engineered, plant-derived EVs of the invention increased the expression of activation markers CD25 (FIG. **12**B) and HLA DR in lymphocytes (FIG. **12**C).

[0178] The stimulation with the plant-derived EVs of the invention induced an increased expression of both activation markers by CD4+ lymphocytes, demonstrating immune cell activation.

[0179] The stimulation was compared to negative controls, untreated cells (NT) and cells treated with native plant-derived EVs. As expected, positive controls stimulated lymphocyte proliferation: human T-Activator CD3/CD28 (CTR+) and purified proteins (SARS-CoV-2 Spike Glycoprotein RBD protein (S protein), or SARS-CoV-2 Nucleocapsid protein (N protein)).

[0180] Overall, the above results demonstrate that engineered, plant-derived EVs of the invention activate immune response following incorporation into APC (such as macrophages) and they can be assayed with different nucleic acids molecules coding for different protein antigens.

[0181] Taken together, all these data demonstrate that engineered, plant-derived EVs of the invention are able to transfer a functional mRNA to APC, which in turn is translated into a correctly folded protein antigen and can specifically activate immune response. Of note, only engineered, plant-derived EVs of the invention, and not native plant-derived EVs, induced lymphocyte activation.

Example 7

[0182] With the aim of demonstrating the suitability of the EVs according to the invention as vehicles for nucleic acid delivery for use as a vaccine, experiments on in vivo mouse model were performed.

[0183] In particular, mice were immunized two times (with a break of three weeks between the two treatments) and the presence of specific antibodies in serum was measured after two weeks following the last dose. Mice were treated with native plant-derived EVs (native EV) or plant-derived EVs of the invention engineered with mRNA molecules coding for a viral protein antigen as example, the SARS-CoV-2 Spike Glycoprotein RBD (S1) (engineered EV), using intramuscular or oral administration routes.

[0184] FIG. **13** shows the measurement of IgA antibodies specific for SARS-CoV-2 Spike Glycoprotein RBD (S1) in mice serum. The vaccination with engineered plant-derived EVs of the invention induced the production of specific antibodies in comparison to the vaccination with native plant-derived EVs following both oral and intramuscular administration. The antibody positive response detected after oral administration demonstrates the protection from gastrointestinal environment of the composition for use according to the invention.

[0185] Taken together, all these data demonstrate that engineered, plant-derived EVs of the invention are suitable for use as a vaccine because can be loaded with nucleic acids, which are transferred to APC, translated to a correctly folded antigen, activate immune response and the production of specific antibodies in vivo. The activation of immune response is specific to the antigen because EVs of the invention do not exert per se neither immunostimulatory nor immunosuppressive effects. Furthermore, engineered, plant-derived EVs of the invention can efficiently protect the nucleic acid from degradation, allowing the vaccine administration using different routes.

SEQUENCE LISTING

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<160> NUMBER OF SEQ ID NOS: 50

<210> SEQ ID NO 1
<211> LENGTH: 261
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
<220> FEATURE:
<223> OTHER INFORMATION: Prostate-specific antigen (PSA)

<400> SEQUENCE: 1

Met Trp Val Pro Val Val Phe Leu Thr Leu Ser Val Thr Trp Ile Gly
1               5                   10                  15

Ala Ala Pro Leu Ile Leu Ser Arg Ile Val Gly Gly Trp Glu Cys Glu
            20                  25                  30

Lys His Ser Gln Pro Trp Gln Val Leu Val Ala Ser Arg Gly Arg Ala
        35                  40                  45

Val Cys Gly Gly Val Leu Val His Pro Gln Trp Val Leu Thr Ala Ala
    50                  55                  60
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His Cys Ile Arg Asn Lys Ser Val Ile Leu Leu Gly Arg His Ser Leu
65                  70                  75                  80

Phe His Pro Glu Asp Thr Gly Gln Val Phe Gln Val Ser His Ser Phe
                85                  90                  95

Pro His Pro Leu Tyr Asp Met Ser Leu Leu Lys Asn Arg Phe Leu Arg
            100                 105                 110

Pro Gly Asp Asp Ser Ser His Asp Leu Met Leu Leu Arg Leu Ser Glu
        115                 120                 125

Pro Ala Glu Leu Thr Asp Ala Val Lys Val Met Asp Leu Pro Thr Gln
    130                 135                 140

Glu Pro Ala Leu Gly Thr Thr Cys Tyr Ala Ser Gly Trp Gly Ser Ile
145                 150                 155                 160

Glu Pro Glu Glu Phe Leu Thr Pro Lys Lys Leu Gln Cys Val Asp Leu
                165                 170                 175

His Val Ile Ser Asn Asp Val Cys Ala Gln Val His Pro Gln Lys Val
            180                 185                 190

Thr Lys Phe Met Leu Cys Ala Gly Arg Trp Thr Gly Gly Lys Ser Thr
        195                 200                 205

Cys Ser Gly Asp Ser Gly Gly Pro Leu Val Cys Asn Gly Val Leu Gln
    210                 215                 220

Gly Ile Thr Ser Trp Gly Ser Glu Pro Cys Ala Leu Pro Glu Arg Pro
225                 230                 235                 240

Ser Leu Tyr Thr Lys Val Val His Tyr Arg Lys Trp Ile Lys Asp Thr
                245                 250                 255

Ile Val Ala Asn Pro
            260
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<210> SEQ ID NO 2
<211> LENGTH: 123
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
<220> FEATURE:
<223> OTHER INFORMATION: Prostate stem cell antigen (PSCA)

<400> SEQUENCE: 2

Met Lys Ala Val Leu Leu Ala Leu Leu Met Ala Gly Leu Ala Leu Gln
1               5                   10                  15

Pro Gly Thr Ala Leu Leu Cys Tyr Ser Cys Lys Ala Gln Val Ser Asn
            20                  25                  30

Glu Asp Cys Leu Gln Val Glu Asn Cys Thr Gln Leu Gly Glu Gln Cys
        35                  40                  45

Trp Thr Ala Arg Ile Arg Ala Val Gly Leu Leu Thr Val Ile Ser Lys
    50                  55                  60

Gly Cys Ser Leu Asn Cys Val Asp Asp Ser Gln Asp Tyr Tyr Val Gly
65                  70                  75                  80

Lys Lys Asn Ile Thr Cys Cys Asp Thr Asp Leu Cys Asn Ala Ser Gly
                85                  90                  95

Ala His Ala Leu Gln Pro Ala Ala Ala Ile Leu Ala Leu Leu Pro Ala
            100                 105                 110

Leu Gly Leu Leu Leu Trp Gly Pro Gly Gln Leu
        115                 120


<210> SEQ ID NO 3
<211> LENGTH: 750
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<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
<220> FEATURE:
<223> OTHER INFORMATION: Prostate specific membrane antigen (PSMA)

<400> SEQUENCE: 3

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Met Trp Asn Leu Leu His Glu Thr Asp Ser Ala Val Ala Thr Ala Arg
1               5                   10                  15

Arg Pro Arg Trp Leu Cys Ala Gly Ala Leu Val Leu Ala Gly Gly Phe
            20                  25                  30

Phe Leu Leu Gly Phe Leu Phe Gly Trp Phe Ile Lys Ser Ser Asn Glu
        35                  40                  45

Ala Thr Asn Ile Thr Pro Lys His Asn Met Lys Ala Phe Leu Asp Glu
    50                  55                  60

Leu Lys Ala Glu Asn Ile Lys Lys Phe Leu Tyr Asn Phe Thr Gln Ile
65                  70                  75                  80

Pro His Leu Ala Gly Thr Glu Gln Asn Phe Gln Leu Ala Lys Gln Ile
                85                  90                  95

Gln Ser Gln Trp Lys Glu Phe Gly Leu Asp Ser Val Glu Leu Ala His
            100                 105                 110

Tyr Asp Val Leu Leu Ser Tyr Pro Asn Lys Thr His Pro Asn Tyr Ile
        115                 120                 125

Ser Ile Ile Asn Glu Asp Gly Asn Glu Ile Phe Asn Thr Ser Leu Phe
    130                 135                 140

Glu Pro Pro Pro Pro Gly Tyr Glu Asn Val Ser Asp Ile Val Pro Pro
145                 150                 155                 160

Phe Ser Ala Phe Ser Pro Gln Gly Met Pro Glu Gly Asp Leu Val Tyr
                165                 170                 175

Val Asn Tyr Ala Arg Thr Glu Asp Phe Phe Lys Leu Glu Arg Asp Met
            180                 185                 190

Lys Ile Asn Cys Ser Gly Lys Ile Val Ile Ala Arg Tyr Gly Lys Val
        195                 200                 205

Phe Arg Gly Asn Lys Val Lys Asn Ala Gln Leu Ala Gly Ala Lys Gly
    210                 215                 220

Val Ile Leu Tyr Ser Asp Pro Ala Asp Tyr Phe Ala Pro Gly Val Lys
225                 230                 235                 240

Ser Tyr Pro Asp Gly Trp Asn Leu Pro Gly Gly Gly Val Gln Arg Gly
                245                 250                 255

Asn Ile Leu Asn Leu Asn Gly Ala Gly Asp Pro Leu Thr Pro Gly Tyr
            260                 265                 270

Pro Ala Asn Glu Tyr Ala Tyr Arg Arg Gly Ile Ala Glu Ala Val Gly
        275                 280                 285

Leu Pro Ser Ile Pro Val His Pro Ile Gly Tyr Tyr Asp Ala Gln Lys
    290                 295                 300

Leu Leu Glu Lys Met Gly Gly Ser Ala Pro Pro Asp Ser Ser Trp Arg
305                 310                 315                 320

Gly Ser Leu Lys Val Pro Tyr Asn Val Gly Pro Gly Phe Thr Gly Asn
                325                 330                 335

Phe Ser Thr Gln Lys Val Lys Met His Ile His Ser Thr Asn Glu Val
            340                 345                 350

Thr Arg Ile Tyr Asn Val Ile Gly Thr Leu Arg Gly Ala Val Glu Pro
        355                 360                 365
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Asp Arg Tyr Val Ile Leu Gly Gly His Arg Asp Ser Trp Val Phe Gly
370 375 380

Gly Ile Asp Pro Gln Ser Gly Ala Ala Val Val His Glu Ile Val Arg
385 390 395 400

Ser Phe Gly Thr Leu Lys Lys Glu Gly Trp Arg Pro Arg Arg Thr Ile
405 410 415

Leu Phe Ala Ser Trp Asp Ala Glu Glu Phe Gly Leu Leu Gly Ser Thr
420 425 430

Glu Trp Ala Glu Glu Asn Ser Arg Leu Leu Gln Glu Arg Gly Val Ala
435 440 445

Tyr Ile Asn Ala Asp Ser Ser Ile Glu Gly Asn Tyr Thr Leu Arg Val
450 455 460

Asp Cys Thr Pro Leu Met Tyr Ser Leu Val His Asn Leu Thr Lys Glu
465 470 475 480

Leu Lys Ser Pro Asp Glu Gly Phe Glu Gly Lys Ser Leu Tyr Glu Ser
485 490 495

Trp Thr Lys Lys Ser Pro Ser Pro Glu Phe Ser Gly Met Pro Arg Ile
500 505 510

Ser Lys Leu Gly Ser Gly Asn Asp Phe Glu Val Phe Phe Gln Arg Leu
515 520 525

Gly Ile Ala Ser Gly Arg Ala Arg Tyr Thr Lys Asn Trp Glu Thr Asn
530 535 540

Lys Phe Ser Gly Tyr Pro Leu Tyr His Ser Val Tyr Glu Thr Tyr Glu
545 550 555 560

Leu Val Glu Lys Phe Tyr Asp Pro Met Phe Lys Tyr His Leu Thr Val
565 570 575

Ala Gln Val Arg Gly Gly Met Val Phe Glu Leu Ala Asn Ser Ile Val
580 585 590

Leu Pro Phe Asp Cys Arg Asp Tyr Ala Val Val Leu Arg Lys Tyr Ala
595 600 605

Asp Lys Ile Tyr Ser Ile Ser Met Lys His Pro Gln Glu Met Lys Thr
610 615 620

Tyr Ser Val Ser Phe Asp Ser Leu Phe Ser Ala Val Lys Asn Phe Thr
625 630 635 640

Glu Ile Ala Ser Lys Phe Ser Glu Arg Leu Gln Asp Phe Asp Lys Ser
645 650 655

Asn Pro Ile Val Leu Arg Met Met Asn Asp Gln Leu Met Phe Leu Glu
660 665 670

Arg Ala Phe Ile Asp Pro Leu Gly Leu Pro Asp Arg Pro Phe Tyr Arg
675 680 685

His Val Ile Tyr Ala Pro Ser Ser His Asn Lys Tyr Ala Gly Glu Ser
690 695 700

Phe Pro Gly Ile Tyr Asp Ala Leu Phe Asp Ile Glu Ser Lys Val Asp
705 710 715 720

Pro Ser Lys Ala Trp Gly Glu Val Lys Arg Gln Ile Tyr Val Ala Ala
725 730 735

Phe Thr Val Gln Ala Ala Ala Glu Thr Leu Ser Glu Val Ala
740 745 750

<210> SEQ ID NO 4
<211> LENGTH: 339
<212> TYPE: PRT

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<213> ORGANISM: Homo sapiens
<220> FEATURE:
<223> OTHER INFORMATION: Six-transmembrane epithelial antigen of the
      prostate (STEAP1)

<400> SEQUENCE: 4

Met Glu Ser Arg Lys Asp Ile Thr Asn Gln Glu Glu Leu Trp Lys Met
1               5                   10                  15

Lys Pro Arg Arg Asn Leu Glu Glu Asp Asp Tyr Leu His Lys Asp Thr
            20                  25                  30

Gly Glu Thr Ser Met Leu Lys Arg Pro Val Leu Leu His Leu His Gln
        35                  40                  45

Thr Ala His Ala Asp Glu Phe Asp Cys Pro Ser Glu Leu Gln His Thr
    50                  55                  60

Gln Glu Leu Phe Pro Gln Trp His Leu Pro Ile Lys Ile Ala Ala Ile
65                  70                  75                  80

Ile Ala Ser Leu Thr Phe Leu Tyr Thr Leu Leu Arg Glu Val Ile His
                85                  90                  95

Pro Leu Ala Thr Ser His Gln Gln Tyr Phe Tyr Lys Ile Pro Ile Leu
            100                 105                 110

Val Ile Asn Lys Val Leu Pro Met Val Ser Ile Thr Leu Leu Ala Leu
        115                 120                 125

Val Tyr Leu Pro Gly Val Ile Ala Ala Ile Val Gln Leu His Asn Gly
    130                 135                 140

Thr Lys Tyr Lys Lys Phe Pro His Trp Leu Asp Lys Trp Met Leu Thr
145                 150                 155                 160

Arg Lys Gln Phe Gly Leu Leu Ser Phe Phe Ala Val Leu His Ala
                165                 170                 175

Ile Tyr Ser Leu Ser Tyr Pro Met Arg Arg Ser Tyr Arg Tyr Lys Leu
            180                 185                 190

Leu Asn Trp Ala Tyr Gln Gln Val Gln Gln Asn Lys Glu Asp Ala Trp
        195                 200                 205

Ile Glu His Asp Val Trp Arg Met Glu Ile Tyr Val Ser Leu Gly Ile
    210                 215                 220

Val Gly Leu Ala Ile Leu Ala Leu Leu Ala Val Thr Ser Ile Pro Ser
225                 230                 235                 240

Val Ser Asp Ser Leu Thr Trp Arg Glu Phe His Tyr Ile Gln Ser Lys
                245                 250                 255

Leu Gly Ile Val Ser Leu Leu Leu Gly Thr Ile His Ala Leu Ile Phe
            260                 265                 270

Ala Trp Asn Lys Trp Ile Asp Ile Lys Gln Phe Val Trp Tyr Thr Pro
        275                 280                 285

Pro Thr Phe Met Ile Ala Val Phe Leu Pro Ile Val Val Leu Ile Phe
    290                 295                 300

Lys Ser Ile Leu Phe Leu Pro Cys Leu Arg Lys Lys Ile Leu Lys Ile
305                 310                 315                 320

Arg His Gly Trp Glu Asp Val Thr Lys Ile Asn Lys Thr Glu Ile Cys
                325                 330                 335

Ser Gln Leu


<210> SEQ ID NO 5
<211> LENGTH: 1255
<212> TYPE: PRT
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<213> ORGANISM: Homo sapiens
<220> FEATURE:
<223> OTHER INFORMATION: Receptor tyrosine-protein kinase erbB-2

<400> SEQUENCE: 5

```
Met Glu Leu Ala Ala Leu Cys Arg Trp Gly Leu Leu Leu Ala Leu Leu
1               5                   10                  15

Pro Pro Gly Ala Ala Ser Thr Gln Val Cys Thr Gly Thr Asp Met Lys
            20                  25                  30

Leu Arg Leu Pro Ala Ser Pro Glu Thr His Leu Asp Met Leu Arg His
        35                  40                  45

Leu Tyr Gln Gly Cys Gln Val Val Gln Gly Asn Leu Glu Leu Thr Tyr
    50                  55                  60

Leu Pro Thr Asn Ala Ser Leu Ser Phe Leu Gln Asp Ile Gln Glu Val
65                  70                  75                  80

Gln Gly Tyr Val Leu Ile Ala His Asn Gln Val Arg Gln Val Pro Leu
                85                  90                  95

Gln Arg Leu Arg Ile Val Arg Gly Thr Gln Leu Phe Glu Asp Asn Tyr
            100                 105                 110

Ala Leu Ala Val Leu Asp Asn Gly Asp Pro Leu Asn Asn Thr Thr Pro
        115                 120                 125

Val Thr Gly Ala Ser Pro Gly Gly Leu Arg Glu Leu Gln Leu Arg Ser
    130                 135                 140

Leu Thr Glu Ile Leu Lys Gly Gly Val Leu Ile Gln Arg Asn Pro Gln
145                 150                 155                 160

Leu Cys Tyr Gln Asp Thr Ile Leu Trp Lys Asp Ile Phe His Lys Asn
                165                 170                 175

Asn Gln Leu Ala Leu Thr Leu Ile Asp Thr Asn Arg Ser Arg Ala Cys
            180                 185                 190

His Pro Cys Ser Pro Met Cys Lys Gly Ser Arg Cys Trp Gly Glu Ser
        195                 200                 205

Ser Glu Asp Cys Gln Ser Leu Thr Arg Thr Val Cys Ala Gly Gly Cys
    210                 215                 220

Ala Arg Cys Lys Gly Pro Leu Pro Thr Asp Cys Cys His Glu Gln Cys
225                 230                 235                 240

Ala Ala Gly Cys Thr Gly Pro Lys His Ser Asp Cys Leu Ala Cys Leu
                245                 250                 255

His Phe Asn His Ser Gly Ile Cys Glu Leu His Cys Pro Ala Leu Val
            260                 265                 270

Thr Tyr Asn Thr Asp Thr Phe Glu Ser Met Pro Asn Pro Glu Gly Arg
        275                 280                 285

Tyr Thr Phe Gly Ala Ser Cys Val Thr Ala Cys Pro Tyr Asn Tyr Leu
    290                 295                 300

Ser Thr Asp Val Gly Ser Cys Thr Leu Val Cys Pro Leu His Asn Gln
305                 310                 315                 320

Glu Val Thr Ala Glu Asp Gly Thr Gln Arg Cys Glu Lys Cys Ser Lys
                325                 330                 335

Pro Cys Ala Arg Val Cys Tyr Gly Leu Gly Met Glu His Leu Arg Glu
            340                 345                 350

Val Arg Ala Val Thr Ser Ala Asn Ile Gln Glu Phe Ala Gly Cys Lys
        355                 360                 365

Lys Ile Phe Gly Ser Leu Ala Phe Leu Pro Glu Ser Phe Asp Gly Asp
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    370                 375                 380

Pro Ala Ser Asn Thr Ala Pro Leu Gln Pro Glu Gln Leu Gln Val Phe
385                 390                 395                 400

Glu Thr Leu Glu Glu Ile Thr Gly Tyr Leu Tyr Ile Ser Ala Trp Pro
                405                 410                 415

Asp Ser Leu Pro Asp Leu Ser Val Phe Gln Asn Leu Gln Val Ile Arg
            420                 425                 430

Gly Arg Ile Leu His Asn Gly Ala Tyr Ser Leu Thr Leu Gln Gly Leu
        435                 440                 445

Gly Ile Ser Trp Leu Gly Leu Arg Ser Leu Arg Glu Leu Gly Ser Gly
    450                 455                 460

Leu Ala Leu Ile His His Asn Thr His Leu Cys Phe Val His Thr Val
465                 470                 475                 480

Pro Trp Asp Gln Leu Phe Arg Asn Pro His Gln Ala Leu Leu His Thr
                485                 490                 495

Ala Asn Arg Pro Glu Asp Glu Cys Val Gly Glu Gly Leu Ala Cys His
            500                 505                 510

Gln Leu Cys Ala Arg Gly His Cys Trp Gly Pro Gly Pro Thr Gln Cys
        515                 520                 525

Val Asn Cys Ser Gln Phe Leu Arg Gly Gln Glu Cys Val Glu Glu Cys
    530                 535                 540

Arg Val Leu Gln Gly Leu Pro Arg Glu Tyr Val Asn Ala Arg His Cys
545                 550                 555                 560

Leu Pro Cys His Pro Glu Cys Gln Pro Gln Asn Gly Ser Val Thr Cys
                565                 570                 575

Phe Gly Pro Glu Ala Asp Gln Cys Val Ala Cys Ala His Tyr Lys Asp
            580                 585                 590

Pro Pro Phe Cys Val Ala Arg Cys Pro Ser Gly Val Lys Pro Asp Leu
        595                 600                 605

Ser Tyr Met Pro Ile Trp Lys Phe Pro Asp Glu Glu Gly Ala Cys Gln
    610                 615                 620

Pro Cys Pro Ile Asn Cys Thr His Ser Cys Val Asp Leu Asp Asp Lys
625                 630                 635                 640

Gly Cys Pro Ala Glu Gln Arg Ala Ser Pro Leu Thr Ser Ile Ile Ser
                645                 650                 655

Ala Val Val Gly Ile Leu Leu Val Val Val Leu Gly Val Val Phe Gly
            660                 665                 670

Ile Leu Ile Lys Arg Arg Gln Gln Lys Ile Arg Lys Tyr Thr Met Arg
        675                 680                 685

Arg Leu Leu Gln Glu Thr Glu Leu Val Glu Pro Leu Thr Pro Ser Gly
    690                 695                 700

Ala Met Pro Asn Gln Ala Gln Met Arg Ile Leu Lys Glu Thr Glu Leu
705                 710                 715                 720

Arg Lys Val Lys Val Leu Gly Ser Gly Ala Phe Gly Thr Val Tyr Lys
                725                 730                 735

Gly Ile Trp Ile Pro Asp Gly Glu Asn Val Lys Ile Pro Val Ala Ile
            740                 745                 750

Lys Val Leu Arg Glu Asn Thr Ser Pro Lys Ala Asn Lys Glu Ile Leu
        755                 760                 765

Asp Glu Ala Tyr Val Met Ala Gly Val Gly Ser Pro Tyr Val Ser Arg
    770                 775                 780
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Leu Leu Gly Ile Cys Leu Thr Ser Thr Val Gln Leu Val Thr Gln Leu
785                 790                 795                 800

Met Pro Tyr Gly Cys Leu Leu Asp His Val Arg Glu Asn Arg Gly Arg
                805                 810                 815

Leu Gly Ser Gln Asp Leu Leu Asn Trp Cys Met Gln Ile Ala Lys Gly
            820                 825                 830

Met Ser Tyr Leu Glu Asp Val Arg Leu Val His Arg Asp Leu Ala Ala
        835                 840                 845

Arg Asn Val Leu Val Lys Ser Pro Asn His Val Lys Ile Thr Asp Phe
    850                 855                 860

Gly Leu Ala Arg Leu Leu Asp Ile Asp Glu Thr Glu Tyr His Ala Asp
865                 870                 875                 880

Gly Gly Lys Val Pro Ile Lys Trp Met Ala Leu Glu Ser Ile Leu Arg
                885                 890                 895

Arg Arg Phe Thr His Gln Ser Asp Val Trp Ser Tyr Gly Val Thr Val
            900                 905                 910

Trp Glu Leu Met Thr Phe Gly Ala Lys Pro Tyr Asp Gly Ile Pro Ala
        915                 920                 925

Arg Glu Ile Pro Asp Leu Leu Glu Lys Gly Glu Arg Leu Pro Gln Pro
    930                 935                 940

Pro Ile Cys Thr Ile Asp Val Tyr Met Ile Met Val Lys Cys Trp Met
945                 950                 955                 960

Ile Asp Ser Glu Cys Arg Pro Arg Phe Arg Glu Leu Val Ser Glu Phe
                965                 970                 975

Ser Arg Met Ala Arg Asp Pro Gln Arg Phe Val Val Ile Gln Asn Glu
            980                 985                 990

Asp Leu Gly Pro Ala Ser Pro Leu Asp Ser Thr Phe Tyr Arg Ser Leu
        995                 1000                1005

Leu Glu Asp Asp Asp Met Gly Asp Leu Val Asp Ala Glu Glu Tyr Leu
    1010                1015                1020

Val Pro Gln Gln Gly Phe Phe Cys Pro Asp Pro Ala Pro Gly Ala Gly
1025                1030                1035                1040

Gly Met Val His His Arg His Arg Ser Ser Ser Thr Arg Ser Gly Gly
                1045                1050                1055

Gly Asp Leu Thr Leu Gly Leu Glu Pro Ser Glu Glu Glu Ala Pro Arg
            1060                1065                1070

Ser Pro Leu Ala Pro Ser Glu Gly Ala Gly Ser Asp Val Phe Asp Gly
        1075                1080                1085

Asp Leu Gly Met Gly Ala Ala Lys Gly Leu Gln Ser Leu Pro Thr His
    1090                1095                1100

Asp Pro Ser Pro Leu Gln Arg Tyr Ser Glu Asp Pro Thr Val Pro Leu
1105                1110                1115                1120

Pro Ser Glu Thr Asp Gly Tyr Val Ala Pro Leu Thr Cys Ser Pro Gln
                1125                1130                1135

Pro Glu Tyr Val Asn Gln Pro Asp Val Arg Pro Gln Pro Pro Ser Pro
            1140                1145                1150

Arg Glu Gly Pro Leu Pro Ala Ala Arg Pro Ala Gly Ala Thr Leu Glu
        1155                1160                1165

Arg Pro Lys Thr Leu Ser Pro Gly Lys Asn Gly Val Val Lys Asp Val
    1170                1175                1180
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Phe Ala Phe Gly Gly Ala Val Glu Asn Pro Glu Tyr Leu Thr Pro Gln
1185                1190                1195                1200

Gly Gly Ala Ala Pro Gln Pro His Pro Pro Ala Phe Ser Pro Ala
                1205                1210                1215

Phe Asp Asn Leu Tyr Tyr Trp Asp Gln Asp Pro Pro Glu Arg Gly Ala
            1220                1225                1230

Pro Pro Ser Thr Phe Lys Gly Thr Pro Thr Ala Glu Asn Pro Glu Tyr
        1235                1240                1245

Leu Gly Leu Asp Val Pro Val
    1250                1255


<210> SEQ ID NO 6
<211> LENGTH: 1255
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
<220> FEATURE:
<223> OTHER INFORMATION: Mucin 1, cell surface associated protein (MUC1)

<400> SEQUENCE: 6

Met Thr Pro Gly Thr Gln Ser Pro Phe Phe Leu Leu Leu Leu Leu Thr
1               5                   10                  15

Val Leu Thr Val Val Thr Gly Ser Gly His Ala Ser Ser Thr Pro Gly
            20                  25                  30

Gly Glu Lys Glu Thr Ser Ala Thr Gln Arg Ser Ser Val Pro Ser Ser
        35                  40                  45

Thr Glu Lys Asn Ala Val Ser Met Thr Ser Ser Val Leu Ser Ser His
    50                  55                  60

Ser Pro Gly Ser Gly Ser Ser Thr Thr Gln Gly Gln Asp Val Thr Leu
65                  70                  75                  80

Ala Pro Ala Thr Glu Pro Ala Ser Gly Ser Ala Ala Thr Trp Gly Gln
                85                  90                  95

Asp Val Thr Ser Val Pro Val Thr Arg Pro Ala Leu Gly Ser Thr Thr
            100                 105                 110

Pro Pro Ala His Asp Val Thr Ser Ala Pro Asp Asn Lys Pro Ala Pro
        115                 120                 125

Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr
    130                 135                 140

Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser
145                 150                 155                 160

Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His
                165                 170                 175

Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala
            180                 185                 190

Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro
        195                 200                 205

Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr
    210                 215                 220

Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser
225                 230                 235                 240

Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His
                245                 250                 255

Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala
            260                 265                 270
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Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro
        275                 280                 285

Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr
    290                 295                 300

Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser
305                 310                 315                 320

Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His
                325                 330                 335

Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala
            340                 345                 350

Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro
        355                 360                 365

Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr
    370                 375                 380

Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser
385                 390                 395                 400

Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His
                405                 410                 415

Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala
            420                 425                 430

Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro
        435                 440                 445

Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr
    450                 455                 460

Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser
465                 470                 475                 480

Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His
                485                 490                 495

Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala
            500                 505                 510

Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro
        515                 520                 525

Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr
    530                 535                 540

Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser
545                 550                 555                 560

Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His
                565                 570                 575

Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala
            580                 585                 590

Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro
        595                 600                 605

Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr
    610                 615                 620

Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser
625                 630                 635                 640

Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His
                645                 650                 655

Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala
            660                 665                 670

Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro
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        675                 680                 685

Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr
    690                 695                 700

Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser
705                 710                 715                 720

Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His
                725                 730                 735

Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala
            740                 745                 750

Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro
        755                 760                 765

Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr
    770                 775                 780

Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser
785                 790                 795                 800

Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His
                805                 810                 815

Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala
            820                 825                 830

Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro
        835                 840                 845

Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr
    850                 855                 860

Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser
865                 870                 875                 880

Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala Pro Pro Ala His
                885                 890                 895

Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro Gly Ser Thr Ala
            900                 905                 910

Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Thr Arg Pro Ala Pro
        915                 920                 925

Gly Ser Thr Ala Pro Pro Ala His Gly Val Thr Ser Ala Pro Asp Asn
    930                 935                 940

Arg Pro Ala Leu Gly Ser Thr Ala Pro Pro Val His Asn Val Thr Ser
945                 950                 955                 960

Ala Ser Gly Ser Ala Ser Gly Ser Ala Ser Thr Leu Val His Asn Gly
                965                 970                 975

Thr Ser Ala Arg Ala Thr Thr Thr Pro Ala Ser Lys Ser Thr Pro Phe
            980                 985                 990

Ser Ile Pro Ser His His Ser Asp Thr Pro Thr Thr Leu Ala Ser His
        995                 1000                1005

Ser Thr Lys Thr Asp Ala Ser Ser Thr His His Ser Ser Val Pro Pro
    1010                1015                1020

Leu Thr Ser Ser Asn His Ser Thr Ser Pro Gln Leu Ser Thr Gly Val
1025                1030                1035                1040

Ser Phe Phe Phe Leu Ser Phe His Ile Ser Asn Leu Gln Phe Asn Ser
                1045                1050                1055

Ser Leu Glu Asp Pro Ser Thr Asp Tyr Tyr Gln Glu Leu Gln Arg Asp
            1060                1065                1070

Ile Ser Glu Met Phe Leu Gln Ile Tyr Lys Gln Gly Gly Phe Leu Gly
        1075                1080                1085
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Leu Ser Asn Ile Lys Phe Arg Pro Gly Ser Val Val Val Gln Leu Thr
    1090   1095   1100

Leu Ala Phe Arg Glu Gly Thr Ile Asn Val His Asp Val Glu Thr Gln
1105   1110   1115   1120

Phe Asn Gln Tyr Lys Thr Glu Ala Ala Ser Arg Tyr Asn Leu Thr Ile
                1125   1130   1135

Ser Asp Val Ser Val Ser Asp Val Pro Phe Pro Phe Ser Ala Gln Ser
            1140   1145   1150

Gly Ala Gly Val Pro Gly Trp Gly Ile Ala Leu Leu Val Leu Val Cys
        1155   1160   1165

Val Leu Val Ala Leu Ala Ile Val Tyr Leu Ile Ala Leu Ala Val Cys
    1170   1175   1180

Gln Cys Arg Arg Lys Asn Tyr Gly Gln Leu Asp Ile Phe Pro Ala Arg
1185   1190   1195   1200

Asp Thr Tyr His Pro Met Ser Glu Tyr Pro Thr Tyr His Thr His Gly
                1205   1210   1215

Arg Tyr Val Pro Pro Ser Ser Thr Asp Arg Ser Pro Tyr Glu Lys Val
            1220   1225   1230

Ser Ala Gly Asn Gly Gly Ser Ser Leu Ser Tyr Thr Asn Pro Ala Val
        1235   1240   1245

Ala Ala Thr Ser Ala Asn Leu
    1250   1255

<210> SEQ ID NO 7
<211> LENGTH: 519
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
<220> FEATURE:
<223> OTHER INFORMATION: Tyrosinase-related protein 2 (TRP-2)

<400> SEQUENCE: 7

Met Ser Pro Leu Trp Trp Gly Phe Leu Leu Ser Cys Leu Gly Cys Lys
1   5   10   15

Ile Leu Pro Gly Ala Gln Gly Gln Phe Pro Arg Val Cys Met Thr Val
            20   25   30

Asp Ser Leu Val Asn Lys Glu Cys Cys Pro Arg Leu Gly Ala Glu Ser
        35   40   45

Ala Asn Val Cys Gly Ser Gln Gln Gly Arg Gly Gln Cys Thr Glu Val
    50   55   60

Arg Ala Asp Thr Arg Pro Trp Ser Gly Pro Tyr Ile Leu Arg Asn Gln
65   70   75   80

Asp Asp Arg Glu Leu Trp Pro Arg Lys Phe Phe His Arg Thr Cys Lys
                85   90   95

Cys Thr Gly Asn Phe Ala Gly Tyr Asn Cys Gly Asp Cys Lys Phe Gly
            100   105   110

Trp Thr Gly Pro Asn Cys Glu Arg Lys Lys Pro Pro Val Ile Arg Gln
        115   120   125

Asn Ile His Ser Leu Ser Pro Gln Glu Arg Glu Gln Phe Leu Gly Ala
    130   135   140

Leu Asp Leu Ala Lys Lys Arg Val His Pro Asp Tyr Val Ile Thr Thr
145   150   155   160

Gln His Trp Leu Gly Leu Leu Gly Pro Asn Gly Thr Gln Pro Gln Phe
                165   170   175

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Ala Asn Cys Ser Val Tyr Asp Phe Phe Val Trp Leu His Tyr Tyr Ser
            180                 185                 190

Val Arg Asp Thr Leu Leu Gly Pro Gly Arg Pro Tyr Arg Ala Ile Asp
        195                 200                 205

Phe Ser His Gln Gly Pro Ala Phe Val Thr Trp His Arg Tyr His Leu
    210                 215                 220

Leu Cys Leu Glu Arg Asp Leu Gln Arg Leu Ile Gly Asn Glu Ser Phe
225                 230                 235                 240

Ala Leu Pro Tyr Trp Asn Phe Ala Thr Gly Arg Asn Glu Cys Asp Val
                245                 250                 255

Cys Thr Asp Gln Leu Phe Gly Ala Ala Arg Pro Asp Asp Pro Thr Leu
            260                 265                 270

Ile Ser Arg Asn Ser Arg Phe Ser Ser Trp Glu Thr Val Cys Asp Ser
        275                 280                 285

Leu Asp Asp Tyr Asn His Leu Val Thr Leu Cys Asn Gly Thr Tyr Glu
    290                 295                 300

Gly Leu Leu Arg Arg Asn Gln Met Gly Arg Asn Ser Met Lys Leu Pro
305                 310                 315                 320

Thr Leu Lys Asp Ile Arg Asp Cys Leu Ser Leu Gln Lys Phe Asp Asn
                325                 330                 335

Pro Pro Phe Phe Gln Asn Ser Thr Phe Ser Phe Arg Asn Ala Leu Glu
            340                 345                 350

Gly Phe Asp Lys Ala Asp Gly Thr Leu Asp Ser Gln Val Met Ser Leu
        355                 360                 365

His Asn Leu Val His Ser Phe Leu Asn Gly Thr Asn Ala Leu Pro His
    370                 375                 380

Ser Ala Ala Asn Asp Pro Ile Phe Val Val Leu His Ser Phe Thr Asp
385                 390                 395                 400

Ala Ile Phe Asp Glu Trp Met Lys Arg Phe Asn Pro Pro Ala Asp Ala
                405                 410                 415

Trp Pro Gln Glu Leu Ala Pro Ile Gly His Asn Arg Met Tyr Asn Met
            420                 425                 430

Val Pro Phe Phe Pro Pro Val Thr Asn Glu Glu Leu Phe Leu Thr Ser
        435                 440                 445

Asp Gln Leu Gly Tyr Ser Tyr Ala Ile Asp Leu Pro Val Ser Val Glu
    450                 455                 460

Glu Thr Pro Gly Trp Pro Thr Thr Leu Leu Val Val Met Gly Thr Leu
465                 470                 475                 480

Val Ala Leu Val Gly Leu Phe Val Leu Leu Ala Phe Leu Gln Tyr Arg
                485                 490                 495

Arg Leu Arg Lys Gly Tyr Thr Pro Leu Met Glu Thr His Leu Ser Ser
            500                 505                 510

Lys Arg Tyr Thr Glu Glu Ala
        515
```

<210> SEQ ID NO 8
<211> LENGTH: 766
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
<220> FEATURE:
<223> OTHER INFORMATION: Serine/threonine-protein kinase B-raf

<400> SEQUENCE: 8

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Met Ala Ala Leu Ser Gly Gly Gly Gly Gly Gly Ala Glu Pro Gly Gln
1 5 10 15

Ala Leu Phe Asn Gly Asp Met Glu Pro Glu Ala Gly Ala Gly Ala Gly
20 25 30

Ala Ala Ala Ser Ser Ala Ala Asp Pro Ala Ile Pro Glu Glu Val Trp
35 40 45

Asn Ile Lys Gln Met Ile Lys Leu Thr Gln Glu His Ile Glu Ala Leu
50 55 60

Leu Asp Lys Phe Gly Gly Glu His Asn Pro Pro Ser Ile Tyr Leu Glu
65 70 75 80

Ala Tyr Glu Glu Tyr Thr Ser Lys Leu Asp Ala Leu Gln Gln Arg Glu
85 90 95

Gln Gln Leu Leu Glu Ser Leu Gly Asn Gly Thr Asp Phe Ser Val Ser
100 105 110

Ser Ser Ala Ser Met Asp Thr Val Thr Ser Ser Ser Ser Ser Ser Leu
115 120 125

Ser Val Leu Pro Ser Ser Leu Ser Val Phe Gln Asn Pro Thr Asp Val
130 135 140

Ala Arg Ser Asn Pro Lys Ser Pro Gln Lys Pro Ile Val Arg Val Phe
145 150 155 160

Leu Pro Asn Lys Gln Arg Thr Val Val Pro Ala Arg Cys Gly Val Thr
165 170 175

Val Arg Asp Ser Leu Lys Lys Ala Leu Met Met Arg Gly Leu Ile Pro
180 185 190

Glu Cys Cys Ala Val Tyr Arg Ile Gln Asp Gly Glu Lys Lys Pro Ile
195 200 205

Gly Trp Asp Thr Asp Ile Ser Trp Leu Thr Gly Glu Glu Leu His Val
210 215 220

Glu Val Leu Glu Asn Val Pro Leu Thr Thr His Asn Phe Val Arg Lys
225 230 235 240

Thr Phe Phe Thr Leu Ala Phe Cys Asp Phe Cys Arg Lys Leu Leu Phe
245 250 255

Gln Gly Phe Arg Cys Gln Thr Cys Gly Tyr Lys Phe His Gln Arg Cys
260 265 270

Ser Thr Glu Val Pro Leu Met Cys Val Asn Tyr Asp Gln Leu Asp Leu
275 280 285

Leu Phe Val Ser Lys Phe Phe Glu His His Pro Ile Pro Gln Glu Glu
290 295 300

Ala Ser Leu Ala Glu Thr Ala Leu Thr Ser Gly Ser Ser Pro Ser Ala
305 310 315 320

Pro Ala Ser Asp Ser Ile Gly Pro Gln Ile Leu Thr Ser Pro Ser Pro
325 330 335

Ser Lys Ser Ile Pro Ile Pro Gln Pro Phe Arg Pro Ala Asp Glu Asp
340 345 350

His Arg Asn Gln Phe Gly Gln Arg Asp Arg Ser Ser Ser Ala Pro Asn
355 360 365

Val His Ile Asn Thr Ile Glu Pro Val Asn Ile Asp Asp Leu Ile Arg
370 375 380

Asp Gln Gly Phe Arg Gly Asp Gly Gly Ser Thr Thr Gly Leu Ser Ala
385 390 395 400

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Thr Pro Pro Ala Ser Leu Pro Gly Ser Leu Thr Asn Val Lys Ala Leu
                405                 410                 415

Gln Lys Ser Pro Gly Pro Gln Arg Glu Arg Lys Ser Ser Ser Ser Ser
            420                 425                 430

Glu Asp Arg Asn Arg Met Lys Thr Leu Gly Arg Arg Asp Ser Ser Asp
        435                 440                 445

Asp Trp Glu Ile Pro Asp Gly Gln Ile Thr Val Gly Gln Arg Ile Gly
    450                 455                 460

Ser Gly Ser Phe Gly Thr Val Tyr Lys Gly Lys Trp His Gly Asp Val
465                 470                 475                 480

Ala Val Lys Met Leu Asn Val Thr Ala Pro Thr Pro Gln Gln Leu Gln
                485                 490                 495

Ala Phe Lys Asn Glu Val Gly Val Leu Arg Lys Thr Arg His Val Asn
            500                 505                 510

Ile Leu Leu Phe Met Gly Tyr Ser Thr Lys Pro Gln Leu Ala Ile Val
        515                 520                 525

Thr Gln Trp Cys Glu Gly Ser Ser Leu Tyr His His Leu His Ile Ile
    530                 535                 540

Glu Thr Lys Phe Glu Met Ile Lys Leu Ile Asp Ile Ala Arg Gln Thr
545                 550                 555                 560

Ala Gln Gly Met Asp Tyr Leu His Ala Lys Ser Ile Ile His Arg Asp
                565                 570                 575

Leu Lys Ser Asn Asn Ile Phe Leu His Glu Asp Leu Thr Val Lys Ile
            580                 585                 590

Gly Asp Phe Gly Leu Ala Thr Val Lys Ser Arg Trp Ser Gly Ser His
        595                 600                 605

Gln Phe Glu Gln Leu Ser Gly Ser Ile Leu Trp Met Ala Pro Glu Val
    610                 615                 620

Ile Arg Met Gln Asp Lys Asn Pro Tyr Ser Phe Gln Ser Asp Val Tyr
625                 630                 635                 640

Ala Phe Gly Ile Val Leu Tyr Glu Leu Met Thr Gly Gln Leu Pro Tyr
                645                 650                 655

Ser Asn Ile Asn Asn Arg Asp Gln Ile Ile Phe Met Val Gly Arg Gly
            660                 665                 670

Tyr Leu Ser Pro Asp Leu Ser Lys Val Arg Ser Asn Cys Pro Lys Ala
        675                 680                 685

Met Lys Arg Leu Met Ala Glu Cys Leu Lys Lys Lys Arg Asp Glu Arg
    690                 695                 700

Pro Leu Phe Pro Gln Ile Leu Ala Ser Ile Glu Leu Leu Ala Arg Ser
705                 710                 715                 720

Leu Pro Lys Ile His Arg Ser Ala Ser Glu Pro Ser Leu Asn Arg Ala
                725                 730                 735

Gly Phe Gln Thr Glu Asp Phe Ser Leu Tyr Ala Cys Ala Ser Pro Lys
            740                 745                 750

Thr Pro Ile Gln Ala Gly Gly Tyr Gly Ala Phe Pro Val His
        755                 760                 765
```

<210> SEQ ID NO 9
<211> LENGTH: 976
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
<220> FEATURE:
<223> OTHER INFORMATION: Mast/stem cell growth factor receptor KIT

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<400> SEQUENCE: 9

```
Met Arg Gly Ala Arg Gly Ala Trp Asp Phe Leu Cys Val Leu Leu Leu
1               5                   10                  15

Leu Leu Arg Val Gln Thr Gly Ser Ser Gln Pro Ser Val Ser Pro Gly
            20                  25                  30

Glu Pro Ser Pro Pro Ser Ile His Pro Gly Lys Ser Asp Leu Ile Val
        35                  40                  45

Arg Val Gly Asp Glu Ile Arg Leu Leu Cys Thr Asp Pro Gly Phe Val
    50                  55                  60

Lys Trp Thr Phe Glu Ile Leu Asp Glu Thr Asn Glu Asn Lys Gln Asn
65                  70                  75                  80

Glu Trp Ile Thr Glu Lys Ala Glu Ala Thr Asn Thr Gly Lys Tyr Thr
                85                  90                  95

Cys Thr Asn Lys His Gly Leu Ser Asn Ser Ile Tyr Val Phe Val Arg
            100                 105                 110

Asp Pro Ala Lys Leu Phe Leu Val Asp Arg Ser Leu Tyr Gly Lys Glu
        115                 120                 125

Asp Asn Asp Thr Leu Val Arg Cys Pro Leu Thr Asp Pro Glu Val Thr
    130                 135                 140

Asn Tyr Ser Leu Lys Gly Cys Gln Gly Lys Pro Leu Pro Lys Asp Leu
145                 150                 155                 160

Arg Phe Ile Pro Asp Pro Lys Ala Gly Ile Met Ile Lys Ser Val Lys
                165                 170                 175

Arg Ala Tyr His Arg Leu Cys Leu His Cys Ser Val Asp Gln Glu Gly
            180                 185                 190

Lys Ser Val Leu Ser Glu Lys Phe Ile Leu Lys Val Arg Pro Ala Phe
        195                 200                 205

Lys Ala Val Pro Val Val Ser Val Ser Lys Ala Ser Tyr Leu Leu Arg
    210                 215                 220

Glu Gly Glu Glu Phe Thr Val Thr Cys Thr Ile Lys Asp Val Ser Ser
225                 230                 235                 240

Ser Val Tyr Ser Thr Trp Lys Arg Glu Asn Ser Gln Thr Lys Leu Gln
                245                 250                 255

Glu Lys Tyr Asn Ser Trp His His Gly Asp Phe Asn Tyr Glu Arg Gln
            260                 265                 270

Ala Thr Leu Thr Ile Ser Ser Ala Arg Val Asn Asp Ser Gly Val Phe
        275                 280                 285

Met Cys Tyr Ala Asn Asn Thr Phe Gly Ser Ala Asn Val Thr Thr Thr
    290                 295                 300

Leu Glu Val Val Asp Lys Gly Phe Ile Asn Ile Phe Pro Met Ile Asn
305                 310                 315                 320

Thr Thr Val Phe Val Asn Asp Gly Glu Asn Val Asp Leu Ile Val Glu
                325                 330                 335

Tyr Glu Ala Phe Pro Lys Pro Glu His Gln Gln Trp Ile Tyr Met Asn
            340                 345                 350

Arg Thr Phe Thr Asp Lys Trp Glu Asp Tyr Pro Lys Ser Glu Asn Glu
        355                 360                 365

Ser Asn Ile Arg Tyr Val Ser Glu Leu His Leu Thr Arg Leu Lys Gly
    370                 375                 380

Thr Glu Gly Gly Thr Tyr Thr Phe Leu Val Ser Asn Ser Asp Val Asn
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385                 390                 395                 400

Ala Ala Ile Ala Phe Asn Val Tyr Val Asn Thr Lys Pro Glu Ile Leu
                405                 410                 415

Thr Tyr Asp Arg Leu Val Asn Gly Met Leu Gln Cys Val Ala Ala Gly
            420                 425                 430

Phe Pro Glu Pro Thr Ile Asp Trp Tyr Phe Cys Pro Gly Thr Glu Gln
        435                 440                 445

Arg Cys Ser Ala Ser Val Leu Pro Val Asp Val Gln Thr Leu Asn Ser
    450                 455                 460

Ser Gly Pro Pro Phe Gly Lys Leu Val Val Gln Ser Ser Ile Asp Ser
465                 470                 475                 480

Ser Ala Phe Lys His Asn Gly Thr Val Glu Cys Lys Ala Tyr Asn Asp
                485                 490                 495

Val Gly Lys Thr Ser Ala Tyr Phe Asn Phe Ala Phe Lys Gly Asn Asn
            500                 505                 510

Lys Glu Gln Ile His Pro His Thr Leu Phe Thr Pro Leu Leu Ile Gly
        515                 520                 525

Phe Val Ile Val Ala Gly Met Met Cys Ile Ile Val Met Ile Leu Thr
    530                 535                 540

Tyr Lys Tyr Leu Gln Lys Pro Met Tyr Glu Val Gln Trp Lys Val Val
545                 550                 555                 560

Glu Glu Ile Asn Gly Asn Asn Tyr Val Tyr Ile Asp Pro Thr Gln Leu
                565                 570                 575

Pro Tyr Asp His Lys Trp Glu Phe Pro Arg Asn Arg Leu Ser Phe Gly
            580                 585                 590

Lys Thr Leu Gly Ala Gly Ala Phe Gly Lys Val Val Glu Ala Thr Ala
        595                 600                 605

Tyr Gly Leu Ile Lys Ser Asp Ala Ala Met Thr Val Ala Val Lys Met
    610                 615                 620

Leu Lys Pro Ser Ala His Leu Thr Glu Arg Glu Ala Leu Met Ser Glu
625                 630                 635                 640

Leu Lys Val Leu Ser Tyr Leu Gly Asn His Met Asn Ile Val Asn Leu
                645                 650                 655

Leu Gly Ala Cys Thr Ile Gly Gly Pro Thr Leu Val Ile Thr Glu Tyr
            660                 665                 670

Cys Cys Tyr Gly Asp Leu Leu Asn Phe Leu Arg Arg Lys Arg Asp Ser
        675                 680                 685

Phe Ile Cys Ser Lys Gln Glu Asp His Ala Glu Ala Ala Leu Tyr Lys
    690                 695                 700

Asn Leu Leu His Ser Lys Glu Ser Ser Cys Ser Asp Ser Thr Asn Glu
705                 710                 715                 720

Tyr Met Asp Met Lys Pro Gly Val Ser Tyr Val Val Pro Thr Lys Ala
                725                 730                 735

Asp Lys Arg Arg Ser Val Arg Ile Gly Ser Tyr Ile Glu Arg Asp Val
            740                 745                 750

Thr Pro Ala Ile Met Glu Asp Asp Glu Leu Ala Leu Asp Leu Glu Asp
        755                 760                 765

Leu Leu Ser Phe Ser Tyr Gln Val Ala Lys Gly Met Ala Phe Leu Ala
    770                 775                 780

Ser Lys Asn Cys Ile His Arg Asp Leu Ala Ala Arg Asn Ile Leu Leu
785                 790                 795                 800
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```
Thr His Gly Arg Ile Thr Lys Ile Cys Asp Phe Gly Leu Ala Arg Asp
                805                 810                 815

Ile Lys Asn Asp Ser Asn Tyr Val Val Lys Gly Asn Ala Arg Leu Pro
            820                 825                 830

Val Lys Trp Met Ala Pro Glu Ser Ile Phe Asn Cys Val Tyr Thr Phe
        835                 840                 845

Glu Ser Asp Val Trp Ser Tyr Gly Ile Phe Leu Trp Glu Leu Phe Ser
    850                 855                 860

Leu Gly Ser Ser Pro Tyr Pro Gly Met Pro Val Asp Ser Lys Phe Tyr
865                 870                 875                 880

Lys Met Ile Lys Glu Gly Phe Arg Met Leu Ser Pro Glu His Ala Pro
                885                 890                 895

Ala Glu Met Tyr Asp Ile Met Lys Thr Cys Trp Asp Ala Asp Pro Leu
            900                 905                 910

Lys Arg Pro Thr Phe Lys Gln Ile Val Gln Leu Ile Glu Lys Gln Ile
        915                 920                 925

Ser Glu Ser Thr Asn His Ile Tyr Ser Asn Leu Ala Asn Cys Ser Pro
    930                 935                 940

Asn Arg Gln Lys Pro Val Val Asp His Ser Val Arg Ile Asn Ser Val
945                 950                 955                 960

Gly Ser Thr Ala Ser Ser Ser Gln Pro Leu Leu Val His Asp Asp Val
                965                 970                 975


<210> SEQ ID NO 10
<211> LENGTH: 189
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
<220> FEATURE:
<223> OTHER INFORMATION: GTPase NRas

<400> SEQUENCE: 10

Met Thr Glu Tyr Lys Leu Val Val Val Gly Ala Gly Gly Val Gly Lys
1               5                   10                  15

Ser Ala Leu Thr Ile Gln Leu Ile Gln Asn His Phe Val Asp Glu Tyr
            20                  25                  30

Asp Pro Thr Ile Glu Asp Ser Tyr Arg Lys Gln Val Val Ile Asp Gly
        35                  40                  45

Glu Thr Cys Leu Leu Asp Ile Leu Asp Thr Ala Gly Gln Glu Glu Tyr
    50                  55                  60

Ser Ala Met Arg Asp Gln Tyr Met Arg Thr Gly Glu Gly Phe Leu Cys
65                  70                  75                  80

Val Phe Ala Ile Asn Asn Ser Lys Ser Phe Ala Asp Ile Asn Leu Tyr
                85                  90                  95

Arg Glu Gln Ile Lys Arg Val Lys Asp Ser Asp Asp Val Pro Met Val
            100                 105                 110

Leu Val Gly Asn Lys Cys Asp Leu Pro Thr Arg Thr Val Asp Thr Lys
        115                 120                 125

Gln Ala His Glu Leu Ala Lys Ser Tyr Gly Ile Pro Phe Ile Glu Thr
    130                 135                 140

Ser Ala Lys Thr Arg Gln Gly Val Glu Asp Ala Phe Tyr Thr Leu Val
145                 150                 155                 160

Arg Glu Ile Arg Gln Tyr Arg Met Lys Lys Leu Asn Ser Ser Asp Asp
                165                 170                 175
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Gly Thr Gln Gly Cys Met Gly Leu Pro Cys Val Val Met
            180                 185


<210> SEQ ID NO 11
<211> LENGTH: 309
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
<220> FEATURE:
<223> OTHER INFORMATION: Melanoma-associated antigen 1

<400> SEQUENCE: 11

Met Ser Leu Glu Gln Arg Ser Leu His Cys Lys Pro Glu Glu Ala Leu
1               5                   10                  15

Glu Ala Gln Gln Glu Ala Leu Gly Leu Val Cys Val Gln Ala Ala Thr
            20                  25                  30

Ser Ser Ser Ser Pro Leu Val Leu Gly Thr Leu Glu Glu Val Pro Thr
        35                  40                  45

Ala Gly Ser Thr Asp Pro Pro Gln Ser Pro Gln Gly Ala Ser Ala Phe
    50                  55                  60

Pro Thr Thr Ile Asn Phe Thr Arg Gln Arg Gln Pro Ser Glu Gly Ser
65                  70                  75                  80

Ser Ser Arg Glu Glu Glu Gly Pro Ser Thr Ser Cys Ile Leu Glu Ser
                85                  90                  95

Leu Phe Arg Ala Val Ile Thr Lys Lys Val Ala Asp Leu Val Gly Phe
            100                 105                 110

Leu Leu Leu Lys Tyr Arg Ala Arg Glu Pro Val Thr Lys Ala Glu Met
        115                 120                 125

Leu Glu Ser Val Ile Lys Asn Tyr Lys His Cys Phe Pro Glu Ile Phe
    130                 135                 140

Gly Lys Ala Ser Glu Ser Leu Gln Leu Val Phe Gly Ile Asp Val Lys
145                 150                 155                 160

Glu Ala Asp Pro Thr Gly His Ser Tyr Val Leu Val Thr Cys Leu Gly
                165                 170                 175

Leu Ser Tyr Asp Gly Leu Leu Gly Asp Asn Gln Ile Met Pro Lys Thr
            180                 185                 190

Gly Phe Leu Ile Ile Val Leu Val Met Ile Ala Met Glu Gly Gly His
        195                 200                 205

Ala Pro Glu Glu Glu Ile Trp Glu Glu Leu Ser Val Met Glu Val Tyr
    210                 215                 220

Asp Gly Arg Glu His Ser Ala Tyr Gly Glu Pro Arg Lys Leu Leu Thr
225                 230                 235                 240

Gln Asp Leu Val Gln Glu Lys Tyr Leu Glu Tyr Arg Gln Val Pro Asp
                245                 250                 255

Ser Asp Pro Ala Arg Tyr Glu Phe Leu Trp Gly Pro Arg Ala Leu Ala
            260                 265                 270

Glu Thr Ser Tyr Val Lys Val Leu Glu Tyr Val Ile Lys Val Ser Ala
        275                 280                 285

Arg Val Arg Phe Phe Phe Pro Ser Leu Arg Glu Ala Ala Leu Arg Glu
    290                 295                 300

Glu Glu Glu Gly Val
305


<210> SEQ ID NO 12
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<211> LENGTH: 314
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
<220> FEATURE:
<223> OTHER INFORMATION: Melanoma-associated antigen 1

<400> SEQUENCE: 12

```
Met Pro Leu Glu Gln Arg Ser Gln His Cys Lys Pro Glu Glu Gly Leu
1               5                   10                  15

Glu Ala Arg Gly Glu Ala Leu Gly Leu Val Gly Ala Gln Ala Pro Ala
            20                  25                  30

Thr Glu Glu Gln Glu Ala Ala Ser Ser Ser Ser Thr Leu Val Glu Val
        35                  40                  45

Thr Leu Gly Glu Val Pro Ala Ala Glu Ser Pro Asp Pro Pro Gln Ser
    50                  55                  60

Pro Gln Gly Ala Ser Ser Leu Pro Thr Thr Met Asn Tyr Pro Leu Trp
65                  70                  75                  80

Ser Gln Ser Tyr Glu Asp Ser Ser Asn Gln Glu Glu Glu Gly Pro Ser
                85                  90                  95

Thr Phe Pro Asp Leu Glu Ser Glu Phe Gln Ala Ala Leu Ser Arg Lys
            100                 105                 110

Val Ala Glu Leu Val His Phe Leu Leu Leu Lys Tyr Arg Ala Arg Glu
        115                 120                 125

Pro Val Thr Lys Ala Glu Met Leu Gly Ser Val Val Gly Asn Trp Gln
    130                 135                 140

Tyr Phe Phe Pro Val Ile Phe Ser Lys Ala Ser Ser Ser Leu Gln Leu
145                 150                 155                 160

Val Phe Gly Ile Glu Leu Met Glu Val Asp Pro Ile Gly His Leu Tyr
                165                 170                 175

Ile Phe Ala Thr Cys Leu Gly Leu Ser Tyr Asp Gly Leu Leu Gly Asp
            180                 185                 190

Asn Gln Ile Met Pro Lys Ala Gly Leu Leu Ile Ile Val Leu Ala Ile
        195                 200                 205

Ile Ala Arg Glu Gly Asp Cys Ala Pro Glu Glu Lys Ile Trp Glu Glu
    210                 215                 220

Leu Ser Val Leu Glu Val Phe Glu Gly Arg Glu Asp Ser Ile Leu Gly
225                 230                 235                 240

Asp Pro Lys Lys Leu Leu Thr Gln His Phe Val Gln Glu Asn Tyr Leu
                245                 250                 255

Glu Tyr Arg Gln Val Pro Gly Ser Asp Pro Ala Cys Tyr Glu Phe Leu
            260                 265                 270

Trp Gly Pro Arg Ala Leu Val Glu Thr Ser Tyr Val Lys Val Leu His
        275                 280                 285

His Met Val Lys Ile Ser Gly Gly Pro His Ile Ser Tyr Pro Pro Leu
    290                 295                 300

His Glu Trp Val Leu Arg Glu Gly Glu Glu
305                 310
```

<210> SEQ ID NO 13
<211> LENGTH: 180
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
<220> FEATURE:
<223> OTHER INFORMATION: NY-ESO-1 protein

-continued

<400> SEQUENCE: 13

```
Met Gln Ala Glu Gly Arg Gly Thr Gly Gly Ser Thr Gly Asp Ala Asp
1               5                   10                  15

Gly Pro Gly Gly Pro Gly Ile Pro Asp Gly Pro Gly Gly Asn Ala Gly
            20                  25                  30

Gly Pro Gly Glu Ala Gly Ala Thr Gly Gly Arg Gly Pro Arg Gly Ala
        35                  40                  45

Gly Ala Ala Arg Ala Ser Gly Pro Gly Gly Gly Ala Pro Arg Gly Pro
    50                  55                  60

His Gly Gly Ala Ala Ser Gly Leu Asn Gly Cys Cys Arg Cys Gly Ala
65                  70                  75                  80

Arg Gly Pro Glu Ser Arg Leu Leu Glu Phe Tyr Leu Ala Met Pro Phe
                85                  90                  95

Ala Thr Pro Met Glu Ala Glu Leu Ala Arg Arg Ser Leu Ala Gln Asp
            100                 105                 110

Ala Pro Pro Leu Pro Val Pro Gly Val Leu Leu Lys Glu Phe Thr Val
        115                 120                 125

Ser Gly Asn Ile Leu Thr Ile Arg Leu Thr Ala Ala Asp His Arg Gln
    130                 135                 140

Leu Gln Leu Ser Ile Ser Ser Cys Leu Gln Gln Leu Ser Leu Leu Met
145                 150                 155                 160

Trp Ile Thr Gln Cys Phe Leu Pro Val Phe Leu Ala Gln Pro Pro Ser
                165                 170                 175

Gly Gln Arg Arg
            180
```

<210> SEQ ID NO 14
<211> LENGTH: 3822
<212> TYPE: RNA
<213> ORGANISM: SARS coronavirus
<220> FEATURE:
<223> OTHER INFORMATION: mRNA encoding SARS-CoV-2 Spike (S) protein

<400> SEQUENCE: 14

```
auguuuguuu uucuuguuuu auugccacua gucucuaguc agugugguuaa ucuuacaacc     60

agaacucaau uacccccugc auacacuaau ucuuucacac guggguguuua uuacccugac    120

aaaguuuuca gauccucagu uuuacauuca acucaggacu uguucuuacc uuucuuuucc    180

aauguuacuu gguuccaugc uauacauguc ucugggacca augguacuaa gagguuugau    240

aacccugucc uaccauuuaa ugauggguu uauuuugcuu ccacugagaa gucuaacaua    300

auaagaggcu ggauuuuugg uacuacuuua gauucgaaga cccagucccu acuuauuguu    360

aauaacgcua cuaauguugu uauuaaaguc ugugaauuuc aauuuuguaa ugauccauuu    420

uuggguguuu auuaccacaa aaacaacaaa aguuggaugg aaagugaguu cagaguuuau    480

ucuagugcga auaauugcac uuuugaauau gucucucagc cuuuucuuau ggaccuugaa    540

ggaaaacagg guaauuucaa aaaucuuagg gaauuugugu uuaagaauau ugaugguuau    600

uuuaaaauau auucuaagca cacgccuauu aauuuagugc gugaucuccc ucaggguuuu    660

ucggcuuuag aaccauuggu agauuugcca auagguauua acaucacuag guuucaaacu    720

uuacuugcuu uacauagaag uuauuugacu ccuggugauu cuucuucagg uuggacagcu    780

ggugcugcag cuuauuaugu ggguuaucuu caaccuagga cuuuucuauu aaaauauaau    840
```

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```
gaaaauggaa ccauuacaga ugcuguagac ugugcacuug acccucucuc agaaacaaag    900
uguacguuga aauccuucac uguagaaaaa ggaaucuauc aaacuucuaa cuuuagaguc    960
caaccaacag aaucuauugu uagauuuccu aauauuacaa acuugugccc uuuuggugaa   1020
guuuuuaacg ccaccagauu ugcaucuguu uaugcuugga acaggaagag aaucagcaac   1080
uguguugcug auuauucugu ccuauauaau uccgcaucau uuuccacuuu uaaguguuau   1140
ggagugucuc cuacuaaauu aaaugaucuc ugcuuuacua augucuaugc agauucauuu   1200
guaauuagag gugaugaagu cagacaaauc gcuccagggc aaacuggaaa gauugcugau   1260
uauaauuaua aauuaccaga ugauuuuaca ggcugcguua uagcuuggaa uucuaacaau   1320
cuugauucua agguuggugg uaauuauaau uaccuguaua gauuguuuag gaagucuaau   1380
cucaaaccuu uugagagaga uauuucaacu gaaaucuauc aggccgguag cacaccuugu   1440
aaugguguug aagguuuuaa uuguuacuuu ccuuuacaau cauaugguuu ccaacccacu   1500
aaugguguug guuaccaacc auacagagua guaguacuuu cuuuugaacu ucuacaugca   1560
ccagcaacug uuuguggacc uaaaaagucu acuaauuugg uuaaaaacaa augugucaau   1620
uucaacuuca augguuuaac aggcacaggu guucuuacug agucuaacaa aaaguuucug   1680
ccuuuccaac aauuuggcag agacauugcu gacacuacug augcuguccg ugauccacag   1740
acacuugaga uucuugacau uacaccaugu ucuuuuggug gugucagugu uauaacacca   1800
ggaacaaaua cuucuaacca gguugcuguu cuuuaucagg auguuaacug cacagaaguc   1860
ccuguugcua uucaugcaga ucaacuuacu ccuacuuggc guguuuauuc uacagguucu   1920
aauguuuuuc aaacacgugc aggcuguuua auaggggcug aacaugucaa caacucauau   1980
gagugugaca uacccauugg ugcagguaua ugcgcuaguu aucagacuca gacuaauucu   2040
ccucggcggg cacguagugu agcuagucaa uccaucauug ccuacacuau gucacuuggu   2100
gcagaaaauu caguugcuua cucuaauaac ucuauugcca uacccacaaa uuuuacuauu   2160
aguguuacca cagaaauucu accagugucu augaccaaga caucaguaga uuguacaaug   2220
uacauuugug gugauucaac ugaaugcagc aaucuuuugu ugcaauaugg caguuuuugu   2280
acacaauuaa accgugcuuu aacuggaaua gcuguugaac aagacaaaaa cacccaagaa   2340
guuuuugcac aagucaaaca aauuuacaaa acaccaccaa uuaaagauuu ugguggguuu   2400
aauuuuucac aaauauuacc agauccauca aaaccaagca agaggucauu uauugaagau   2460
cuacuuuuca acaaagugac acuugcagau gcuggcuuca ucaaacaaua uggugauugc   2520
cuuggugaua uugcugcuag agaccucauu ugugcacaaa aguuuaacgg ccuuacuguu   2580
uugccaccuu ugcucacaga ugaaaugauu gcucaauaca cuucugcacu guuagcgggu   2640
acaaucacuu cugguuggac cuuuggugca ggugcugcau uacaaauacc auuugcuaug   2700
caaauggcuu auagguuuaa ugguauugga guuacacaga auguucucua ugagaaccaa   2760
aaauugauug ccaaccaauu uaauagugcu auuggcaaaa uucaagacuc acuuucuucc   2820
acagcaagug cacuuggaaa acuucaagau guggucaacc aaaaugcaca agcuuuaaac   2880
acgcuuguua aacaacuuag cuccaauuuu ggugcaauuu caaguguuuu aaaugauauc   2940
cuuucacguc uugacaaagu ugaggcugaa gugcaaauug auagguugau cacaggcaga   3000
cuucaaaguu ugcagacaua ugugacucaa caauuaauua gagcugcaga aaucagagcu   3060
ucugcuaauc uugcugcuac uaaaauguca gaguguguac uuggacaauc aaaaagaguu   3120
```

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```
gauuuuugug gaaagggcua ucaucuuaug uccuucccuc agucagcacc ucauggugua   3180

gucuucuugc augugacuua ugucccugca caagaaaaga acuucacaac ugcuccugcc   3240

auuugucaug auggaaaagc acacuuuccu cgugaaggug ucuuuguuuc aaauggcaca   3300

cacugguuug uaacacaaag gaauuuuuau gaaccacaaa ucauuacuac agacaacaca   3360

uuugugucug guaacuguga uguuguaaua ggaauuguca acaacacagu uuaugauccu   3420

uugcaaccug aauuagacuc auucaaggag gaguuagaua aauauuuuaa gaaucauaca   3480

ucaccagaug uugauuuagg ugacaucucu ggcauuaaug cuucaguugu aaacauucaa   3540

aaagaaauug accgccucaa ugagguugcc aagaauuuaa augaaucucu caucgaucuc   3600

caagaacuug gaaaguauga gcaguauaua aaauggccau gguacauuug gcuagguuuu   3660

auagcuggcu ugauugccau aguaauggug acaauuaugc uuugcuguau gaccaguugc   3720

uguaguuguc ucaagggcug uuguucuugu ggauccugcu gcaaauuuga ugaagacgac   3780

ucugagccag ugcucaaagg agucaaauua cauuacacau aa                      3822
```

<210> SEQ ID NO 15
<211> LENGTH: 254
<212> TYPE: PRT
<213> ORGANISM: SARS coronavirus
<220> FEATURE:
<223> OTHER INFORMATION: SARS-CoV-2 Spike (S) protein

<400> SEQUENCE: 15

```
Pro Asn Ile Thr Asn Leu Cys Pro Phe Gly Glu Val Phe Asn Ala Thr
1               5                   10                  15

Arg Phe Ala Ser Val Tyr Ala Trp Asn Arg Lys Arg Ile Ser Asn Cys
            20                  25                  30

Val Ala Asp Tyr Ser Val Leu Tyr Asn Ser Ala Ser Phe Ser Thr Phe
        35                  40                  45

Lys Cys Tyr Gly Val Ser Pro Thr Lys Leu Asn Asp Leu Cys Phe Thr
    50                  55                  60

Asn Val Tyr Ala Asp Ser Phe Val Ile Arg Gly Asp Glu Val Arg Gln
65                  70                  75                  80

Ile Ala Pro Gly Gln Thr Gly Lys Ile Ala Asp Tyr Asn Tyr Lys Leu
                85                  90                  95

Pro Asp Asp Phe Thr Gly Cys Val Ile Ala Trp Asn Ser Asn Asn Leu
            100                 105                 110

Asp Ser Lys Val Gly Gly Asn Tyr Asn Tyr Leu Tyr Arg Leu Phe Arg
        115                 120                 125

Lys Ser Asn Leu Lys Pro Phe Glu Arg Asp Ile Ser Thr Glu Ile Tyr
    130                 135                 140

Gln Ala Gly Ser Thr Pro Cys Asn Gly Val Glu Gly Phe Asn Cys Tyr
145                 150                 155                 160

Phe Pro Leu Gln Ser Tyr Gly Phe Gln Pro Thr Asn Gly Val Gly Tyr
                165                 170                 175

Gln Pro Tyr Arg Val Val Val Leu Ser Phe Glu Leu Leu His Ala Pro
            180                 185                 190

Ala Thr Val Cys Gly Pro Lys Lys Ser Thr Asn Leu Val Lys Asn Lys
        195                 200                 205

Cys Val Asn Phe Asn Phe Asn Gly Leu Thr Gly Thr Gly Val Leu Thr
    210                 215                 220
```

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```
Glu Ser Asn Lys Lys Phe Leu Pro Phe Gln Gln Phe Gly Arg Asp Ile
225                 230                 235                 240

Ala Asp Thr Thr Asp Ala Val Arg Asp Pro Gln Thr Leu Glu
                245                 250


<210> SEQ ID NO 16
<211> LENGTH: 609
<212> TYPE: PRT
<213> ORGANISM: SARS coronavirus
<220> FEATURE:
<223> OTHER INFORMATION: SARS-CoV-2 Spike (S) protein

<400> SEQUENCE: 16

Cys Asp Ile Pro Ile Gly Ala Gly Ile Cys Ala Ser Tyr Gln Thr Gln
1               5                   10                  15

Thr Asn Ser Pro Arg Arg Ala Arg Ser Val Ala Ser Gln Ser Ile Ile
            20                  25                  30

Ala Tyr Thr Met Ser Leu Gly Ala Glu Asn Ser Val Ala Tyr Ser Asn
        35                  40                  45

Asn Ser Ile Ala Ile Pro Thr Asn Phe Thr Ile Ser Val Thr Thr Glu
    50                  55                  60

Ile Leu Pro Val Ser Met Thr Lys Thr Ser Val Asp Cys Thr Met Tyr
65                  70                  75                  80

Ile Cys Gly Asp Ser Thr Glu Cys Ser Asn Leu Leu Leu Gln Tyr Gly
                85                  90                  95

Ser Phe Cys Thr Gln Leu Asn Arg Ala Leu Thr Gly Ile Ala Val Glu
            100                 105                 110

Gln Asp Lys Asn Thr Gln Glu Val Phe Ala Gln Val Lys Gln Ile Tyr
        115                 120                 125

Lys Thr Pro Pro Ile Lys Asp Phe Gly Gly Phe Asn Phe Ser Gln Ile
    130                 135                 140

Leu Pro Asp Pro Ser Lys Pro Ser Lys Arg Ser Phe Ile Glu Asp Leu
145                 150                 155                 160

Leu Phe Asn Lys Val Thr Leu Ala Asp Ala Gly Phe Ile Lys Gln Tyr
                165                 170                 175

Gly Asp Cys Leu Gly Asp Ile Ala Ala Arg Asp Leu Ile Cys Ala Gln
            180                 185                 190

Lys Phe Asn Gly Leu Thr Val Leu Pro Pro Leu Leu Thr Asp Glu Met
        195                 200                 205

Ile Ala Gln Tyr Thr Ser Ala Leu Leu Ala Gly Thr Ile Thr Ser Gly
    210                 215                 220

Trp Thr Phe Gly Ala Gly Ala Ala Leu Gln Ile Pro Phe Ala Met Gln
225                 230                 235                 240

Met Ala Tyr Arg Phe Asn Gly Ile Gly Val Thr Gln Asn Val Leu Tyr
                245                 250                 255

Glu Asn Gln Lys Leu Ile Ala Asn Gln Phe Asn Ser Ala Ile Gly Lys
            260                 265                 270

Ile Gln Asp Ser Leu Ser Ser Thr Ala Ser Ala Leu Gly Lys Leu Gln
        275                 280                 285

Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn Thr Leu Val Lys Gln
    290                 295                 300

Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val Leu Asn Asp Ile Leu
305                 310                 315                 320
```

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```
Ser Arg Leu Asp Lys Val Glu Ala Glu Val Gln Ile Asp Arg Leu Ile
                325                 330                 335

Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val Thr Gln Gln Leu Ile
            340                 345                 350

Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn Leu Ala Ala Thr Lys Met
        355                 360                 365

Ser Glu Cys Val Leu Gly Gln Ser Lys Arg Val Asp Phe Cys Gly Lys
    370                 375                 380

Gly Tyr His Leu Met Ser Phe Pro Gln Ser Ala Pro His Gly Val Val
385                 390                 395                 400

Phe Leu His Val Thr Tyr Val Pro Ala Gln Glu Lys Asn Phe Thr Thr
                405                 410                 415

Ala Pro Ala Ile Cys His Asp Gly Lys Ala His Phe Pro Arg Glu Gly
            420                 425                 430

Val Phe Val Ser Asn Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe
        435                 440                 445

Tyr Glu Pro Gln Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn
    450                 455                 460

Cys Asp Val Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu
465                 470                 475                 480

Gln Pro Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys
                485                 490                 495

Asn His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn
            500                 505                 510

Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu Val
        515                 520                 525

Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu Gly Lys
    530                 535                 540

Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu Gly Phe Ile
545                 550                 555                 560

Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met Leu Cys Cys Met
                565                 570                 575

Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys Ser Cys Gly Ser Cys
            580                 585                 590

Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro Val Leu Lys Gly Val Lys
        595                 600                 605

Leu
```

<210> SEQ ID NO 17
<211> LENGTH: 1260
<212> TYPE: RNA
<213> ORGANISM: SARS coronavirus
<220> FEATURE:
<223> OTHER INFORMATION: mRNA encoding SARS-CoV-2 nucleocapsid (N) protein

<400> SEQUENCE: 17

```
augucugaua auggacccca aaaucagcga aaugcacccc gcauuacguu ugguggaccc      60

ucagauucaa cuggcaguaa ccagaaugga gaacgcagug gggcgcgauc aaaacaacgu     120

cggccccaag guuuacccaa uaauacugcg ucuugguuca ccgcucucac ucaacauggc     180

aaggaagacc uuaaauuccc ucgaggacaa ggcguuccaa uuaacaccaa uagcagucca     240

gaugaccaaa uuggcuacua ccgaagagcu accagacgaa uucguggugg ugacgguaaa     300
```

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```
augaaagauc ucaguccaag augguauuuc uacuaccuag gaacugggcc agaagcugga    360

cuucccuaug gugcuaacaa agacggcauc auaugggguug caacugaggg agccuugaau    420

acaccaaaag aucacauugg cacccgcaau ccugcuaaca augcugcaau cgugcuacaa    480

cuuccucaag gaacaacauu gccaaaaggc uucuacgcag aagggagcag aggcggcagu    540

caagccucuu cucguuccuc aucacguagu cgcaacaguu caagaaauuc aacuccaggc    600

agcaguaggg gaacuucucc ugcuagaaug gcuggcaaug gcggugaugc ugcucuugcu    660

uugcugcugc uugacagauu gaaccagcuu gagagcaaaa ugucugguaa aggccaacaa    720

caacaaggcc aaacugucac uaagaaaucu gcugcugagg cuucuaagaa gccucggcaa    780

aaacguacug ccacuaaagc auacaaugua acacaagcuu ucggcagacg ugguccagaa    840

caaacccaag gaaauuuugg ggaccaggaa cuaaucagac aaggaacuga uuacaaacau    900

uggccgcaaa uugcacaauu ugcccccagc gcuucagcgu ucuucggaau gucgcgcauu    960

ggcauggaag ucacaccuuc gggaacgugg uugaccuaca caggugccau caaauuggau   1020

gacaaagauc caaauuucaa agaucaaguc auuuugcuga auaagcauau ugacgcauac   1080

aaaacauucc caccaacaga gccuaaaaag gacaaaaaga agaaggcuga ugaaacucaa   1140

gccuuaccgc agagacagaa gaaacagcaa acugugacuc uucuuccugc ugcagauuug   1200

gaugauuucu ccaaacaauu gcaacaaucc augagcagug cugacucaac ucaggccuaa   1260
```

```
<210> SEQ ID NO 18
<211> LENGTH: 355
<212> TYPE: PRT
<213> ORGANISM: SARS coronavirus
<220> FEATURE:
<223> OTHER INFORMATION: SARS-CoV-2 nucleocapsid (N) protein

<400> SEQUENCE: 18

Arg Ile Thr Phe Gly Gly Pro Ser Asp Ser Thr Gly Ser Asn Gln Asn
1               5                   10                  15

Gly Glu Arg Ser Gly Ala Arg Ser Lys Gln Arg Arg Pro Gln Gly Leu
            20                  25                  30

Pro Asn Asn Thr Ala Ser Trp Phe Thr Ala Leu Thr Gln His Gly Lys
        35                  40                  45

Glu Asp Leu Lys Phe Pro Arg Gly Gln Gly Val Pro Ile Asn Thr Asn
    50                  55                  60

Ser Ser Pro Asp Asp Gln Ile Gly Tyr Tyr Arg Arg Ala Thr Arg Arg
65                  70                  75                  80

Ile Arg Gly Gly Asp Gly Lys Met Lys Asp Leu Ser Pro Arg Trp Tyr
                85                  90                  95

Phe Tyr Tyr Leu Gly Thr Gly Pro Glu Ala Gly Leu Pro Tyr Gly Ala
            100                 105                 110

Asn Lys Asp Gly Ile Ile Trp Val Ala Thr Glu Gly Ala Leu Asn Thr
        115                 120                 125

Pro Lys Asp His Ile Gly Thr Arg Asn Pro Ala Asn Asn Ala Ala Ile
    130                 135                 140

Val Leu Gln Leu Pro Gln Gly Thr Thr Leu Pro Lys Gly Phe Tyr Ala
145                 150                 155                 160

Glu Gly Ser Arg Gly Gly Ser Gln Ala Ser Ser Arg Ser Ser Ser Arg
                165                 170                 175
```

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```
Ser Arg Asn Ser Ser Arg Asn Ser Thr Pro Gly Ser Ser Arg Gly Thr
            180                 185                 190

Ser Pro Ala Arg Met Ala Gly Asn Gly Gly Asp Ala Ala Leu Ala Leu
        195                 200                 205

Leu Leu Leu Asp Arg Leu Asn Gln Leu Glu Ser Lys Met Ser Gly Lys
    210                 215                 220

Gly Gln Gln Gln Gln Gly Gln Thr Val Thr Lys Lys Ser Ala Ala Glu
225                 230                 235                 240

Ala Ser Lys Lys Pro Arg Gln Lys Arg Thr Ala Thr Lys Ala Tyr Asn
                245                 250                 255

Val Thr Gln Ala Phe Gly Arg Arg Gly Pro Glu Gln Thr Gln Gly Asn
            260                 265                 270

Phe Gly Asp Gln Glu Leu Ile Arg Gln Gly Thr Asp Tyr Lys His Trp
        275                 280                 285

Pro Gln Ile Ala Gln Phe Ala Pro Ser Ala Ser Ala Phe Phe Gly Met
    290                 295                 300

Ser Arg Ile Gly Met Glu Val Thr Pro Ser Gly Thr Trp Leu Thr Tyr
305                 310                 315                 320

Thr Gly Ala Ile Lys Leu Asp Asp Lys Asp Pro Asn Phe Lys Asp Gln
                325                 330                 335

Val Ile Leu Leu Asn Lys His Ile Asp Ala Tyr Lys Thr Phe Pro Pro
            340                 345                 350

Thr Glu Pro
        355
```

<210> SEQ ID NO 19
<211> LENGTH: 669
<212> TYPE: RNA
<213> ORGANISM: SARS coronavirus
<220> FEATURE:
<223> OTHER INFORMATION: mRNA encoding SARS-CoV-2 membrane (M) protein

<400> SEQUENCE: 19

```
auggcagauu ccaacgguac uauuaccguu gaagagcuua aaaagcuccu ugaacaaugg     60

aaccuaguaa uagguuuccu auuccuuaca uggauuuguc uucuacaauu ugccuaugcc    120

aacaggaaua gguuuuugua uauaauuaag uuaauuuucc ucuggcuguu auggccagua    180

acuuuagcuu guuuugugcu ugcugcuguu uacagaauaa auuggaucac cgguggaauu    240

gcuaucgcaa uggcuugucu uguaggcuug auguggcuca gcuacuucau ugcuucuuuc    300

agacuguuug cgcguacgcg uuccaugugg ucauucaauc cagaaacuaa cauucuucuc    360

aacgugccac uccauggcac uauucugacc agaccgcuuc uagaaaguga acucguaauc    420

ggagcuguga uccuucgugg acaucuucgu auugcuggac accaucuagg acgcugugac    480

aucaaggacc ugccuaaaga aaucacuguu gcuacaucac gaacgcuuuc uuauuacaaa    540

uugggagcuu cgcagcgugu agcaggugac ucagguuuug cugcauacag ucgcuacagg    600

auuggcaacu auaaauuaaa cacagaccau uccaguagca gugacaauau ugcuuugcuu    660

guacaguaa                                                            669
```

<210> SEQ ID NO 20
<211> LENGTH: 218
<212> TYPE: PRT
<213> ORGANISM: SARS coronavirus
<220> FEATURE:

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<223> OTHER INFORMATION: SARS-CoV-2 membrane (M) protein

<400> SEQUENCE: 20

```
Ser Asn Gly Thr Ile Thr Val Glu Glu Leu Lys Lys Leu Leu Glu Gln
1               5                   10                  15

Trp Asn Leu Val Ile Gly Phe Leu Phe Leu Thr Trp Ile Cys Leu Leu
            20                  25                  30

Gln Phe Ala Tyr Ala Asn Arg Asn Arg Phe Leu Tyr Ile Ile Lys Leu
        35                  40                  45

Ile Phe Leu Trp Leu Leu Trp Pro Val Thr Leu Ala Cys Phe Val Leu
    50                  55                  60

Ala Ala Val Tyr Arg Ile Asn Trp Ile Thr Gly Gly Ile Ala Ile Ala
65                  70                  75                  80

Met Ala Cys Leu Val Gly Leu Met Trp Leu Ser Tyr Phe Ile Ala Ser
                85                  90                  95

Phe Arg Leu Phe Ala Arg Thr Arg Ser Met Trp Ser Phe Asn Pro Glu
            100                 105                 110

Thr Asn Ile Leu Leu Asn Val Pro Leu His Gly Thr Ile Leu Thr Arg
        115                 120                 125

Pro Leu Leu Glu Ser Glu Leu Val Ile Gly Ala Val Ile Leu Arg Gly
    130                 135                 140

His Leu Arg Ile Ala Gly His His Leu Gly Arg Cys Asp Ile Lys Asp
145                 150                 155                 160

Leu Pro Lys Glu Ile Thr Val Ala Thr Ser Arg Thr Leu Ser Tyr Tyr
                165                 170                 175

Lys Leu Gly Ala Ser Gln Arg Val Ala Gly Asp Ser Gly Phe Ala Ala
            180                 185                 190

Tyr Ser Arg Tyr Arg Ile Gly Asn Tyr Lys Leu Asn Thr Asp His Ser
        195                 200                 205

Ser Ser Ser Asp Asn Ile Ala Leu Leu Val
    210                 215
```

<210> SEQ ID NO 21
<211> LENGTH: 195
<212> TYPE: PRT
<213> ORGANISM: Influenza A virus
<220> FEATURE:
<223> OTHER INFORMATION: Hemagglutinin (HA) protein of influenza A virus
H5N1

<400> SEQUENCE: 21

```
Asn Asp Gln Gly Ser Gly Tyr Ala Ala Asp Lys Glu Ser Thr Gln Lys
1               5                   10                  15

Ala Phe Asn Gly Ile Thr Asn Lys Val Asn Ser Val Ile Glu Lys Met
            20                  25                  30

Asn Thr Gln Phe Glu Ala Val Gly Lys Glu Phe Ser Asn Leu Glu Lys
        35                  40                  45

Arg Leu Glu Asn Leu Asn Lys Lys Met Glu Asp Gly Phe Leu Asp Val
    50                  55                  60

Trp Thr Tyr Asn Ala Glu Leu Leu Val Leu Met Glu Asn Glu Arg Thr
65                  70                  75                  80

Leu Asp Phe His Asp Ser Asn Val Lys Asn Leu Tyr Asp Lys Val Arg
                85                  90                  95

Met Gln Leu Arg Asp Asn Val Lys Glu Leu Gly Asn Gly Cys Phe Glu
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            100                 105                 110

Phe Tyr His Lys Cys Asp Asn Glu Cys Met Asp Ser Val Lys Asn Gly
        115                 120                 125

Thr Tyr Asp Tyr Pro Lys Tyr Glu Glu Glu Ser Lys Leu Asn Arg Asn
    130                 135                 140

Glu Ile Lys Gly Val Lys Leu Ser Ser Met Gly Val Tyr Gln Ile Leu
145                 150                 155                 160

Ala Ile Tyr Ala Thr Val Ala Gly Ser Leu Ser Leu Ala Ile Met Met
                165                 170                 175

Ala Gly Ile Ser Phe Trp Met Cys Ser Asn Gly Ser Leu Gln Cys Arg
            180                 185                 190

Ile Cys Ile
        195


<210> SEQ ID NO 22
<211> LENGTH: 538
<212> TYPE: PRT
<213> ORGANISM: Influenza A virus
<220> FEATURE:
<223> OTHER INFORMATION: Hemagglutinin (HA) protein of influenza A virus
      H3N2

<400> SEQUENCE: 22

Leu Cys Leu Gly His His Ala Val Pro Asn Gly Thr Ile Val Lys Thr
1               5                   10                  15

Ile Thr Asn Asp Gln Ile Glu Val Thr Asn Ala Thr Glu Leu Val Gln
            20                  25                  30

Ser Ser Ser Thr Gly Gly Ile Cys Asp Ser Pro His Gln Ile Leu Asp
        35                  40                  45

Gly Glu Asn Cys Thr Leu Ile Asp Ala Leu Leu Gly Asp Pro Gln Cys
    50                  55                  60

Asp Gly Phe Gln Asn Lys Lys Trp Asp Leu Phe Val Glu Arg Ser Lys
65                  70                  75                  80

Ala Tyr Ser Asn Cys Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Ser Leu
                85                  90                  95

Arg Ser Leu Val Ala Ser Ser Gly Thr Leu Glu Phe Asn Asn Glu Ser
            100                 105                 110

Phe Asn Trp Thr Gly Val Thr Gln Asn Gly Thr Ser Ser Ala Cys Lys
        115                 120                 125

Arg Arg Ser Asn Asn Ser Phe Phe Ser Arg Leu Asn Trp Leu Thr His
    130                 135                 140

Leu Lys Phe Lys Tyr Pro Ala Leu Asn Val Thr Met Pro Asn Asn Glu
145                 150                 155                 160

Lys Phe Asp Lys Leu Tyr Ile Trp Gly Val His His Pro Gly Thr Asp
                165                 170                 175

Asn Asp Gln Ile Ser Leu Tyr Ala Gln Ala Ser Gly Arg Ile Thr Val
            180                 185                 190

Ser Thr Lys Arg Ser Gln Gln Thr Val Ile Pro Ser Ile Gly Ser Arg
        195                 200                 205

Pro Arg Ile Arg Asp Val Pro Ser Arg Ile Ser Ile Tyr Trp Thr Ile
    210                 215                 220

Val Lys Pro Gly Asp Ile Leu Leu Ile Asn Ser Thr Gly Asn Leu Ile
225                 230                 235                 240
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Ala Pro Arg Gly Tyr Phe Lys Ile Arg Ser Gly Lys Ser Ser Ile Met
                245                 250                 255

Arg Ser Asp Ala Pro Ile Gly Lys Cys Asn Ser Glu Cys Ile Thr Pro
            260                 265                 270

Asn Gly Ser Ile Pro Asn Asp Lys Pro Phe Gln Asn Val Asn Arg Ile
        275                 280                 285

Thr Tyr Gly Ala Cys Pro Arg Tyr Val Lys Gln Asn Thr Leu Lys Leu
    290                 295                 300

Ala Thr Gly Met Arg Asn Val Pro Glu Lys Gln Thr Arg Gly Ile Phe
305                 310                 315                 320

Gly Ala Ile Ala Gly Phe Ile Glu Asn Gly Trp Glu Gly Met Val Asp
                325                 330                 335

Gly Trp Tyr Gly Phe Arg His Gln Asn Ser Glu Gly Thr Gly Gln Ala
            340                 345                 350

Ala Asp Leu Lys Ser Thr Gln Ala Ala Ile Asn Gln Ile Asn Gly Lys
        355                 360                 365

Leu Asn Arg Leu Ile Gly Lys Thr Asn Glu Lys Phe His Gln Ile Glu
    370                 375                 380

Lys Glu Phe Ser Glu Val Glu Gly Arg Ile Gln Asp Leu Glu Lys Tyr
385                 390                 395                 400

Val Glu Asp Thr Lys Ile Asp Leu Trp Ser Tyr Asn Ala Glu Leu Leu
                405                 410                 415

Val Ala Leu Glu Asn Gln His Thr Ile Asp Leu Thr Asp Ser Glu Met
            420                 425                 430

Asn Lys Leu Phe Glu Arg Thr Lys Lys Gln Leu Arg Glu Asn Ala Glu
        435                 440                 445

Asp Met Gly Asn Gly Cys Phe Lys Ile Tyr His Lys Cys Asp Asn Ala
    450                 455                 460

Cys Ile Gly Ser Ile Arg Asn Gly Thr Tyr Asp His Asp Val Tyr Arg
465                 470                 475                 480

Asp Glu Ala Leu Asn Asn Arg Phe Gln Ile Lys Gly Val Glu Leu Lys
                485                 490                 495

Ser Gly Tyr Lys Asp Trp Ile Leu Trp Ile Ser Phe Ala Ile Ser Cys
            500                 505                 510

Phe Leu Leu Cys Val Ala Leu Leu Gly Phe Ile Met Trp Ala Cys Gln
        515                 520                 525

Lys Gly Asn Ile Arg Cys Asn Ile Cys Ile
    530                 535
```

<210> SEQ ID NO 23
<211> LENGTH: 538
<212> TYPE: PRT
<213> ORGANISM: Influenza A virus
<220> FEATURE:
<223> OTHER INFORMATION: Hemagglutinin (HA) protein of influenza A virus H1N1

<400> SEQUENCE: 23

```
Leu Cys Leu Gly His His Ala Val Pro Asn Gly Thr Ile Val Lys Thr
1               5                   10                  15

Ile Thr Asn Asp Gln Ile Glu Val Thr Asn Ala Thr Glu Leu Val Gln
            20                  25                  30

Ser Ser Ser Thr Gly Gly Ile Cys Asp Ser Pro His Gln Ile Leu Asp
        35                  40                  45
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Gly Glu Asn Cys Thr Leu Ile Asp Ala Leu Leu Gly Asp Pro Gln Cys
    50                  55                  60

Asp Gly Phe Gln Asn Lys Lys Trp Asp Leu Phe Val Glu Arg Ser Lys
65                  70                  75                  80

Ala Tyr Ser Asn Cys Tyr Pro Tyr Asp Val Pro Asp Tyr Ala Ser Leu
                85                  90                  95

Arg Ser Leu Val Ala Ser Ser Gly Thr Leu Glu Phe Asn Asn Glu Ser
            100                 105                 110

Phe Asn Trp Thr Gly Val Thr Gln Asn Gly Thr Ser Ser Ala Cys Lys
        115                 120                 125

Arg Arg Ser Asn Asn Ser Phe Phe Ser Arg Leu Asn Trp Leu Thr His
    130                 135                 140

Leu Lys Phe Lys Tyr Pro Ala Leu Asn Val Thr Met Pro Asn Asn Glu
145                 150                 155                 160

Lys Phe Asp Lys Leu Tyr Ile Trp Gly Val His His Pro Gly Thr Asp
                165                 170                 175

Asn Asp Gln Ile Ser Leu Tyr Ala Gln Ala Ser Gly Arg Ile Thr Val
            180                 185                 190

Ser Thr Lys Arg Ser Gln Gln Thr Val Ile Pro Ser Ile Gly Ser Arg
        195                 200                 205

Pro Arg Ile Arg Asp Val Pro Ser Arg Ile Ser Ile Tyr Trp Thr Ile
    210                 215                 220

Val Lys Pro Gly Asp Ile Leu Leu Ile Asn Ser Thr Gly Asn Leu Ile
225                 230                 235                 240

Ala Pro Arg Gly Tyr Phe Lys Ile Arg Ser Gly Lys Ser Ser Ile Met
                245                 250                 255

Arg Ser Asp Ala Pro Ile Gly Lys Cys Asn Ser Glu Cys Ile Thr Pro
            260                 265                 270

Asn Gly Ser Ile Pro Asn Asp Lys Pro Phe Gln Asn Val Asn Arg Ile
        275                 280                 285

Thr Tyr Gly Ala Cys Pro Arg Tyr Val Lys Gln Asn Thr Leu Lys Leu
    290                 295                 300

Ala Thr Gly Met Arg Asn Val Pro Glu Lys Gln Thr Arg Gly Ile Phe
305                 310                 315                 320

Gly Ala Ile Ala Gly Phe Ile Glu Asn Gly Trp Glu Gly Met Val Asp
                325                 330                 335

Gly Trp Tyr Gly Phe Arg His Gln Asn Ser Glu Gly Thr Gly Gln Ala
            340                 345                 350

Ala Asp Leu Lys Ser Thr Gln Ala Ala Ile Asn Gln Ile Asn Gly Lys
        355                 360                 365

Leu Asn Arg Leu Ile Gly Lys Thr Asn Glu Lys Phe His Gln Ile Glu
    370                 375                 380

Lys Glu Phe Ser Glu Val Glu Gly Arg Ile Gln Asp Leu Glu Lys Tyr
385                 390                 395                 400

Val Glu Asp Thr Lys Ile Asp Leu Trp Ser Tyr Asn Ala Glu Leu Leu
                405                 410                 415

Val Ala Leu Glu Asn Gln His Thr Ile Asp Leu Thr Asp Ser Glu Met
            420                 425                 430

Asn Lys Leu Phe Glu Arg Thr Lys Lys Gln Leu Arg Glu Asn Ala Glu
        435                 440                 445
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Asp Met Gly Asn Gly Cys Phe Lys Ile Tyr His Lys Cys Asp Asn Ala
    450                 455                 460

Cys Ile Gly Ser Ile Arg Asn Gly Thr Tyr Asp His Asp Val Tyr Arg
465                 470                 475                 480

Asp Glu Ala Leu Asn Asn Arg Phe Gln Ile Lys Gly Val Glu Leu Lys
                485                 490                 495

Ser Gly Tyr Lys Asp Trp Ile Leu Trp Ile Ser Phe Ala Ile Ser Cys
            500                 505                 510

Phe Leu Leu Cys Val Ala Leu Leu Gly Phe Ile Met Trp Ala Cys Gln
        515                 520                 525

Lys Gly Asn Ile Arg Cys Asn Ile Cys Ile
    530                 535


<210> SEQ ID NO 24
<211> LENGTH: 540
<212> TYPE: PRT
<213> ORGANISM: Influenza A virus
<220> FEATURE:
<223> OTHER INFORMATION: Hemagglutinin (HA) protein of influenza A virus
      H7N9

<400> SEQUENCE: 24

Ile Cys Leu Gly His His Ala Val Ser Asn Gly Thr Lys Val Asn Thr
1               5                   10                  15

Leu Thr Glu Arg Gly Val Glu Val Val Asn Ala Thr Glu Thr Val Glu
            20                  25                  30

Arg Thr Asn Ile Pro Arg Ile Cys Ser Lys Gly Lys Arg Thr Val Asp
        35                  40                  45

Leu Gly Gln Cys Gly Leu Leu Gly Thr Ile Thr Gly Pro Pro Gln Cys
    50                  55                  60

Asp Gln Phe Leu Glu Phe Ser Ala Asp Leu Ile Ile Glu Arg Arg Glu
65                  70                  75                  80

Gly Ser Asp Val Cys Tyr Pro Gly Lys Phe Val Asn Glu Glu Ala Leu
                85                  90                  95

Arg Gln Ile Leu Arg Glu Ser Gly Gly Ile Asp Lys Glu Ala Met Gly
            100                 105                 110

Phe Thr Tyr Ser Gly Ile Arg Thr Asn Gly Ala Thr Ser Ala Cys Arg
        115                 120                 125

Arg Ser Gly Ser Ser Phe Tyr Ala Glu Met Lys Trp Leu Leu Ser Asn
    130                 135                 140

Thr Asp Asn Ala Ala Phe Pro Gln Met Thr Lys Ser Tyr Lys Asn Thr
145                 150                 155                 160

Arg Lys Ser Pro Ala Leu Ile Val Trp Gly Ile His His Ser Val Ser
                165                 170                 175

Thr Ala Glu Gln Thr Lys Leu Tyr Gly Ser Gly Asn Lys Leu Val Thr
            180                 185                 190

Val Gly Ser Ser Asn Tyr Gln Gln Ser Phe Val Pro Ser Pro Gly Ala
        195                 200                 205

Arg Pro Gln Val Asn Gly Leu Ser Gly Arg Ile Asp Phe His Trp Leu
    210                 215                 220

Met Leu Asn Pro Asn Asp Thr Val Thr Phe Ser Phe Asn Gly Ala Phe
225                 230                 235                 240

Ile Ala Pro Asp Arg Ala Ser Phe Leu Arg Gly Lys Ser Met Gly Ile
                245                 250                 255
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Gln Ser Gly Val Gln Val Asp Ala Asn Cys Glu Gly Asp Cys Tyr His
            260                 265                 270

Ser Gly Gly Thr Ile Ile Ser Asn Leu Pro Phe Gln Asn Ile Asp Ser
        275                 280                 285

Arg Ala Val Gly Lys Cys Pro Arg Tyr Val Lys Gln Arg Ser Leu Leu
    290                 295                 300

Leu Ala Thr Gly Met Lys Asn Val Pro Glu Ile Pro Lys Gly Arg Gly
305                 310                 315                 320

Leu Phe Gly Ala Ile Ala Gly Phe Ile Glu Asn Gly Trp Glu Gly Leu
                325                 330                 335

Ile Asp Gly Trp Tyr Gly Phe Arg His Gln Asn Ala Gln Gly Glu Gly
            340                 345                 350

Thr Ala Ala Asp Tyr Lys Ser Thr Gln Ser Ala Ile Asp Gln Ile Thr
        355                 360                 365

Gly Lys Leu Asn Arg Leu Ile Glu Lys Thr Asn Gln Gln Phe Glu Leu
    370                 375                 380

Ile Asp Asn Glu Phe Asn Glu Val Glu Lys Gln Ile Gly Asn Val Ile
385                 390                 395                 400

Asn Trp Thr Arg Asp Ser Ile Thr Glu Val Trp Ser Tyr Asn Ala Glu
                405                 410                 415

Leu Leu Val Ala Met Glu Asn Gln His Thr Ile Asp Leu Ala Asp Ser
            420                 425                 430

Glu Met Asp Lys Leu Tyr Glu Arg Val Lys Arg Gln Leu Arg Glu Asn
        435                 440                 445

Ala Glu Glu Asp Gly Thr Gly Cys Phe Glu Ile Phe His Lys Cys Asp
    450                 455                 460

Asp Asp Cys Met Ala Ser Ile Arg Asn Asn Thr Tyr Asp His Ser Lys
465                 470                 475                 480

Tyr Arg Glu Glu Ala Met Gln Asn Arg Ile Gln Ile Asp Pro Val Lys
                485                 490                 495

Leu Ser Ser Gly Tyr Lys Asp Val Ile Leu Trp Phe Ser Phe Gly Ala
            500                 505                 510

Ser Cys Phe Ile Leu Leu Ala Ile Val Met Gly Leu Val Phe Ile Cys
        515                 520                 525

Val Lys Asn Gly Asn Met Arg Cys Thr Ile Cys Ile
    530                 535                 540


<210> SEQ ID NO 25
<211> LENGTH: 540
<212> TYPE: PRT
<213> ORGANISM: Influenza A virus
<220> FEATURE:
<223> OTHER INFORMATION: Hemagglutinin (HA) protein of influenza A virus
      H1N1

<400> SEQUENCE: 25

Ile Cys Ile Gly His Gln Ser Thr Asn Ser Thr Glu Thr Val Asp Thr
1               5                   10                  15

Leu Thr Glu Thr Asn Val Pro Val Thr His Ala Lys Glu Leu Leu His
            20                  25                  30

Thr Glu His Asn Gly Met Leu Cys Ala Thr Ser Leu Gly His Pro Leu
        35                  40                  45

Ile Leu Asp Thr Cys Thr Ile Glu Gly Leu Val Tyr Gly Asn Pro Ser
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    50                  55                  60

Cys Asp Leu Leu Leu Gly Gly Arg Glu Trp Ser Tyr Ile Val Glu Arg
65                  70                  75                  80

Ser Ser Ala Val Asn Gly Thr Cys Tyr Pro Gly Asn Val Glu Asn Leu
                85                  90                  95

Glu Glu Leu Arg Thr Leu Phe Ser Ser Ala Ser Ser Tyr Gln Arg Ile
            100                 105                 110

Gln Ile Phe Pro Asp Thr Thr Trp Asn Val Thr Tyr Thr Gly Thr Ser
        115                 120                 125

Arg Ala Cys Ser Gly Ser Phe Tyr Arg Ser Met Arg Trp Leu Thr Gln
    130                 135                 140

Lys Ser Gly Phe Tyr Pro Val Gln Asp Ala Gln Tyr Thr Asn Asn Arg
145                 150                 155                 160

Gly Lys Ser Ile Leu Phe Val Trp Gly Ile His His Pro Pro Thr Tyr
                165                 170                 175

Thr Glu Gln Thr Asn Leu Tyr Ile Arg Asn Asp Thr Thr Thr Ser Val
            180                 185                 190

Thr Thr Glu Asp Leu Asn Arg Thr Phe Lys Pro Val Ile Gly Pro Arg
        195                 200                 205

Pro Leu Val Asn Gly Leu Gln Gly Arg Ile Asp Tyr Tyr Trp Ser Val
    210                 215                 220

Leu Lys Pro Gly Gln Thr Leu Arg Val Arg Ser Asn Gly Asn Leu Ile
225                 230                 235                 240

Ala Pro Trp Tyr Gly His Val Leu Ser Gly Gly Ser His Gly Arg Ile
                245                 250                 255

Leu Lys Thr Asp Leu Lys Gly Gly Asn Cys Val Val Gln Cys Gln Thr
            260                 265                 270

Glu Lys Gly Gly Leu Asn Ser Thr Leu Pro Phe His Asn Ile Ser Lys
        275                 280                 285

Tyr Ala Phe Gly Thr Cys Pro Lys Tyr Val Arg Val Asn Ser Leu Lys
    290                 295                 300

Leu Ala Val Gly Leu Arg Asn Val Pro Ala Arg Ser Ser Arg Gly Leu
305                 310                 315                 320

Phe Gly Ala Ile Ala Gly Phe Ile Glu Gly Gly Trp Pro Gly Leu Val
                325                 330                 335

Ala Gly Trp Tyr Gly Phe Gln His Ser Asn Asp Gln Gly Val Gly Met
            340                 345                 350

Ala Ala Asp Arg Asp Ser Thr Gln Lys Ala Ile Asp Lys Ile Thr Ser
        355                 360                 365

Lys Val Asn Asn Ile Val Asp Lys Met Asn Lys Gln Tyr Glu Ile Ile
    370                 375                 380

Asp His Glu Phe Ser Glu Val Glu Thr Arg Leu Asn Met Ile Asn Asn
385                 390                 395                 400

Lys Ile Asp Asp Gln Ile Gln Asp Val Trp Ala Tyr Asn Ala Glu Leu
                405                 410                 415

Leu Val Leu Leu Glu Asn Gln Lys Thr Leu Asp Glu His Asp Ala Asn
            420                 425                 430

Val Asn Asn Leu Tyr Asn Lys Val Lys Arg Ala Leu Gly Ser Asn Ala
        435                 440                 445

Met Glu Asp Gly Lys Gly Cys Phe Glu Leu Tyr His Lys Cys Asp Asp
    450                 455                 460
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Gln Cys Met Glu Thr Ile Arg Asn Gly Thr Tyr Asn Arg Arg Lys Tyr
465                 470                 475                 480

Arg Glu Glu Ser Arg Leu Glu Arg Gln Lys Ile Glu Gly Val Lys Leu
                485                 490                 495

Glu Ser Glu Gly Thr Tyr Lys Ile Leu Thr Ile Tyr Ser Thr Val Ala
            500                 505                 510

Ser Ser Leu Val Leu Ala Met Gly Phe Ala Ala Phe Leu Phe Trp Ala
        515                 520                 525

Met Ser Asn Gly Ser Cys Arg Cys Asn Ile Cys Ile
    530                 535                 540


<210> SEQ ID NO 26
<211> LENGTH: 545
<212> TYPE: PRT
<213> ORGANISM: Influenza A virus
<220> FEATURE:
<223> OTHER INFORMATION: Hemagglutinin (HA) protein of influenza A virus
      H2N2

<400> SEQUENCE: 26

Ile Cys Ile Gly Tyr His Ala Asn Asn Ser Thr Glu Lys Val Asp Thr
1               5                   10                  15

Ile Leu Glu Arg Asn Val Thr Val Thr His Ala Lys Asp Ile Leu Glu
            20                  25                  30

Lys Thr His Asn Gly Lys Leu Cys Lys Leu Asn Gly Ile Pro Pro Leu
        35                  40                  45

Glu Leu Gly Asp Cys Ser Ile Ala Gly Trp Leu Leu Gly Asn Pro Glu
    50                  55                  60

Cys Asp Arg Leu Leu Ser Val Pro Glu Trp Ser Tyr Ile Met Glu Lys
65                  70                  75                  80

Glu Asn Pro Arg Tyr Ser Leu Cys Tyr Pro Gly Ser Phe Asn Asp Tyr
                85                  90                  95

Glu Glu Leu Lys His Leu Leu Ser Ser Val Lys His Phe Glu Lys Val
            100                 105                 110

Lys Ile Leu Pro Lys Asp Arg Trp Thr Gln His Thr Thr Thr Gly Gly
        115                 120                 125

Ser Trp Ala Cys Ala Val Ser Gly Lys Pro Ser Phe Phe Arg Asn Met
    130                 135                 140

Val Trp Leu Thr Arg Lys Gly Ser Asn Tyr Pro Val Ala Lys Gly Ser
145                 150                 155                 160

Tyr Asn Asn Thr Ser Gly Glu Gln Met Leu Ile Ile Trp Gly Val His
                165                 170                 175

His Pro Asn Asp Glu Ala Glu Gln Arg Ala Leu Tyr Gln Asn Val Gly
            180                 185                 190

Thr Tyr Val Ser Val Ala Thr Ser Thr Leu Tyr Lys Arg Ser Ile Pro
        195                 200                 205

Glu Ile Ala Ala Arg Pro Lys Val Asn Gly Leu Gly Arg Arg Met Glu
    210                 215                 220

Phe Ser Trp Thr Leu Leu Asp Met Trp Asp Thr Ile Asn Phe Glu Ser
225                 230                 235                 240

Thr Gly Asn Leu Val Ala Pro Glu Tyr Gly Phe Lys Ile Ser Lys Arg
                245                 250                 255

Gly Ser Ser Gly Ile Met Lys Thr Glu Gly Thr Leu Glu Asn Cys Glu
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            260                 265                 270

Thr Lys Cys Gln Thr Pro Leu Gly Ala Ile Asn Thr Thr Leu Pro Phe
        275                 280                 285

His Asn Val His Pro Leu Thr Ile Gly Glu Cys Pro Lys Tyr Val Lys
    290                 295                 300

Ser Glu Lys Leu Val Leu Ala Thr Gly Leu Arg Asn Val Pro Gln Ile
305                 310                 315                 320

Glu Ser Arg Gly Leu Phe Gly Ala Ile Ala Gly Phe Ile Glu Gly Gly
                325                 330                 335

Trp Gln Gly Met Val Asp Gly Trp Tyr Gly Tyr His His Ser Asn Asp
            340                 345                 350

Gln Gly Ser Gly Tyr Ala Ala Asp Lys Glu Ser Thr Gln Lys Ala Phe
        355                 360                 365

Asn Gly Ile Thr Asn Lys Val Asn Ser Val Ile Glu Lys Met Asn Thr
    370                 375                 380

Gln Phe Glu Ala Val Gly Lys Glu Phe Ser Asn Leu Glu Lys Arg Leu
385                 390                 395                 400

Glu Asn Leu Asn Lys Lys Met Glu Asp Gly Phe Leu Asp Val Trp Thr
                405                 410                 415

Tyr Asn Ala Glu Leu Leu Val Leu Met Glu Asn Glu Arg Thr Leu Asp
            420                 425                 430

Phe His Asp Ser Asn Val Lys Asn Leu Tyr Asp Lys Val Arg Met Gln
        435                 440                 445

Leu Arg Asp Asn Val Lys Glu Leu Gly Asn Gly Cys Phe Glu Phe Tyr
    450                 455                 460

His Lys Cys Asp Asn Glu Cys Met Asp Ser Val Lys Asn Gly Thr Tyr
465                 470                 475                 480

Asp Tyr Pro Lys Tyr Glu Glu Glu Ser Lys Leu Asn Arg Asn Glu Ile
                485                 490                 495

Lys Gly Val Lys Leu Ser Ser Met Gly Val Tyr Gln Ile Leu Ala Ile
            500                 505                 510

Tyr Ala Thr Val Ala Gly Ser Leu Ser Leu Ala Ile Met Met Ala Gly
        515                 520                 525

Ile Ser Phe Trp Met Cys Ser Asn Gly Ser Leu Gln Cys Arg Ile Cys
    530                 535                 540

Ile
545


&lt;210&gt; SEQ ID NO 27
&lt;211&gt; LENGTH: 567
&lt;212&gt; TYPE: PRT
&lt;213&gt; ORGANISM: Influenza B virus
&lt;220&gt; FEATURE:
&lt;223&gt; OTHER INFORMATION: Hemagglutinin (HA) protein

&lt;400&gt; SEQUENCE: 27

Ile Cys Thr Gly Ile Thr Ser Ser Asn Ser Pro His Val Val Lys Thr
1               5                   10                  15

Ala Thr Gln Gly Glu Val Asn Val Thr Gly Val Ile Pro Leu Thr Thr
            20                  25                  30

Thr Pro Thr Lys Ser His Phe Ala Asn Leu Lys Gly Thr Gln Thr Arg
        35                  40                  45

Gly Lys Leu Cys Pro Asn Cys Phe Asn Cys Thr Asp Leu Asp Val Ala
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    50                  55                  60

Leu Gly Arg Pro Lys Cys Met Gly Asn Thr Pro Ser Ala Lys Val Ser
65                  70                  75                  80

Ile Leu His Glu Val Lys Pro Ala Thr Ser Gly Cys Phe Pro Ile Met
                85                  90                  95

His Asp Arg Thr Lys Ile Arg Gln Leu Pro Asn Leu Leu Arg Gly Tyr
            100                 105                 110

Glu Asn Ile Arg Leu Ser Thr Ser Asn Val Ile Asn Thr Glu Thr Ala
        115                 120                 125

Pro Gly Gly Pro Tyr Lys Val Gly Thr Ser Gly Ser Cys Pro Asn Val
    130                 135                 140

Ala Asn Gly Asn Gly Phe Phe Asn Thr Met Ala Trp Val Ile Pro Lys
145                 150                 155                 160

Asp Asn Asn Lys Thr Ala Ile Asn Pro Val Thr Val Glu Val Pro Tyr
                165                 170                 175

Ile Cys Ser Glu Gly Glu Asp Gln Ile Thr Val Trp Gly Phe His Ser
            180                 185                 190

Asp Asp Lys Thr Gln Met Glu Arg Leu Tyr Gly Asp Ser Asn Pro Gln
        195                 200                 205

Lys Phe Thr Ser Ser Ala Asn Gly Val Thr Thr His Tyr Val Ser Gln
    210                 215                 220

Ile Gly Gly Phe Pro Asn Gln Thr Glu Asp Glu Gly Leu Lys Gln Ser
225                 230                 235                 240

Gly Arg Ile Val Val Asp Tyr Met Val Gln Lys Pro Gly Lys Thr Gly
                245                 250                 255

Thr Ile Val Tyr Gln Arg Gly Ile Leu Leu Pro Gln Lys Val Trp Cys
            260                 265                 270

Ala Ser Gly Arg Ser Lys Val Ile Lys Gly Ser Leu Pro Leu Ile Gly
        275                 280                 285

Glu Ala Asp Cys Leu His Glu Lys Tyr Gly Gly Leu Asn Lys Ser Lys
    290                 295                 300

Pro Tyr Tyr Thr Gly Glu His Ala Lys Ala Ile Gly Asn Cys Pro Ile
305                 310                 315                 320

Trp Val Lys Thr Pro Leu Lys Leu Ala Asn Gly Thr Lys Tyr Arg Pro
                325                 330                 335

Pro Ala Lys Leu Leu Lys Glu Arg Gly Phe Phe Gly Ala Ile Ala Gly
            340                 345                 350

Phe Leu Glu Gly Gly Trp Glu Gly Met Ile Ala Gly Trp His Gly Tyr
        355                 360                 365

Thr Ser His Gly Ala His Gly Val Ala Val Ala Ala Asp Leu Lys Ser
    370                 375                 380

Thr Gln Glu Ala Ile Asn Lys Ile Thr Lys Asn Leu Asn Tyr Leu Ser
385                 390                 395                 400

Glu Leu Glu Val Lys Asn Leu Gln Arg Leu Ser Gly Ala Met Asn Glu
                405                 410                 415

Leu His Asp Glu Ile Leu Glu Leu Asp Glu Lys Val Asp Asp Leu Arg
            420                 425                 430

Ala Asp Thr Ile Ser Ser Gln Ile Glu Leu Ala Val Leu Leu Ser Asn
        435                 440                 445

Glu Gly Ile Ile Asn Ser Glu Asp Glu His Leu Leu Ala Leu Glu Arg
    450                 455                 460
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Lys Leu Lys Lys Met Leu Gly Pro Ser Ala Val Glu Ile Gly Asn Gly
465 470 475 480

Cys Phe Glu Thr Lys His Lys Cys Asn Gln Thr Cys Leu Asp Arg Ile
485 490 495

Ala Ala Gly Thr Phe Asn Ala Gly Asp Phe Ser Leu Pro Thr Phe Asp
500 505 510

Ser Leu Asn Ile Thr Ala Ala Ser Leu Asn Asp Asp Gly Leu Asp Asn
515 520 525

His Thr Ile Leu Leu Tyr Tyr Ser Thr Ala Ala Ser Ser Leu Ala Val
530 535 540

Thr Leu Met Ile Ala Ile Phe Ile Val Tyr Met Val Ser Arg Asp Asn
545 550 555 560

Val Ser Cys Ser Ile Cys Leu
565

<210> SEQ ID NO 28
<211> LENGTH: 381
<212> TYPE: PRT
<213> ORGANISM: Influenza A virus
<220> FEATURE:
<223> OTHER INFORMATION: Neuraminidase (NA) protein of influenza A virus H5N1

<400> SEQUENCE: 28

Leu Ala Gly Asn Ser Ser Leu Cys Pro Ile Ser Gly Trp Ala Val His
1 5 10 15

Ser Lys Asp Asn Gly Ile Arg Ile Gly Ser Lys Gly Asp Val Phe Val
20 25 30

Ile Arg Glu Pro Phe Ile Ser Cys Ser His Leu Glu Cys Arg Thr Phe
35 40 45

Phe Leu Thr Gln Gly Ala Leu Leu Asn Asp Lys His Ser Asn Gly Thr
50 55 60

Val Lys Asp Arg Ser Pro His Arg Thr Leu Met Ser Cys Pro Val Gly
65 70 75 80

Glu Ala Pro Ser Pro Tyr Asn Ser Arg Phe Glu Ser Val Ala Trp Ser
85 90 95

Ala Ser Ala Cys His Asp Gly Thr Ser Trp Leu Thr Ile Gly Ile Ser
100 105 110

Gly Pro Asp Asn Gly Ala Val Ala Val Leu Lys Tyr Asn Gly Ile Ile
115 120 125

Thr Asp Thr Ile Lys Ser Trp Arg Asn Asn Ile Leu Arg Thr Gln Glu
130 135 140

Ser Glu Cys Ala Cys Val Asn Gly Ser Cys Phe Thr Val Met Thr Asp
145 150 155 160

Gly Pro Ser Asn Gly Gln Ala Ser Tyr Lys Ile Phe Lys Met Glu Lys
165 170 175

Gly Lys Val Val Lys Ser Val Glu Leu Asn Ala Pro Asn Tyr His Tyr
180 185 190

Glu Glu Cys Ser Cys Tyr Pro Asp Ala Gly Glu Ile Thr Cys Val Cys
195 200 205

Arg Asp Asn Trp His Gly Ser Asn Arg Pro Trp Val Ser Phe Asn Gln
210 215 220

Asn Leu Glu Tyr Gln Ile Gly Tyr Ile Cys Ser Gly Val Phe Gly Asp

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225                 230                 235                 240

Asn Pro Arg Pro Asn Asp Gly Thr Gly Ser Cys Gly Pro Val Ser Pro
                245                 250                 255

Asn Gly Ala Tyr Gly Val Lys Gly Phe Ser Phe Lys Tyr Gly Asn Gly
            260                 265                 270

Val Trp Ile Gly Arg Thr Lys Ser Thr Asn Ser Arg Ser Gly Phe Glu
        275                 280                 285

Met Ile Trp Asp Pro Asn Gly Trp Thr Gly Thr Asp Ser Ser Phe Ser
    290                 295                 300

Val Lys Gln Asp Ile Val Ala Ile Thr Asp Trp Ser Gly Tyr Ser Gly
305                 310                 315                 320

Ser Phe Val Gln His Pro Glu Leu Thr Gly Leu Asp Cys Ile Arg Pro
                325                 330                 335

Cys Phe Trp Val Glu Leu Ile Arg Gly Arg Pro Lys Glu Ser Thr Ile
            340                 345                 350

Trp Thr Ser Gly Ser Ser Ile Ser Phe Cys Gly Val Asn Ser Asp Thr
        355                 360                 365

Val Gly Trp Ser Trp Pro Asp Asp Ala Glu Leu Pro Phe
    370                 375                 380


<210> SEQ ID NO 29
<211> LENGTH: 383
<212> TYPE: PRT
<213> ORGANISM: Influenza A virus
<220> FEATURE:
<223> OTHER INFORMATION: Neuraminidase (NA) protein of influenza A virus
      H1N1

<400> SEQUENCE: 29

Val Ile Leu Thr Gly Asn Ser Ser Leu Cys Pro Ile Arg Gly Trp Ala
1               5                   10                  15

Ile Tyr Ser Lys Asp Asn Ser Ile Arg Ile Gly Ser Lys Gly Asp Val
            20                  25                  30

Phe Val Ile Arg Glu Pro Phe Ile Ser Cys Ser His Leu Glu Cys Arg
        35                  40                  45

Thr Phe Phe Leu Thr Gln Gly Ala Leu Leu Asn Asp Arg His Ser Asn
    50                  55                  60

Gly Thr Val Lys Asp Arg Ser Pro Tyr Arg Ala Leu Met Ser Cys Pro
65                  70                  75                  80

Val Gly Glu Ala Pro Ser Pro Tyr Asn Ser Arg Phe Glu Ser Val Ala
                85                  90                  95

Trp Ser Ala Ser Ala Cys His Asp Gly Met Gly Trp Leu Thr Ile Gly
            100                 105                 110

Ile Ser Gly Pro Asp Asn Gly Ala Val Ala Val Leu Lys Tyr Asn Gly
        115                 120                 125

Ile Ile Thr Glu Thr Ile Lys Ser Trp Arg Lys Lys Ile Leu Arg Thr
    130                 135                 140

Gln Glu Ser Glu Cys Ala Cys Val Asn Gly Ser Cys Phe Thr Ile Met
145                 150                 155                 160

Thr Asp Gly Pro Ser Asp Gly Leu Ala Ser Tyr Lys Ile Phe Lys Ile
                165                 170                 175

Glu Lys Gly Lys Val Thr Lys Ser Ile Glu Leu Asn Ala Pro Asn Ser
            180                 185                 190
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His Tyr Glu Glu Cys Ser Cys Tyr Pro Asp Thr Gly Lys Val Met Cys
195 200 205

Val Cys Arg Asp Asn Trp His Gly Ser Asn Arg Pro Trp Val Ser Phe
210 215 220

Asp Gln Asn Leu Asp Tyr Gln Ile Gly Tyr Ile Cys Ser Gly Val Phe
225 230 235 240

Gly Asp Asn Pro Arg Pro Lys Asp Gly Thr Gly Ser Cys Gly Pro Val
245 250 255

Tyr Val Asp Gly Ala Asn Gly Val Lys Gly Phe Ser Tyr Arg Tyr Gly
260 265 270

Asn Gly Val Trp Ile Gly Arg Thr Lys Ser His Ser Ser Arg His Gly
275 280 285

Phe Glu Met Ile Trp Asp Pro Asn Gly Trp Thr Glu Thr Asp Ser Lys
290 295 300

Phe Ser Val Arg Gln Asp Val Val Ala Met Thr Asp Trp Ser Gly Tyr
305 310 315 320

Ser Gly Ser Phe Val Gln His Pro Glu Leu Thr Gly Leu Asp Cys Ile
325 330 335

Arg Pro Cys Phe Trp Val Glu Leu Ile Arg Gly Arg Pro Lys Glu Lys
340 345 350

Thr Ile Trp Thr Ser Ala Ser Ser Ile Ser Phe Cys Gly Val Asn Ser
355 360 365

Asp Thr Val Asp Trp Ser Trp Pro Asp Gly Ala Glu Leu Pro Phe
370 375 380

<210> SEQ ID NO 30
<211> LENGTH: 384
<212> TYPE: PRT
<213> ORGANISM: Influenza A virus
<220> FEATURE:
<223> OTHER INFORMATION: Neuraminidase (NA) protein of influenza A virus H3N2

<400> SEQUENCE: 30

Glu Tyr Arg Asn Trp Ser Lys Pro Gln Cys Asp Ile Thr Gly Phe Ala
1 5 10 15

Pro Phe Ser Lys Asp Asn Ser Ile Arg Leu Ser Ala Gly Gly Asp Ile
20 25 30

Trp Val Thr Arg Glu Pro Tyr Val Ser Cys Asp Pro Asp Lys Cys Tyr
35 40 45

Gln Phe Ala Leu Gly Gln Gly Thr Thr Leu Asn Asn Val His Ser Asn
50 55 60

Asp Thr Val His Asp Arg Thr Pro Tyr Arg Thr Leu Leu Met Asn Glu
65 70 75 80

Leu Gly Val Pro Phe His Leu Gly Thr Lys Gln Val Cys Ile Ala Trp
85 90 95

Ser Ser Ser Ser Cys His Asp Gly Lys Ala Trp Leu His Val Cys Val
100 105 110

Thr Gly Asp Asp Lys Asn Ala Thr Ala Ser Phe Ile Tyr Asn Gly Arg
115 120 125

Leu Val Asp Ser Ile Val Ser Trp Ser Lys Lys Ile Leu Arg Thr Gln
130 135 140

Glu Ser Glu Cys Val Cys Ile Asn Gly Thr Cys Thr Val Val Met Thr
145 150 155 160

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Asp Gly Ser Ala Ser Gly Lys Ala Asp Thr Lys Ile Leu Phe Ile Glu
                165                 170                 175

Glu Gly Lys Ile Ile His Thr Ser Thr Leu Ser Gly Ser Ala Gln His
            180                 185                 190

Val Glu Glu Cys Ser Cys Tyr Pro Arg Tyr Pro Gly Val Arg Cys Val
        195                 200                 205

Cys Arg Asp Asn Trp Lys Gly Ser Asn Arg Pro Ile Val Asp Ile Asn
    210                 215                 220

Ile Lys Asp Tyr Ser Ile Val Ser Ser Tyr Val Cys Ser Gly Leu Val
225                 230                 235                 240

Gly Asp Thr Pro Arg Lys Asn Asp Ser Ser Ser Ser Ser His Cys Leu
                245                 250                 255

Asp Pro Asn Asn Glu Glu Gly Gly His Gly Val Lys Gly Trp Ala Phe
            260                 265                 270

Asp Asp Gly Asn Asp Val Trp Met Gly Arg Thr Ile Ser Glu Lys Leu
        275                 280                 285

Arg Ser Gly Tyr Glu Thr Phe Lys Val Ile Glu Gly Trp Ser Lys Pro
    290                 295                 300

Asn Ser Lys Leu Gln Ile Asn Arg Gln Val Ile Val Asp Arg Gly Asn
305                 310                 315                 320

Arg Ser Gly Tyr Ser Gly Ile Phe Ser Val Glu Gly Lys Ser Cys Ile
                325                 330                 335

Asn Arg Cys Phe Tyr Val Glu Leu Ile Arg Gly Arg Lys Glu Glu Thr
            340                 345                 350

Glu Val Leu Trp Thr Ser Asn Ser Ile Val Val Phe Cys Gly Thr Ser
        355                 360                 365

Gly Thr Tyr Gly Thr Gly Ser Trp Pro Asp Gly Ala Asp Ile Asn Leu
    370                 375                 380
```

<210> SEQ ID NO 31
<211> LENGTH: 384
<212> TYPE: PRT
<213> ORGANISM: Influenza A virus
<220> FEATURE:
<223> OTHER INFORMATION: Neuraminidase (NA) protein of influenza A virus H7N9

<400> SEQUENCE: 31

```
Phe Asn Asn Leu Thr Lys Gly Leu Cys Thr Ile Asn Ser Trp His Ile
1               5                   10                  15

Tyr Gly Lys Asp Asn Ala Val Arg Ile Gly Glu Ser Ser Asp Val Leu
            20                  25                  30

Val Thr Arg Glu Pro Tyr Val Ser Cys Asp Pro Asp Glu Cys Arg Phe
        35                  40                  45

Tyr Ala Leu Ser Gln Gly Thr Thr Ile Arg Gly Lys His Ser Asn Gly
    50                  55                  60

Thr Ile His Asp Arg Ser Gln Tyr Arg Ala Leu Ile Ser Trp Pro Leu
65                  70                  75                  80

Ser Ser Pro Pro Thr Val Tyr Asn Ser Arg Val Glu Cys Ile Gly Trp
                85                  90                  95

Ser Ser Thr Ser Cys His Asp Gly Lys Ser Arg Met Ser Ile Cys Ile
            100                 105                 110

Ser Gly Pro Asn Asn Asn Ala Ser Ala Val Val Trp Tyr Asn Arg Arg
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        115                 120                 125

Pro Val Ala Glu Ile Asn Thr Trp Ala Arg Asn Ile Leu Arg Thr Gln
    130                 135                 140

Glu Ser Glu Cys Val Cys His Asn Gly Val Cys Pro Val Val Phe Thr
145                 150                 155                 160

Asp Gly Ser Ala Thr Gly Pro Ala Asp Thr Arg Ile Tyr Tyr Phe Lys
                165                 170                 175

Glu Gly Lys Ile Leu Lys Trp Glu Ser Leu Thr Gly Thr Ala Lys His
            180                 185                 190

Ile Glu Glu Cys Ser Cys Tyr Gly Glu Arg Thr Gly Ile Thr Cys Thr
        195                 200                 205

Cys Arg Asp Asn Trp Gln Gly Ser Asn Arg Pro Val Ile Gln Ile Asp
    210                 215                 220

Pro Val Ala Met Thr His Thr Ser Gln Tyr Ile Cys Ser Pro Val Leu
225                 230                 235                 240

Thr Asp Asn Pro Arg Pro Asn Asp Pro Asn Ile Gly Lys Cys Asn Asp
                245                 250                 255

Pro Tyr Pro Gly Asn Asn Asn Asn Gly Val Lys Gly Phe Ser Tyr Leu
            260                 265                 270

Asp Gly Ala Asn Thr Trp Leu Gly Arg Thr Ile Ser Thr Ala Ser Arg
        275                 280                 285

Ser Gly Tyr Glu Met Leu Lys Val Pro Asn Ala Leu Thr Asp Asp Arg
    290                 295                 300

Ser Lys Pro Ile Gln Gly Gln Thr Ile Val Leu Asn Ala Asp Trp Ser
305                 310                 315                 320

Gly Tyr Ser Gly Ser Phe Met Asp Tyr Trp Ala Glu Gly Asp Cys Tyr
                325                 330                 335

Arg Ala Cys Phe Tyr Val Glu Leu Ile Arg Gly Arg Pro Lys Glu Asp
            340                 345                 350

Lys Val Trp Trp Thr Ser Asn Ser Ile Val Ser Met Cys Ser Ser Thr
        355                 360                 365

Glu Phe Leu Gly Gln Trp Asn Trp Pro Asp Gly Ala Lys Ile Glu Tyr
    370                 375                 380


<210> SEQ ID NO 32
<211> LENGTH: 384
<212> TYPE: PRT
<213> ORGANISM: Influenza A virus
<220> FEATURE:
<223> OTHER INFORMATION: Neuraminidase (NA) protein of influenza A virus
      H9N2

<400> SEQUENCE: 32

Glu Tyr Lys Asn Trp Ser Lys Pro Gln Cys Gln Ile Thr Gly Phe Ala
1               5                   10                  15

Pro Phe Ser Lys Asp Asn Ser Ile Arg Leu Ser Ala Gly Gly Asp Ile
            20                  25                  30

Trp Val Thr Arg Glu Pro Tyr Val Ser Cys Gly Leu Gly Lys Cys Tyr
        35                  40                  45

Gln Phe Ala Leu Gly Gln Gly Thr Thr Leu Asn Asn Lys His Ser Asn
    50                  55                  60

Gly Thr Ile His Asp Arg Ser Pro His Arg Thr Leu Leu Met Asn Glu
65                  70                  75                  80
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```
Leu Gly Val Pro Phe His Leu Gly Thr Lys Gln Val Cys Ile Ala Trp
                85                  90                  95

Ser Ser Ser Ser Cys His Asp Gly Lys Ala Trp Leu His Val Cys Val
            100                 105                 110

Thr Gly Asp Asp Arg Asn Ala Thr Ala Ser Ile Ile Tyr Asp Gly Met
        115                 120                 125

Leu Thr Asp Ser Ile Gly Ser Trp Ser Lys Asn Ile Leu Arg Thr Gln
    130                 135                 140

Glu Ser Glu Cys Val Cys Ile Asn Gly Thr Cys Thr Val Val Met Thr
145                 150                 155                 160

Asp Gly Ser Ala Ser Gly Arg Ala Asp Thr Lys Ile Leu Phe Ile Arg
                165                 170                 175

Glu Gly Lys Ile Val His Ile Gly Pro Leu Ser Gly Ser Ala Gln His
            180                 185                 190

Val Glu Glu Cys Ser Cys Tyr Pro Arg Tyr Pro Glu Val Arg Cys Val
        195                 200                 205

Cys Arg Asp Asn Trp Lys Gly Ser Asn Arg Pro Val Leu Tyr Ile Asn
    210                 215                 220

Val Ala Asp Tyr Ser Val Asp Ser Ser Tyr Val Cys Ser Gly Leu Val
225                 230                 235                 240

Gly Asp Thr Pro Arg Asn Asp Asp Ser Ser Ser Ser Ser Asn Cys Arg
                245                 250                 255

Asp Pro Asn Asn Glu Arg Gly Gly Pro Gly Val Lys Gly Trp Ala Phe
            260                 265                 270

Asp Asn Gly Asn Asp Val Trp Met Gly Arg Thr Ile Lys Lys Asp Ser
        275                 280                 285

Arg Ser Gly Tyr Glu Thr Phe Arg Val Val Gly Gly Trp Thr Thr Ala
    290                 295                 300

Asn Ser Lys Ser Gln Ile Asn Arg Gln Val Ile Val Asp Ser Asp Asn
305                 310                 315                 320

Trp Ser Gly Tyr Ser Gly Ile Phe Ser Val Glu Gly Lys Thr Cys Ile
                325                 330                 335

Asn Arg Cys Phe Tyr Val Glu Leu Ile Arg Gly Arg Pro Gln Glu Thr
            340                 345                 350

Arg Val Trp Trp Thr Ser Asn Ser Ile Ile Val Phe Cys Gly Thr Ser
        355                 360                 365

Gly Thr Tyr Gly Thr Gly Ser Trp Pro Asp Gly Ala Asn Ile Asn Phe
    370                 375                 380


<210> SEQ ID NO 33
<211> LENGTH: 384
<212> TYPE: PRT
<213> ORGANISM: Influenza A virus
<220> FEATURE:
<223> OTHER INFORMATION: Neuraminidase (NA) protein of influenza A virus
      H2N2

<400> SEQUENCE: 33

Glu Tyr Arg Asn Trp Ser Lys Pro Gln Cys Gln Ile Thr Gly Phe Ala
1               5                   10                  15

Pro Phe Ser Lys Asp Asn Ser Ile Arg Leu Ser Ala Gly Gly Asp Ile
            20                  25                  30

Trp Val Thr Arg Glu Pro Tyr Val Ser Cys Asp Pro Gly Lys Cys Tyr
        35                  40                  45
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```
Gln Phe Ala Leu Gly Gln Gly Thr Thr Leu Asp Asn Lys His Ser Asn
    50                  55                  60

Asp Thr Ile His Asp Arg Ile Pro His Arg Thr Leu Leu Met Asn Glu
65                  70                  75                  80

Leu Gly Val Pro Phe His Leu Gly Thr Arg Gln Val Cys Val Ala Trp
                85                  90                  95

Ser Ser Ser Ser Cys His Asp Gly Lys Ala Trp Leu His Val Cys Val
            100                 105                 110

Thr Gly Asp Asp Lys Asn Ala Thr Ala Ser Phe Ile Tyr Asp Gly Arg
        115                 120                 125

Leu Met Asp Ser Ile Gly Ser Trp Ser Gln Asn Ile Leu Arg Thr Gln
    130                 135                 140

Glu Ser Glu Cys Val Cys Ile Asn Gly Thr Cys Thr Val Val Met Thr
145                 150                 155                 160

Asp Gly Ser Ala Ser Gly Arg Ala Asp Thr Arg Ile Leu Phe Ile Glu
                165                 170                 175

Glu Gly Lys Ile Val His Ile Ser Pro Leu Ser Gly Ser Ala Gln His
            180                 185                 190

Val Glu Glu Cys Ser Cys Tyr Pro Arg Tyr Pro Asp Val Arg Cys Ile
        195                 200                 205

Cys Arg Asp Asn Trp Lys Gly Ser Asn Arg Pro Val Ile Asp Ile Asn
    210                 215                 220

Met Glu Asp Tyr Ser Ile Asp Ser Ser Tyr Val Cys Ser Gly Leu Val
225                 230                 235                 240

Gly Asp Thr Pro Arg Asn Asp Asp Arg Ser Ser Asn Ser Asn Cys Arg
                245                 250                 255

Asn Pro Asn Asn Glu Arg Gly Asn Pro Gly Val Lys Gly Trp Ala Phe
            260                 265                 270

Asp Asn Gly Asp Asp Val Trp Met Gly Arg Thr Ile Ser Lys Asp Leu
        275                 280                 285

Arg Ser Gly Tyr Glu Thr Phe Lys Val Ile Gly Gly Trp Ser Thr Pro
    290                 295                 300

Asn Ser Lys Ser Gln Ile Asn Arg Gln Val Ile Val Asp Ser Asn Asn
305                 310                 315                 320

Trp Ser Gly Tyr Ser Gly Ile Phe Ser Val Glu Gly Lys Arg Cys Ile
                325                 330                 335

Asn Arg Cys Phe Tyr Val Glu Leu Ile Arg Gly Arg Gln Gln Glu Thr
            340                 345                 350

Arg Val Trp Trp Thr Ser Asn Ser Ile Val Val Phe Cys Gly Thr Ser
        355                 360                 365

Gly Thr Tyr Gly Thr Gly Ser Trp Pro Asp Gly Ala Asn Ile Asn Phe
    370                 375                 380
```

<210> SEQ ID NO 34
<211> LENGTH: 381
<212> TYPE: PRT
<213> ORGANISM: Influenza A virus
<220> FEATURE:
<223> OTHER INFORMATION: Neuraminidase (NA) protein of influenza A virus
      H1N1

<400> SEQUENCE: 34

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Leu Ala Gly Asn Ser Ser Leu Cys Pro Val Ser Gly Trp Ala Ile Tyr
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```
1               5                   10                  15

Ser Lys Asp Asn Ser Val Arg Ile Gly Ser Lys Gly Asp Val Phe Val
            20                  25                  30

Ile Arg Glu Pro Phe Ile Ser Cys Ser Pro Leu Glu Cys Arg Thr Phe
        35                  40                  45

Phe Leu Thr Gln Gly Ala Leu Leu Asn Asp Lys His Ser Asn Gly Thr
    50                  55                  60

Ile Lys Asp Arg Ser Pro Tyr Arg Thr Leu Met Ser Cys Pro Ile Gly
65                  70                  75                  80

Glu Val Pro Ser Pro Tyr Asn Ser Arg Phe Glu Ser Val Ala Trp Ser
                85                  90                  95

Ala Ser Ala Cys His Asp Gly Ile Asn Trp Leu Thr Ile Gly Ile Ser
            100                 105                 110

Gly Pro Asp Asn Gly Ala Val Ala Val Leu Lys Tyr Asn Gly Ile Ile
        115                 120                 125

Thr Asp Thr Ile Lys Ser Trp Arg Asn Asn Ile Leu Arg Thr Gln Glu
    130                 135                 140

Ser Glu Cys Ala Cys Val Asn Gly Ser Cys Phe Thr Val Met Thr Asp
145                 150                 155                 160

Gly Pro Ser Asn Gly Gln Ala Ser Tyr Lys Ile Phe Arg Ile Glu Lys
                165                 170                 175

Gly Lys Ile Val Lys Ser Val Glu Met Asn Ala Pro Asn Tyr His Tyr
            180                 185                 190

Glu Glu Cys Ser Cys Tyr Pro Asp Ser Ser Glu Ile Thr Cys Val Cys
        195                 200                 205

Arg Asp Asn Trp His Gly Ser Asn Arg Pro Trp Val Ser Phe Asn Gln
    210                 215                 220

Asn Leu Glu Tyr Gln Ile Gly Tyr Ile Cys Ser Gly Ile Phe Gly Asp
225                 230                 235                 240

Asn Pro Arg Pro Asn Asp Lys Thr Gly Ser Cys Gly Pro Val Ser Ser
                245                 250                 255

Asn Gly Ala Asn Gly Val Lys Gly Phe Ser Phe Lys Tyr Gly Asn Gly
            260                 265                 270

Val Trp Ile Gly Arg Thr Lys Ser Ile Ser Ser Arg Asn Gly Phe Glu
        275                 280                 285

Met Ile Trp Asp Pro Asn Gly Trp Thr Gly Thr Asp Asn Asn Phe Ser
    290                 295                 300

Ile Lys Gln Asp Ile Val Gly Ile Asn Glu Trp Ser Gly Tyr Ser Gly
305                 310                 315                 320

Ser Phe Val Gln His Pro Glu Leu Thr Gly Leu Asp Cys Ile Arg Pro
                325                 330                 335

Cys Phe Trp Val Glu Leu Ile Arg Gly Arg Pro Lys Glu Asn Thr Ile
            340                 345                 350

Trp Thr Ser Gly Ser Ser Ile Ser Phe Cys Gly Val Asn Ser Asp Thr
        355                 360                 365

Val Gly Trp Ser Trp Pro Asp Gly Ala Glu Leu Pro Phe
    370                 375                 380
```

<210> SEQ ID NO 35
<211> LENGTH: 466
<212> TYPE: PRT
<213> ORGANISM: Influenza B virus

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<220> FEATURE:
<223> OTHER INFORMATION: Neuraminidase (NA) protein

<400> SEQUENCE: 35

```
Met Leu Pro Ser Thr Val Gln Thr Leu Thr Leu Leu Leu Thr Ser Gly
1               5                   10                  15

Gly Val Leu Leu Ser Leu Tyr Val Ser Ala Ser Leu Ser Tyr Leu Leu
            20                  25                  30

Tyr Ser Asp Val Leu Leu Lys Phe Ser Ser Thr Lys Thr Thr Ala Pro
        35                  40                  45

Thr Met Ser Leu Glu Cys Thr Asn Ala Ser Asn Ala Gln Thr Val Asn
    50                  55                  60

His Ser Ala Thr Lys Glu Met Thr Phe Pro Pro Pro Glu Pro Glu Trp
65                  70                  75                  80

Thr Tyr Pro Arg Leu Ser Cys Gln Gly Ser Thr Phe Gln Lys Ala Leu
                85                  90                  95

Leu Ile Ser Pro His Arg Phe Gly Glu Ile Lys Gly Asn Ser Ala Pro
            100                 105                 110

Leu Ile Ile Arg Glu Pro Phe Val Ala Cys Gly Pro Lys Glu Cys Arg
        115                 120                 125

His Phe Ala Leu Thr His Tyr Ala Ala Gln Pro Gly Gly Tyr Tyr Asn
    130                 135                 140

Gly Thr Arg Lys Asp Arg Asn Lys Leu Arg His Leu Val Ser Val Lys
145                 150                 155                 160

Leu Gly Lys Ile Pro Thr Val Glu Asn Ser Ile Phe His Met Ala Ala
                165                 170                 175

Trp Ser Gly Ser Ala Cys His Asp Gly Arg Glu Trp Thr Tyr Ile Gly
            180                 185                 190

Val Asp Gly Pro Asp Asn Asp Ala Leu Val Lys Ile Lys Tyr Gly Glu
        195                 200                 205

Ala Tyr Thr Asp Thr Tyr His Ser Tyr Ala His Asn Ile Leu Arg Thr
    210                 215                 220

Gln Glu Ser Ala Cys Asn Cys Ile Gly Gly Asp Cys Tyr Leu Met Ile
225                 230                 235                 240

Thr Asp Gly Ser Ala Ser Gly Ile Ser Lys Cys Arg Phe Leu Lys Ile
                245                 250                 255

Arg Glu Gly Arg Ile Ile Lys Glu Ile Leu Pro Thr Gly Arg Val Glu
            260                 265                 270

His Thr Glu Glu Cys Thr Cys Gly Phe Ala Ser Asn Lys Thr Ile Glu
        275                 280                 285

Cys Ala Cys Arg Asp Asn Ser Tyr Thr Ala Lys Arg Pro Phe Val Lys
    290                 295                 300

Leu Asn Val Glu Thr Asp Thr Ala Glu Ile Arg Leu Met Cys Thr Lys
305                 310                 315                 320

Thr Tyr Leu Asp Thr Pro Arg Pro Asp Asp Gly Ser Ile Ala Gly Pro
                325                 330                 335

Cys Glu Ser Asn Gly Asp Lys Trp Leu Gly Gly Ile Lys Gly Gly Phe
            340                 345                 350

Val His Gln Arg Met Ala Ser Lys Ile Gly Arg Trp Tyr Ser Arg Thr
        355                 360                 365

Met Ser Lys Thr Asn Arg Met Gly Met Glu Leu Tyr Val Lys Tyr Asp
    370                 375                 380
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```
Gly Asp Pro Trp Thr Asp Ser Asp Ala Leu Thr Leu Ser Gly Val Met
385                 390                 395                 400

Val Ser Ile Glu Glu Pro Gly Trp Tyr Ser Phe Gly Phe Glu Ile Lys
                405                 410                 415

Asp Lys Lys Cys Asp Val Pro Cys Ile Gly Ile Glu Met Val His Asp
            420                 425                 430

Gly Gly Lys Asp Thr Trp His Ser Ala Ala Thr Ala Ile Tyr Cys Leu
        435                 440                 445

Met Gly Ser Gly Gln Leu Leu Trp Asp Thr Val Thr Gly Val Asp Met
    450                 455                 460

Ala Leu
465


<210> SEQ ID NO 36
<211> LENGTH: 842
<212> TYPE: PRT
<213> ORGANISM: Human immunodeficiency virus 1
<220> FEATURE:
<223> OTHER INFORMATION: Envelope protein

<400> SEQUENCE: 36

Met Arg Val Arg Gly Met Gln Arg Asn Trp Gln His Leu Gly Lys Trp
1               5                   10                  15

Gly Leu Leu Phe Leu Gly Met Leu Ile Ile Cys Lys Ala Ala Asp Asp
            20                  25                  30

Leu Trp Val Thr Ile Tyr Tyr Gly Val Pro Val Trp Lys Glu Ala Asn
        35                  40                  45

Thr Thr Leu Phe Cys Ala Ser Asp Ala Lys Ser Tyr Glu Lys Glu Val
    50                  55                  60

His Asn Val Trp Ala Thr His Ala Cys Val Pro Thr Asp Pro Asn Pro
65                  70                  75                  80

Gln Glu Val Ala Leu Asn Val Thr Glu Asn Phe Asn Met Trp Glu Asn
                85                  90                  95

Asp Met Val Glu Gln Met His Lys Asp Ile Ile Ser Leu Trp Asp Gln
            100                 105                 110

Ser Leu Lys Pro Cys Val Lys Leu Thr Pro Leu Cys Val Thr Leu Asn
        115                 120                 125

Cys Thr Asn Ala Thr Thr Thr Asn Asp Thr Leu Ser Asp Gln Ser Ser
    130                 135                 140

Thr Leu Lys Glu Glu Pro Gly Ala Ile Gln Asn Cys Ser Phe Asn Met
145                 150                 155                 160

Thr Thr Glu Val Glu Asp Lys Lys Gln Lys Val His Ala Leu Phe Tyr
                165                 170                 175

Arg Leu Asp Ile Glu Pro Ile Ser Asn Asn Asn Ser Arg Glu Glu Tyr
            180                 185                 190

Arg Leu Ile Thr Cys Asn Thr Ser Thr Ile Thr Gln Ala Cys Pro Lys
        195                 200                 205

Val Ser Trp Asp Pro Ile Pro Ile His Tyr Cys Ala Pro Ala Gly Tyr
    210                 215                 220

Ala Ile Leu Lys Cys Lys Asp Lys Arg Phe Asn Gly Thr Gly Pro Cys
225                 230                 235                 240

Arg Asn Val Ser Thr Val Gln Cys Thr His Gly Ile Arg Pro Val Val
                245                 250                 255
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```
Ser Thr Gln Leu Leu Leu Asn Gly Ser Leu Ser Glu Gly Gly Ile Ile
            260                 265                 270

Ile Arg Ser Gln Asn Leu Ser Asp Asn Ala Lys Thr Ile Ile Val His
        275                 280                 285

Leu Asn Glu Ser Val Gln Ile Asn Cys Thr Arg Pro Asn Asn Asn Thr
    290                 295                 300

Arg Lys Ser Ile Arg Ile Gly Pro Gly Gln Ser Phe Tyr Ala Thr Gly
305                 310                 315                 320

Glu Ile Ile Gly Asp Ile Arg Lys Ala His Cys Asn Ile Ser Gly Glu
                325                 330                 335

Gln Trp Asn Lys Thr Leu Asp Arg Val Lys Ala Glu Leu Lys Leu His
            340                 345                 350

Phe Asn Lys Thr Ile Gln Phe Asn Ser Ser Ser Gly Gly Asp Leu Glu
        355                 360                 365

Ile Thr Met His Ser Phe Asn Cys Arg Gly Glu Phe Phe Tyr Cys Asn
    370                 375                 380

Thr Ser Leu Leu Phe Asn Asn Thr Val Pro Asn Asn Gly Thr Ile Thr
385                 390                 395                 400

Leu Pro Cys Arg Ile Lys Gln Phe Val Asn Met Trp Gln Glu Val Gly
                405                 410                 415

Arg Ala Met Tyr Ala Ala Pro Ile Ala Gly Asn Ile Thr Cys Asn Ser
            420                 425                 430

Asn Ile Thr Gly Leu Leu Leu Thr Arg Asp Gly Gly Gln Ser Asn Asn
        435                 440                 445

Ser Asp Ser Glu Thr Phe Arg Pro Gly Gly Gly Asp Met Lys Asp Asn
    450                 455                 460

Trp Arg Ser Glu Leu Tyr Lys Tyr Lys Val Val Glu Ile Glu Pro Leu
465                 470                 475                 480

Gly Val Ala Pro Thr Arg Pro Lys Arg Pro Val Val Arg Arg Glu Arg
                485                 490                 495

Arg Ala Val Ala Ile Gly Ala Val Phe Leu Gly Phe Leu Ser Ala Ala
            500                 505                 510

Gly Ser Thr Met Gly Ala Ala Ser Leu Thr Leu Thr Val Gln Ala Arg
        515                 520                 525

Gln Leu Leu Ser Gly Ile Val Gln Gln Gln Asn Asn Leu Leu Gln Ala
    530                 535                 540

Ile Glu Ala Gln Gln His Met Leu Gln Leu Thr Val Trp Gly Ile Lys
545                 550                 555                 560

Gln Leu Gln Ala Arg Val Leu Ala Val Glu Arg Tyr Leu Lys Asp Gln
                565                 570                 575

Gln Leu Leu Gly Leu Trp Gly Cys Ser Gly Lys Leu Ile Cys Thr Thr
            580                 585                 590

Asn Val Pro Trp Asn Ser Ser Trp Ser Asn Lys Ser Gln Asp Glu Ile
        595                 600                 605

Trp Asn Asn Met Thr Trp Met Gln Trp Glu Lys Glu Ile Ser Asn Tyr
    610                 615                 620

Ser Lys Thr Ile Tyr Met Leu Ile Glu Lys Ser Gln Ser Gln Gln Glu
625                 630                 635                 640

Arg Asn Glu Gln Glu Leu Leu Glu Leu Asp Lys Trp Asp Ser Leu Trp
                645                 650                 655
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Ser Trp Phe Asp Ile Thr Asn Trp Leu Trp Tyr Ile Lys Ile Phe Ile
            660                 665                 670

Met Ile Val Gly Gly Leu Ile Gly Leu Arg Ile Val Phe Ala Val Leu
        675                 680                 685

Ser Ile Val Asn Arg Val Arg Lys Gly Tyr Ser Pro Leu Ser Leu Gln
    690                 695                 700

Thr Leu Ile Pro Ala Pro Thr Glu Pro Asp Arg Pro Glu Gly Ile Glu
705                 710                 715                 720

Glu Gly Gly Gly Glu Gln Gly Lys Asp Arg Ser Val Arg Leu Val Asn
                725                 730                 735

Gly Phe Leu Ala Leu Val Trp Asp Asp Leu Arg Asn Leu Cys Leu Phe
            740                 745                 750

Ser Tyr Arg His Leu Arg Asp Phe Ile Leu Ile Ala Ala Arg Ile Val
        755                 760                 765

Asp Arg Gly Leu Arg Arg Gly Trp Glu Ala Leu Lys Tyr Leu Gly Asn
    770                 775                 780

Ile Ile Gln Tyr Trp Ser Gln Glu Leu Lys Asn Ser Ala Ile Ser Leu
785                 790                 795                 800

Phe Asn Thr Thr Ala Ile Val Val Ala Glu Gly Thr Asp Arg Val Ile
                805                 810                 815

Glu Ala Leu Gln Arg Ala Val Arg Ala Val Leu Asn Ile Pro Arg Arg
            820                 825                 830

Ile Arg Gln Arg Val Glu Arg Ala Leu Ile
        835                 840


<210> SEQ ID NO 37
<211> LENGTH: 850
<212> TYPE: PRT
<213> ORGANISM: Human immunodeficiency virus 2
<220> FEATURE:
<223> OTHER INFORMATION: Envelope protein

<400> SEQUENCE: 37

Met Thr Ser Glu Lys Thr Gln Leu Leu Ile Ala Ile Leu Leu Ala Ser
1               5                   10                  15

Thr Cys Leu Leu Tyr Cys Lys Gln Tyr Val Thr Val Phe Tyr Gly Val
            20                  25                  30

Pro Ala Trp Arg Asn Ala Ser Ile Pro Leu Phe Cys Ala Thr Lys Asn
        35                  40                  45

Arg Asp Thr Trp Gly Thr Ile Gln Cys Leu Pro Asp Asn Asp Asp Tyr
    50                  55                  60

Gln Glu Ile Ala Leu Asn Val Thr Glu Ala Phe Asp Ala Trp Asn Asn
65                  70                  75                  80

Thr Val Thr Glu Gln Ala Ile Asp Asp Val Lys Arg Leu Phe Glu Thr
                85                  90                  95

Ser Ile Lys Pro Cys Val Lys Leu Thr Pro Leu Cys Val Ala Met Asn
            100                 105                 110

Cys Thr Asn Val Thr Ser Thr Ala Asn Thr Thr Ile Thr Ile Ser Ser
        115                 120                 125

Thr Asn Met Ile Val Asn Asp Ser Ser Pro Cys Ala Ser Asn Asp Ser
    130                 135                 140

Cys Pro Gly Met Gly Glu Glu Glu Met Val Gln Cys Asn Phe Ser Met
145                 150                 155                 160
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```
Thr Gly Leu Gln Arg Asp Lys Val Lys Gln Tyr Asn Glu Thr Trp Tyr
                165                 170                 175

Ser Arg Asp Val Val Cys Asp Gln Gly Glu Asn Asp Thr Arg Arg Cys
            180                 185                 190

Tyr Met Asn His Cys Asn Thr Ser Val Ile Thr Glu Ser Cys Asp Lys
        195                 200                 205

His Tyr Trp Asp Asp Met Arg Phe Arg Tyr Cys Ala Pro Pro Gly Tyr
    210                 215                 220

Ala Leu Leu Arg Cys Asn Asp Thr Asn Tyr Ser Gly Phe Met Pro Asn
225                 230                 235                 240

Cys Ser Lys Val Val Ala Ala Thr Cys Thr Arg Met Met Glu Thr Gln
                245                 250                 255

Thr Ser Thr Trp Phe Gly Phe Asn Gly Thr Arg Ala Glu Asn Arg Thr
            260                 265                 270

Tyr Ile Tyr Trp His Ser Lys Ser Asn Arg Thr Ile Ile Ser Leu Asn
        275                 280                 285

Lys Ser Phe Asn Leu Ser Leu His Cys Lys Arg Pro Gly Asn Lys Thr
    290                 295                 300

Val Val Pro Ile Thr Leu Met Ser Gly Leu Val Phe His Ser Gln Pro
305                 310                 315                 320

Ile Asn Thr Arg Pro Arg Gln Ala Trp Cys Trp Phe Glu Gly Glu Trp
                325                 330                 335

Lys Glu Ala Met Gln Glu Val Lys Arg Ala Ile Ala Lys His Pro Arg
            340                 345                 350

Tyr Thr Gly Thr Asn Asp Thr Gly Lys Ile Asn Leu Thr Ala Pro Gly
        355                 360                 365

Arg Gly Ser Asp Pro Glu Val Ser Tyr Met Trp Thr Asn Cys Arg Gly
    370                 375                 380

Glu Phe Leu Tyr Cys Asn Met Thr Trp Phe Leu Asn Trp Val Glu Asn
385                 390                 395                 400

Arg Thr Gly Ile Gln Arg Asn Tyr Ala Pro Cys His Ile Arg Gln Ile
                405                 410                 415

Ile Asn Thr Trp His Lys Val Gly Lys Asn Val Tyr Leu Pro Pro Arg
            420                 425                 430

Glu Gly Glu Leu Val Cys Asn Ser Thr Val Thr Ser Leu Ile Ala Asp
        435                 440                 445

Ile Asp Trp Ile Asp Lys Asn Gly Thr Asn Ile Thr Phe Ser Ala Glu
    450                 455                 460

Val Ser Glu Leu Tyr Arg Leu Glu Leu Gly Asp Tyr Lys Leu Val Glu
465                 470                 475                 480

Ile Thr Pro Ile Gly Phe Ala Pro Thr Thr Glu Lys Arg Tyr Ser Pro
                485                 490                 495

Asn His Gly Arg Pro Lys Arg Gly Val Phe Val Leu Gly Phe Leu Gly
            500                 505                 510

Phe Leu Ala Thr Ala Gly Ser Ala Met Gly Ala Ala Ser Leu Thr Leu
        515                 520                 525

Ser Ala Gln Ser Arg Thr Leu Leu Ala Gly Ile Val Gln Gln Gln Gln
    530                 535                 540

Gln Leu Leu Asp Val Val Lys Arg Gln Gln Glu Met Leu Arg Leu Thr
545                 550                 555                 560

Val Trp Gly Thr Lys Asn Leu Gln Ala Arg Val Thr Ala Ile Glu Lys
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                565                 570                 575

Tyr Leu Gly Asp Gln Ala Arg Leu Asn Ser Trp Gly Cys Ala Phe Arg
            580                 585                 590

Gln Val Cys His Thr Thr Val Pro Trp Val Asn Asp Thr Leu Met Pro
        595                 600                 605

Asp Trp Lys Asn Met Thr Trp Gln Lys Trp Glu Glu Gln Ile Arg Tyr
    610                 615                 620

Leu Glu Ala Asn Ile Ser Asp Ser Leu Glu Gln Ala Gln Ile Gln Gln
625                 630                 635                 640

Glu Lys Asn Thr Tyr Glu Leu Gln Lys Leu Asn Ser Trp Asp Val Phe
                645                 650                 655

Thr Asn Trp Leu Asp Leu Thr Ser Trp Ile Lys Tyr Ile Gln Tyr Gly
            660                 665                 670

Val Tyr Ile Val Val Gly Ile Ile Ala Leu Arg Ile Val Ile Tyr Val
        675                 680                 685

Val Gln Met Leu Ser Arg Leu Arg Lys Gly Tyr Arg Pro Val Phe Ser
    690                 695                 700

Ser Pro Pro Gly Tyr Ile Gln Gln Ile His Ile His Lys Gly Gln Glu
705                 710                 715                 720

Gln Pro Thr Arg Glu Gly Thr Glu Glu Asp Val Gly Asp Asn Gly Gly
                725                 730                 735

Asp Arg Ser Trp Pro Trp Pro Ile Ala Tyr Leu His Phe Leu Ile Arg
            740                 745                 750

Leu Leu Ile Arg Leu Leu Ile Thr Leu Tyr Asn Ser Cys Arg Asp Leu
        755                 760                 765

Leu Ser Arg Ile Phe Gln Thr Leu Gln Pro Ile Leu Arg Asn Leu Arg
    770                 775                 780

Asp Trp Leu Arg Ile Lys Thr Ala Leu Leu Gln Tyr Gly Cys Glu Trp
785                 790                 795                 800

Ile Gln Glu Ala Phe Gln Ala Ala Ala Arg Thr Thr Gly Glu Thr Leu
                805                 810                 815

Ala Gly Ala Cys Arg Gly Leu Trp Arg Thr Leu Gly Arg Ile Gly Arg
            820                 825                 830

Gly Ile Phe Ala Val Pro Arg Arg Ile Arg Gln Gly Ala Glu Ile Ala
        835                 840                 845

Leu Leu
    850


<210> SEQ ID NO 38
<211> LENGTH: 424
<212> TYPE: PRT
<213> ORGANISM: Human papillomavirus
<220> FEATURE:
<223> OTHER INFORMATION: Major Capsid L1 Protein

<400> SEQUENCE: 38

Ala Val Val Ser Thr Asp Glu Tyr Val Ala Arg Thr Asn Ile Tyr Tyr
1               5                   10                  15

His Ala Gly Thr Ser Arg Leu Leu Ala Val Gly His Pro Tyr Phe Pro
            20                  25                  30

Ile Lys Lys Pro Asn Asn Asn Lys Ile Leu Val Pro Lys Val Ser Gly
        35                  40                  45

Leu Gln Tyr Arg Val Phe Arg Ile His Leu Pro Asp Pro Asn Lys Phe
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    50                  55                  60

Gly Phe Pro Asp Thr Ser Phe Tyr Asn Pro Asp Thr Gln Arg Leu Val
65                  70                  75                  80

Trp Ala Cys Val Gly Val Glu Val Gly Arg Gly Gln Pro Leu Gly Val
                85                  90                  95

Gly Ile Ser Gly His Pro Leu Leu Asn Lys Leu Asp Asp Thr Glu Asn
            100                 105                 110

Ala Ser Ala Tyr Ala Ala Asn Ala Gly Val Asp Asn Arg Glu Cys Ile
        115                 120                 125

Ser Met Asp Tyr Lys Gln Thr Gln Leu Cys Leu Ile Gly Cys Lys Pro
    130                 135                 140

Pro Ile Gly Glu His Trp Gly Lys Gly Ser Pro Cys Thr Gln Val Ala
145                 150                 155                 160

Val Gln Pro Gly Asp Cys Pro Pro Leu Glu Leu Ile Asn Thr Val Ile
                165                 170                 175

Gln Asp Gly Asp Met Val Asp Thr Gly Phe Gly Ala Met Asp Phe Thr
            180                 185                 190

Thr Leu Gln Ala Asn Lys Ser Glu Val Pro Leu Asp Ile Cys Thr Ser
        195                 200                 205

Ile Cys Lys Tyr Pro Asp Tyr Ile Lys Met Val Ser Glu Pro Tyr Gly
    210                 215                 220

Asp Ser Leu Phe Phe Tyr Leu Arg Arg Glu Gln Met Phe Val Arg His
225                 230                 235                 240

Leu Phe Asn Arg Ala Gly Thr Val Gly Glu Asn Val Pro Asp Asp Leu
                245                 250                 255

Tyr Ile Lys Gly Ser Gly Ser Thr Ala Asn Leu Ala Ser Ser Asn Tyr
            260                 265                 270

Phe Pro Thr Pro Ser Gly Ser Met Val Thr Ser Asp Ala Gln Ile Phe
        275                 280                 285

Asn Lys Pro Tyr Trp Leu Gln Arg Ala Gln Gly His Asn Asn Gly Ile
    290                 295                 300

Cys Trp Gly Asn Gln Leu Phe Val Thr Val Val Asp Thr Thr Arg Ser
305                 310                 315                 320

Thr Asn Met Ser Leu Cys Ala Ala Ile Ser Thr Ser Glu Thr Thr Tyr
                325                 330                 335

Lys Asn Thr Asn Phe Lys Glu Tyr Leu Arg His Gly Glu Glu Tyr Asp
            340                 345                 350

Leu Gln Phe Ile Phe Gln Leu Cys Lys Ile Thr Leu Thr Ala Asp Val
        355                 360                 365

Met Thr Tyr Ile His Ser Met Asn Ser Thr Ile Leu Glu Asp Trp Asn
    370                 375                 380

Gly Gly Ser Gly Gly Glu Asp Pro Leu Lys Lys Tyr Thr Phe Trp Glu
385                 390                 395                 400

Val Asn Leu Lys Glu Lys Phe Ser Ala Asp Leu Asp Gln Phe Pro Leu
                405                 410                 415

Gly Arg Lys Phe Leu Leu Gln Leu
            420
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<210> SEQ ID NO 39
<211> LENGTH: 523
<212> TYPE: PRT
<213> ORGANISM: Human papillomavirus

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<220> FEATURE:
<223> OTHER INFORMATION: Minor Capsid L2 Protein

<400> SEQUENCE: 39

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Met Ala Arg Ala Arg Arg Thr Lys Arg Asp Ser Val Thr Asn Ile Tyr
1               5                   10                  15

Arg Thr Cys Lys Ala Ala Gly Thr Cys Pro Pro Asp Val Val Asn Lys
            20                  25                  30

Val Glu Gln Thr Thr Val Ala Asp Gln Ile Leu Lys Tyr Gly Ser Thr
        35                  40                  45

Gly Val Phe Phe Gly Gly Leu Gly Ile Gly Thr Gly Arg Gly Thr Gly
    50                  55                  60

Gly Ser Thr Gly Tyr Gly Pro Leu Gly Glu Gly Thr Ser Val Arg Val
65                  70                  75                  80

Gly Asn Thr Pro Thr Val Ile Arg Pro Ala Leu Val Pro Glu Ala Ile
                85                  90                  95

Gly Pro Ser Glu Leu Ile Pro Ile Asp Ser Val Asn Pro Ile Asp Pro
            100                 105                 110

Ser Ala Ser Ser Ile Ile Pro Leu Thr Glu Ser Thr Gly Pro Asp Leu
        115                 120                 125

Leu Pro Gly Glu Ile Glu Thr Ile Ala Glu Val His Pro Ala Pro Asp
    130                 135                 140

Ile Pro Thr Val Asp Thr Pro Val Val Thr Gly Gly Arg Asn Ser Asn
145                 150                 155                 160

Ala Val Leu Glu Val Ala Asp Pro Ser Pro Pro Thr Arg Asn Arg Val
                165                 170                 175

Ser Arg Thr Gln Tyr Asn Asn Pro Ala Phe Gln Ile Ile Ser Glu Thr
            180                 185                 190

Thr Pro Ser Ala Gly Glu Thr Ser Leu Ser Asp Gln Ile Val Val Gln
        195                 200                 205

Ser Phe Asp Gly Gly Gln Tyr Ile Gly Gly Asn Pro Pro Pro Arg Ser
    210                 215                 220

Val Val Glu Ile Glu Leu Gln Glu Ile Pro Ser Gln Tyr Ser Phe Glu
225                 230                 235                 240

Ile Glu Glu Pro Thr Pro Pro Arg Gln Thr Ser Thr Pro Val Arg Gln
                245                 250                 255

Ala Gln Gln Met Ala Ser Ala Leu Arg Arg Ala Leu Tyr Asn Arg Arg
            260                 265                 270

Phe Thr Gln Gln Val Gln Val Glu Asp Pro Met Phe Tyr Ser Arg Pro
        275                 280                 285

Ser Arg Leu Val Arg Phe Gln Phe Asp Asn Pro Val Phe Glu Glu Glu
    290                 295                 300

Val Thr Gln Val Phe Glu Arg Asp Leu Glu Thr Ile Glu Glu Pro Pro
305                 310                 315                 320

Asp Arg Gln Phe Leu Asp Val Gln Lys Leu Gly Arg Pro Thr Tyr Ala
                325                 330                 335

Glu Thr Pro Ala Gly Tyr Ile Arg Val Ser Arg Leu Gly Lys Arg Ala
            340                 345                 350

Thr Ile Arg Thr Arg Ser Gly Thr Thr Ile Gly Gly Gln Val His Phe
        355                 360                 365

Phe Arg Asp Ile Ser Ser Ile Asp Thr Gln Pro Ser Ile Glu Leu Gln
    370                 375                 380
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Val Leu Gly Glu His Ser Gly Asp Ala Thr Ile Val Gln Gly Pro Val
385 390 395 400

Glu Ser Thr Phe Val Asn Ile Asp Leu Glu Glu Leu Pro Asn Leu Glu
405 410 415

Glu Asn Val His Leu Glu Ser Asp Asp Ile Leu Ile Asp Glu Ala Ile
420 425 430

Glu Asp Phe Ser Gly Ala Gln Leu Val Phe Gly Asn Ser Arg Arg Ser
435 440 445

Asn Thr Val Thr Leu Pro Arg Phe Glu Thr Val Arg Glu Thr Ser Leu
450 455 460

Tyr Thr Val Asp Leu Asp Gly Phe His Val Ser Tyr Pro Glu Ser Arg
465 470 475 480

Ala Tyr Pro Glu Val Ile Pro Thr Glu Pro Asp Asn Thr Pro Thr Val
485 490 495

Ile Ile His Thr Glu Asp Phe Ser Gly Asp Tyr Tyr Leu His Pro Ser
500 505 510

Leu Lys Trp Lys Lys Arg Lys Arg Ala Tyr Leu
515 520

<210> SEQ ID NO 40
<211> LENGTH: 237
<212> TYPE: PRT
<213> ORGANISM: Rabies virus
<220> FEATURE:
<223> OTHER INFORMATION: Glycoprotein

<400> SEQUENCE: 40

Met Ala Pro Gln Val Leu Pro Phe Val Leu Leu Leu Ala Leu Ser Met
1 5 10 15

Cys Leu Gly Lys Phe Pro Ile Tyr Thr Ile Pro Asp Lys Leu Gly Pro
20 25 30

Trp Ser Pro Ile Asp Ile His His Leu Ser Cys Pro Asn Asn Leu Val
35 40 45

Val Glu Asp Glu Gly Cys Thr Asn Leu Ser Gly Phe Ser Tyr Met Glu
50 55 60

Leu Lys Val Gly Tyr Ile Ser Ala Ile Lys Val Asn Gly Phe Thr Cys
65 70 75 80

Thr Gly Val Val Thr Glu Ala Glu Thr Tyr Thr Asn Phe Val Gly Tyr
85 90 95

Val Thr Thr Thr Phe Lys Arg Lys His Phe Arg Pro Thr Pro Asp Ala
100 105 110

Cys Arg Ala Ala Tyr Asn Trp Lys Met Ala Gly Asp Pro Arg Tyr Glu
115 120 125

Glu Ser Leu His Asn Pro Tyr Pro Asp Tyr His Trp Leu Arg Thr Val
130 135 140

Lys Thr Thr Lys Glu Ser Leu Val Ile Ile Ser Pro Ser Val Ala Asp
145 150 155 160

Leu Asp Pro Tyr Asp Lys Ser Leu His Ser Arg Val Phe Pro Ser Gly
165 170 175

Lys Cys Ser Gly Ile Thr Ile Ser Ser Thr Tyr Cys Ser Thr Asn His
180 185 190

Asp Tyr Thr Ile Trp Met Pro Glu Asn Pro Arg Leu Gly Thr Ser Cys
195 200 205

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Asp Ile Phe Thr Asn Ser Arg Gly Lys Arg Ala Ser Lys Gly Gly Lys
    210                 215                 220

Thr Cys Gly Phe Val Asp Glu Arg Gly Leu Tyr Lys Ser
225                 230                 235


<210> SEQ ID NO 41
<211> LENGTH: 368
<212> TYPE: PRT
<213> ORGANISM: Cytomegalovirus
<220> FEATURE:
<223> OTHER INFORMATION: Glycoprotein

<400> SEQUENCE: 41

Met Met Thr Met Trp Cys Leu Thr Leu Phe Val Leu Trp Met Leu Arg
1               5                   10                  15

Val Val Gly Met His Val Leu Arg Tyr Gly Tyr Thr Gly Ile Phe Asp
            20                  25                  30

Asp Thr Ser His Met Thr Leu Thr Val Val Gly Ile Phe Asp Gly Gln
        35                  40                  45

His Phe Phe Thr Tyr His Val Asn Ser Ser Asp Lys Ala Ser Ser Arg
    50                  55                  60

Ala Asn Gly Thr Ile Ser Trp Met Ala Asn Val Ser Ala Ala Tyr Pro
65                  70                  75                  80

Thr Tyr Leu Asp Gly Glu Arg Ala Lys Gly Asp Leu Ile Phe Asn Gln
                85                  90                  95

Thr Glu Gln Asn Leu Leu Glu Leu Glu Ile Ala Leu Gly Tyr Arg Ser
            100                 105                 110

Gln Ser Val Leu Thr Trp Thr His Glu Cys Asn Thr Thr Glu Asn Gly
        115                 120                 125

Ser Phe Val Ala Gly Tyr Glu Gly Phe Gly Trp Asp Gly Glu Thr Leu
    130                 135                 140

Met Glu Leu Lys Asp Asn Leu Thr Leu Trp Thr Gly Pro Asn Tyr Glu
145                 150                 155                 160

Ile Ser Trp Leu Lys Gln Asn Lys Thr Tyr Ile Asp Gly Lys Ile Lys
                165                 170                 175

Asn Ile Ser Glu Gly Asp Thr Thr Ile Gln Arg Asn Tyr Leu Lys Gly
            180                 185                 190

Asn Cys Thr Gln Trp Ser Val Ile Tyr Ser Gly Phe Gln Pro Pro Val
        195                 200                 205

Thr His Pro Val Val Lys Gly Gly Val Arg Asn Gln Asn Asp Asn Arg
    210                 215                 220

Ala Glu Ala Phe Cys Thr Ser Tyr Gly Phe Phe Pro Gly Glu Ile Asn
225                 230                 235                 240

Ile Thr Phe Ile His Tyr Gly Asp Lys Val Pro Glu Asp Ser Glu Pro
                245                 250                 255

Gln Cys Asn Pro Leu Leu Pro Thr Leu Asp Gly Thr Phe His Gln Gly
            260                 265                 270

Cys Tyr Val Ala Ile Phe Cys Asn Gln Asn Tyr Thr Cys Arg Val Thr
        275                 280                 285

His Gly Asn Trp Thr Val Glu Ile Pro Ile Ser Val Thr Ser Pro Asp
    290                 295                 300

Asp Ser Ser Ser Gly Glu Val Pro Asp His Pro Thr Ala Asn Lys Arg
305                 310                 315                 320
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Tyr Asn Thr Met Thr Ile Ser Ser Val Leu Leu Ala Leu Leu Leu Cys
325 330 335

Ala Leu Leu Phe Ala Phe Leu His Tyr Phe Thr Thr Leu Lys Gln Tyr
340 345 350

Leu Arg Asn Leu Ala Phe Ala Trp Arg Tyr Arg Lys Val Arg Ser Ser
355 360 365

<210> SEQ ID NO 42
<211> LENGTH: 621
<212> TYPE: PRT
<213> ORGANISM: Hepatitis C virus
<220> FEATURE:
<223> OTHER INFORMATION: Envelope glycoproteins E1E2

<400> SEQUENCE: 42

Ile Arg Leu Asp Gly Val Asn Tyr Ala Thr Gly Asn Leu Pro Gly Cys
1 5 10 15

Ser Phe Ser Ile Phe Leu Leu Ala Leu Leu Ser Cys Leu Thr Ile Pro
20 25 30

Ala Ser Ala Tyr Glu Val Arg Asn Val Ser Gly Val Tyr His Val Thr
35 40 45

Asn Asp Cys Ser Asn Ser Ser Ile Val Tyr Glu Ala Ala Asp Met Ile
50 55 60

Met His Thr Pro Gly Cys Val Pro Cys Val Arg Glu Asn Asn Phe Ser
65 70 75 80

Arg Cys Trp Val Ala Leu Thr Pro Thr Leu Ala Ala Arg Asn Ala Ser
85 90 95

Val Pro Thr Thr Thr Ile Arg Arg His Val Asp Leu Leu Val Gly Ala
100 105 110

Ala Ala Phe Cys Ser Ala Met Tyr Val Gly Asp Leu Cys Gly Ser Val
115 120 125

Phe Leu Val Ser Gln Leu Phe Thr Phe Ser Pro Arg Gln His Glu Thr
130 135 140

Val Gln Asp Cys Asn Cys Ser Ile Tyr Pro Gly His Val Ser Gly His
145 150 155 160

Arg Met Ala Trp Asp Met Met Met Asn Trp Ser Pro Thr Ala Ala Ile
165 170 175

Val Val Ser Gln Leu Leu Arg Ile Pro Gln Ala Val Val Asp Met Val
180 185 190

Ala Gly Ala His Trp Gly Val Leu Ala Gly Leu Ala Tyr Tyr Thr Met
195 200 205

Ile Gly Asn Trp Ala Lys Ala Leu Ile Val Met Leu Leu Phe Ala Gly
210 215 220

Val Asp Gly His Thr His Val Thr Gly Gly Ser Ala Ser Arg Ala Thr
225 230 235 240

Arg Gly Leu Thr Ala Leu Phe Asp Phe Gly Ala Ser Gln Asn Leu Gln
245 250 255

Leu Ile Lys Thr Asn Gly Ser Trp His Ile Asn Arg Thr Ser Leu Asn
260 265 270

Cys Asn Asp Ser Leu Gln Thr Gly Phe Ile Ala Ala Leu Phe Tyr Thr
275 280 285

His Arg Phe Asn Ser Ser Gly Cys Pro Glu Arg Met Ala Ser Cys Arg
290 295 300

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Pro Ile Asp Trp Phe Ala Gln Gly Trp Gly Pro Ile Thr His Asp Lys
305 310 315 320

Gly Gly Ser Ser Asp Gln Arg Pro Tyr Cys Trp His Tyr Ala Pro Arg
325 330 335

Pro Cys Gly Ile Val Pro Ala Ala Gln Val Cys Gly Pro Val Tyr Cys
340 345 350

Phe Thr Pro Ser Pro Val Val Val Gly Thr Thr Asp Arg Phe Gly Val
355 360 365

Pro Thr Tyr Ser Trp Gly Glu Asn Glu Thr Asp Val Leu Leu Leu Asn
370 375 380

Asn Thr Arg Pro Pro Gln Gly Asn Trp Phe Gly Cys Thr Trp Met Asn
385 390 395 400

Ser Thr Gly Phe Thr Lys Thr Cys Gly Gly Pro Pro Cys Asn Ile Gly
405 410 415

Gly Val Gly Asn Asn Thr Leu Ile Cys Pro Thr Asp Cys Phe Arg Lys
420 425 430

His Pro Glu Ala Thr Tyr Thr Lys Cys Gly Ser Gly Pro Trp Leu Thr
435 440 445

Pro Arg Cys Ile Val Asp Tyr Pro Tyr Arg Leu Trp His Tyr Pro Cys
450 455 460

Thr Val Asn Phe Thr Ile Phe Lys Val Arg Met Tyr Val Gly Gly Val
465 470 475 480

Glu His Arg Leu Asn Ala Ala Cys Asn Trp Thr Arg Gly Glu Arg Cys
485 490 495

Asn Leu Glu Asp Arg Asp Arg Ser Glu Leu Ser Pro Leu Leu Leu Ser
500 505 510

Thr Thr Glu Trp Gln Val Leu Pro Cys Ser Phe Thr Thr Leu Pro Ala
515 520 525

Leu Ser Thr Gly Leu Ile His Leu His Gln Asn Ile Val Asp Val Gln
530 535 540

Tyr Leu Tyr Gly Ile Gly Ser Val Val Val Ser Phe Val Ile Lys Trp
545 550 555 560

Glu Tyr Val Val Leu Leu Phe Leu Leu Leu Ala Asp Ala Arg Val Cys
565 570 575

Ala Cys Leu Trp Met Met Leu Leu Ile Ala Gln Ala Glu Ala Ala Leu
580 585 590

Glu Asn Leu Val Val Leu Asn Ala Ala Ser Val Ala Gly Ala His Ser
595 600 605

Ile Leu Ser Phe Leu Val Phe Phe Cys Glu Ala Glu Phe
610 615 620

<210> SEQ ID NO 43
<211> LENGTH: 492
<212> TYPE: PRT
<213> ORGANISM: Respiratory syncytial virus
<220> FEATURE:
<223> OTHER INFORMATION: Fusion (F) protein

<400> SEQUENCE: 43

Thr Glu Glu Phe Tyr Gln Thr Thr Cys Ser Ala Val Ser Lys Gly Tyr
1 5 10 15

Leu Ser Ala Leu Arg Thr Gly Trp Tyr Thr Ser Val Ile Thr Ile Glu
20 25 30

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Leu Ser Asn Ile Lys Glu Asn Lys Cys Asn Gly Thr Asp Ala Lys Val
        35                  40                  45

Lys Leu Ile Lys Gln Glu Leu Asp Lys Tyr Lys Ser Ala Val Thr Glu
    50                  55                  60

Leu Gln Leu Leu Met Gln Ser Thr Pro Ala Thr Asn Asn Arg Ala Arg
65                  70                  75                  80

Arg Glu Leu Pro Arg Phe Met Asn Tyr Thr Leu Asn Asn Thr Lys Asn
                85                  90                  95

Thr Asn Val Thr Leu Ser Lys Lys Arg Lys Arg Arg Phe Leu Gly Phe
            100                 105                 110

Leu Leu Gly Val Gly Ser Ala Ile Ala Ser Gly Ile Ala Val Ser Lys
        115                 120                 125

Val Leu His Leu Glu Gly Glu Val Asn Lys Ile Lys Ser Ala Leu Leu
    130                 135                 140

Ser Thr Asn Lys Ala Val Val Ser Leu Ser Asn Gly Val Ser Val Leu
145                 150                 155                 160

Thr Ser Lys Val Leu Asp Leu Lys Asn Tyr Ile Asp Lys Gln Leu Leu
                165                 170                 175

Pro Ile Val Asn Lys Gln Ser Cys Ser Ile Ser Asn Ile Glu Thr Val
            180                 185                 190

Ile Glu Phe Gln Gln Lys Asn Asn Arg Leu Leu Glu Ile Thr Arg Glu
        195                 200                 205

Phe Ser Val Asn Ala Gly Val Thr Thr Pro Val Ser Thr Tyr Met Leu
    210                 215                 220

Thr Asn Ser Glu Leu Leu Ser Leu Ile Asn Asp Met Pro Ile Thr Asn
225                 230                 235                 240

Asp Gln Lys Lys Leu Met Ser Asn Asn Val Gln Ile Val Arg Gln Gln
                245                 250                 255

Ser Tyr Ser Ile Met Ser Ile Ile Lys Glu Glu Val Leu Ala Tyr Val
            260                 265                 270

Val Gln Leu Pro Leu Tyr Gly Val Ile Asp Thr Pro Cys Trp Lys Leu
        275                 280                 285

His Thr Ser Pro Leu Cys Thr Thr Asn Thr Lys Glu Gly Ser Asn Ile
    290                 295                 300

Cys Leu Thr Arg Thr Asp Arg Gly Trp Tyr Cys Asp Asn Ala Gly Ser
305                 310                 315                 320

Val Ser Phe Phe Pro Leu Ala Glu Thr Cys Lys Val Gln Ser Asn Arg
                325                 330                 335

Val Phe Cys Asp Thr Met Asn Ser Leu Thr Leu Pro Ser Glu Val Asn
            340                 345                 350

Leu Cys Asn Ile Asp Ile Phe Asn Pro Lys Tyr Asp Cys Lys Ile Met
        355                 360                 365

Thr Ser Lys Thr Asp Val Ser Ser Ser Val Ile Thr Ser Leu Gly Ala
    370                 375                 380

Ile Val Ser Cys Tyr Gly Lys Thr Lys Cys Thr Ala Ser Asn Lys Asn
385                 390                 395                 400

Arg Gly Ile Ile Lys Thr Phe Ser Asn Gly Cys Asp Tyr Val Ser Asn
                405                 410                 415

Lys Gly Val Asp Thr Val Ser Val Gly Asn Thr Leu Tyr Tyr Val Asn
            420                 425                 430
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Lys Gln Glu Gly Lys Ser Leu Tyr Val Lys Gly Glu Pro Ile Ile Asn
        435                 440                 445

Phe Tyr Asp Pro Leu Val Phe Pro Ser Asp Glu Phe Asp Ala Ser Ile
    450                 455                 460

Ser Gln Val Asn Glu Lys Ile Asn Gln Ser Leu Ala Phe Ile Arg Lys
465                 470                 475                 480

Ser Asp Glu Leu Leu His Asn Val Asn Ala Gly Lys
                485                 490


<210> SEQ ID NO 44
<211> LENGTH: 334
<212> TYPE: PRT
<213> ORGANISM: Zaire ebolavirus
<220> FEATURE:
<223> OTHER INFORMATION: Spike glycoprotein

<400> SEQUENCE: 44

Lys Arg Thr Ser Phe Phe Leu Trp Val Ile Ile Leu Phe Gln Arg Thr
1               5                   10                  15

Phe Ser Ile Pro Leu Gly Val Ile His Asn Ser Thr Leu Gln Val Ser
            20                  25                  30

Asp Val Asp Lys Leu Val Cys Arg Asp Lys Leu Ser Ser Thr Asn Gln
        35                  40                  45

Leu Arg Ser Val Gly Leu Asn Leu Glu Gly Asn Gly Val Ala Thr Asp
    50                  55                  60

Val Pro Ser Ala Thr Lys Arg Trp Gly Phe Arg Ser Gly Val Pro Pro
65                  70                  75                  80

Lys Val Val Asn Tyr Glu Ala Gly Glu Trp Ala Glu Asn Cys Tyr Asn
                85                  90                  95

Leu Glu Ile Lys Lys Pro Asp Gly Ser Glu Cys Leu Pro Ala Ala Pro
            100                 105                 110

Asp Gly Ile Arg Gly Phe Pro Arg Cys Arg Tyr Val His Lys Val Ser
        115                 120                 125

Gly Thr Gly Pro Cys Ala Gly Asp Phe Ala Phe His Lys Glu Gly Ala
    130                 135                 140

Phe Phe Leu Tyr Asp Arg Leu Ala Ser Thr Val Ile Tyr Arg Gly Thr
145                 150                 155                 160

Thr Phe Ala Glu Gly Val Val Ala Phe Leu Ile Leu Pro Gln Ala Lys
                165                 170                 175

Lys Asp Phe Phe Ser Ser His Pro Leu Arg Glu Pro Val Asn Ala Thr
            180                 185                 190

Glu Asp Pro Ser Ser Gly Tyr Tyr Ser Thr Thr Ile Arg Tyr Gln Ala
        195                 200                 205

Thr Gly Phe Gly Thr Asn Glu Thr Glu Tyr Leu Phe Glu Val Asp Asn
    210                 215                 220

Leu Thr Tyr Val Gln Leu Glu Ser Arg Phe Thr Pro Gln Phe Leu Leu
225                 230                 235                 240

Gln Leu Asn Glu Thr Ile Tyr Thr Ser Gly Lys Arg Ser Asn Thr Thr
                245                 250                 255

Gly Lys Leu Ile Trp Lys Val Asn Pro Glu Ile Asp Thr Thr Ile Gly
            260                 265                 270

Glu Trp Ala Phe Trp Glu Thr Lys Lys Asn Leu Thr Arg Lys Ile Arg
        275                 280                 285
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Ser Glu Glu Leu Ser Phe Thr Val Val Ser Asn Gly Ala Lys Asn Ile
    290                 295                 300

Ser Gly Gln Ser Pro Ala Arg Thr Ser Ser Asp Pro Gly Thr Asn Thr
305                 310                 315                 320

Thr Thr Glu Asp His Lys Ile Met Ala Ser Glu Asn Ser Ser
                325                 330


<210> SEQ ID NO 45
<211> LENGTH: 168
<212> TYPE: PRT
<213> ORGANISM: Zika virus
<220> FEATURE:
<223> OTHER INFORMATION: prM protein

<400> SEQUENCE: 45

Ala Glu Ile Thr Arg Arg Gly Ser Ala Tyr Tyr Met Tyr Leu Asp Arg
1               5                   10                  15

Ser Asp Ala Gly Lys Ala Ile Ser Phe Ala Thr Thr Leu Gly Val Asn
            20                  25                  30

Lys Cys His Val Gln Ile Met Asp Leu Gly His Met Cys Asp Ala Thr
        35                  40                  45

Met Ser Tyr Glu Cys Pro Met Leu Asp Glu Gly Val Glu Pro Asp Asp
    50                  55                  60

Val Asp Cys Trp Cys Asn Thr Thr Ser Thr Trp Val Val Tyr Gly Thr
65                  70                  75                  80

Cys His His Lys Lys Gly Glu Ala Arg Arg Ser Arg Arg Ala Val Thr
                85                  90                  95

Leu Pro Ser His Ser Thr Arg Lys Leu Gln Thr Arg Ser Gln Thr Trp
            100                 105                 110

Leu Glu Ser Arg Glu Tyr Thr Lys His Leu Ile Lys Val Glu Asn Trp
        115                 120                 125

Ile Phe Arg Asn Pro Gly Phe Ala Leu Val Ala Val Ala Ile Ala Trp
    130                 135                 140

Leu Leu Gly Ser Ser Thr Ser Gln Lys Val Ile Tyr Leu Val Met Ile
145                 150                 155                 160

Leu Leu Ile Ala Pro Ala Tyr Ser
                165


<210> SEQ ID NO 46
<211> LENGTH: 617
<212> TYPE: PRT
<213> ORGANISM: Zika virus
<220> FEATURE:
<223> OTHER INFORMATION: Serine protease NS3

<400> SEQUENCE: 46

Ser Gly Ala Leu Trp Asp Val Pro Ala Pro Lys Glu Val Lys Lys Gly
1               5                   10                  15

Glu Thr Thr Asp Gly Val Tyr Arg Val Met Thr Arg Arg Leu Leu Gly
            20                  25                  30

Ser Thr Gln Val Gly Val Gly Val Met Gln Glu Gly Val Phe His Thr
        35                  40                  45

Met Trp His Val Thr Lys Gly Ala Ala Leu Arg Ser Gly Glu Gly Arg
    50                  55                  60

Leu Asp Pro Tyr Trp Gly Asp Val Lys Gln Asp Leu Val Ser Tyr Cys
65                  70                  75                  80
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Gly Pro Trp Lys Leu Asp Ala Ala Trp Asp Gly Leu Ser Glu Val Gln
                85                  90                  95

Leu Leu Ala Val Pro Pro Gly Glu Arg Ala Arg Asn Ile Gln Thr Leu
            100                 105                 110

Pro Gly Ile Phe Lys Thr Lys Asp Gly Asp Ile Gly Ala Val Ala Leu
        115                 120                 125

Asp Tyr Pro Ala Gly Thr Ser Gly Ser Pro Ile Leu Asp Lys Cys Gly
    130                 135                 140

Arg Val Ile Gly Leu Tyr Gly Asn Gly Val Val Ile Lys Asn Gly Ser
145                 150                 155                 160

Tyr Val Ser Ala Ile Thr Gln Gly Lys Arg Glu Glu Glu Thr Pro Val
                165                 170                 175

Glu Cys Phe Glu Pro Ser Met Leu Lys Lys Lys Gln Leu Thr Val Leu
            180                 185                 190

Asp Leu His Pro Gly Ala Gly Lys Thr Arg Arg Val Leu Pro Glu Ile
        195                 200                 205

Val Arg Glu Ala Ile Lys Lys Arg Leu Arg Thr Val Ile Leu Ala Pro
    210                 215                 220

Thr Arg Val Val Ala Ala Glu Met Glu Glu Ala Leu Arg Gly Leu Pro
225                 230                 235                 240

Val Arg Tyr Met Thr Thr Ala Val Asn Val Thr His Ser Gly Thr Glu
                245                 250                 255

Ile Val Asp Leu Met Cys His Ala Thr Phe Thr Ser Arg Leu Leu Gln
            260                 265                 270

Pro Ile Arg Val Pro Asn Tyr Asn Leu Asn Ile Met Asp Glu Ala His
        275                 280                 285

Phe Thr Asp Pro Ser Ser Ile Ala Ala Arg Gly Tyr Ile Ser Thr Arg
    290                 295                 300

Val Glu Met Gly Glu Ala Ala Ala Ile Phe Met Thr Ala Thr Pro Pro
305                 310                 315                 320

Gly Thr Arg Asp Ala Phe Pro Asp Ser Asn Ser Pro Ile Met Asp Thr
                325                 330                 335

Glu Val Glu Val Pro Glu Arg Ala Trp Ser Ser Gly Phe Asp Trp Val
            340                 345                 350

Thr Asp His Ser Gly Lys Thr Val Trp Phe Val Pro Ser Val Arg Asn
        355                 360                 365

Gly Asn Glu Ile Ala Ala Cys Leu Thr Lys Ala Gly Lys Arg Val Ile
    370                 375                 380

Gln Leu Ser Arg Lys Thr Phe Glu Thr Glu Phe Gln Lys Thr Lys Asn
385                 390                 395                 400

Gln Glu Trp Asp Phe Val Ile Thr Thr Asp Ile Ser Glu Met Gly Ala
                405                 410                 415

Asn Phe Lys Ala Asp Arg Val Ile Asp Ser Arg Arg Cys Leu Lys Pro
            420                 425                 430

Val Ile Leu Asp Gly Glu Arg Val Ile Leu Ala Gly Pro Met Pro Val
        435                 440                 445

Thr His Ala Ser Ala Ala Gln Arg Arg Gly Arg Ile Gly Arg Asn Pro
    450                 455                 460

Asn Lys Pro Gly Asp Glu Tyr Met Tyr Gly Gly Gly Cys Ala Glu Thr
465                 470                 475                 480
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-continued

Asp Glu Gly His Ala His Trp Leu Glu Ala Arg Met Leu Leu Asp Asn
485 490 495

Ile Tyr Leu Gln Asp Gly Leu Ile Ala Ser Leu Tyr Arg Pro Glu Ala
500 505 510

Asp Lys Val Ala Ala Ile Glu Gly Glu Phe Lys Leu Arg Thr Glu Gln
515 520 525

Arg Lys Thr Phe Val Glu Leu Met Lys Arg Gly Asp Leu Pro Val Trp
530 535 540

Leu Ala Tyr Gln Val Ala Ser Ala Gly Ile Thr Tyr Thr Asp Arg Arg
545 550 555 560

Trp Cys Phe Asp Gly Thr Thr Asn Asn Thr Ile Met Glu Asp Ser Val
565 570 575

Pro Ala Glu Val Trp Thr Lys Tyr Gly Glu Lys Arg Val Leu Lys Pro
580 585 590

Arg Trp Met Asp Ala Arg Val Cys Ser Asp His Ala Ala Leu Lys Ser
595 600 605

Phe Lys Glu Phe Ala Ala Gly Lys Arg
610 615

<210> SEQ ID NO 47
<211> LENGTH: 130
<212> TYPE: PRT
<213> ORGANISM: Zika virus
<220> FEATURE:
<223> OTHER INFORMATION: Serine protease subunit NS2B

<400> SEQUENCE: 47

Ser Trp Pro Pro Ser Glu Val Leu Thr Ala Val Gly Leu Ile Cys Ala
1 5 10 15

Leu Ala Gly Gly Phe Ala Lys Ala Asp Ile Glu Met Ala Gly Pro Met
20 25 30

Ala Ala Val Gly Leu Leu Ile Val Ser Tyr Val Val Ser Gly Lys Ser
35 40 45

Val Asp Met Tyr Ile Glu Arg Ala Gly Asp Ile Thr Trp Glu Lys Asp
50 55 60

Ala Glu Val Thr Gly Asn Ser Pro Arg Leu Asp Val Ala Leu Asp Glu
65 70 75 80

Ser Gly Asp Phe Ser Leu Val Glu Glu Asp Gly Pro Pro Met Arg Glu
85 90 95

Ile Ile Leu Lys Val Val Leu Met Ala Ile Cys Gly Met Asn Pro Ile
100 105 110

Ala Ile Pro Phe Ala Ala Gly Ala Trp Tyr Val Tyr Val Lys Thr Gly
115 120 125

Lys Arg
130

<210> SEQ ID NO 48
<211> LENGTH: 500
<212> TYPE: PRT
<213> ORGANISM: Zika virus
<220> FEATURE:
<223> OTHER INFORMATION: Envelope protein E

<400> SEQUENCE: 48

Ile Arg Cys Ile Gly Val Ser Asn Arg Asp Phe Val Glu Gly Met Ser
1 5 10 15

-continued

```
Gly Gly Thr Trp Val Asp Val Val Leu Glu His Gly Gly Cys Val Thr
            20                  25                  30

Val Met Ala Gln Asp Lys Pro Thr Val Asp Ile Glu Leu Val Thr Thr
        35                  40                  45

Thr Val Ser Asn Met Ala Glu Val Arg Ser Tyr Cys Tyr Glu Ala Ser
    50                  55                  60

Ile Ser Asp Met Ala Ser Asp Ser Arg Cys Pro Thr Gln Gly Glu Ala
65                  70                  75                  80

Tyr Leu Asp Lys Gln Ser Asp Thr Gln Tyr Val Cys Lys Arg Thr Leu
                85                  90                  95

Val Asp Arg Gly Trp Gly Asn Gly Cys Gly Leu Phe Gly Lys Gly Ser
            100                 105                 110

Leu Val Thr Cys Ala Lys Phe Thr Cys Ser Lys Lys Met Thr Gly Lys
        115                 120                 125

Ser Ile Gln Pro Glu Asn Leu Glu Tyr Arg Ile Met Leu Ser Val His
    130                 135                 140

Gly Ser Gln His Ser Gly Met Ile Gly Tyr Glu Thr Asp Glu Asp Arg
145                 150                 155                 160

Ala Lys Val Glu Val Thr Pro Asn Ser Pro Arg Ala Glu Ala Thr Leu
                165                 170                 175

Gly Gly Phe Gly Ser Leu Gly Leu Asp Cys Glu Pro Arg Thr Gly Leu
            180                 185                 190

Asp Phe Ser Asp Leu Tyr Tyr Leu Thr Met Asn Asn Lys His Trp Leu
        195                 200                 205

Val His Lys Glu Trp Phe His Asp Ile Pro Leu Pro Trp His Ala Gly
    210                 215                 220

Ala Asp Thr Gly Thr Pro His Trp Asn Asn Lys Glu Ala Leu Val Glu
225                 230                 235                 240

Phe Lys Asp Ala His Ala Lys Arg Gln Thr Val Val Val Leu Gly Ser
                245                 250                 255

Gln Glu Gly Ala Val His Thr Ala Leu Ala Gly Ala Leu Glu Ala Glu
            260                 265                 270

Met Asp Gly Ala Lys Gly Arg Leu Phe Ser Gly His Leu Lys Cys Arg
        275                 280                 285

Leu Lys Met Asp Lys Leu Arg Leu Lys Gly Val Ser Tyr Ser Leu Cys
    290                 295                 300

Thr Ala Ala Phe Thr Phe Thr Lys Val Pro Ala Glu Thr Leu His Gly
305                 310                 315                 320

Thr Val Thr Val Glu Val Gln Tyr Ala Gly Thr Asp Gly Pro Cys Lys
                325                 330                 335

Ile Pro Val Gln Met Ala Val Asp Met Gln Thr Leu Thr Pro Val Gly
            340                 345                 350

Arg Leu Ile Thr Ala Asn Pro Val Ile Thr Glu Ser Thr Glu Asn Ser
        355                 360                 365

Lys Met Met Leu Glu Leu Asp Pro Pro Phe Gly Asp Ser Tyr Ile Val
    370                 375                 380

Ile Gly Val Gly Asp Lys Lys Ile Thr His His Trp His Arg Ser Gly
385                 390                 395                 400

Ser Thr Ile Gly Lys Ala Phe Glu Ala Thr Val Arg Gly Ala Lys Arg
                405                 410                 415
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Met Ala Val Leu Gly Asp Thr Ala Trp Asp Phe Gly Ser Val Gly Gly
            420                 425                 430

Val Phe Asn Ser Leu Gly Lys Gly Ile His Gln Ile Phe Gly Ala Ala
        435                 440                 445

Phe Lys Ser Leu Phe Gly Gly Met Ser Trp Phe Ser Gln Ile Leu Ile
    450                 455                 460

Gly Thr Leu Leu Val Trp Leu Gly Leu Asn Thr Lys Asn Gly Ser Ile
465                 470                 475                 480

Ser Leu Thr Cys Leu Ala Leu Gly Gly Val Met Ile Phe Leu Ser Thr
                485                 490                 495

Ala Val Ser Ala
            500
```

<210> SEQ ID NO 49
<211> LENGTH: 104
<212> TYPE: PRT
<213> ORGANISM: Zika virus
<220> FEATURE:
<223> OTHER INFORMATION: Capsid protein C

<400> SEQUENCE: 49

```
Met Lys Asn Pro Lys Glu Glu Ile Arg Arg Ile Arg Ile Val Asn Met
1               5                   10                  15

Leu Lys Arg Gly Val Ala Arg Val Asn Pro Leu Gly Gly Leu Lys Arg
            20                  25                  30

Leu Pro Ala Gly Leu Leu Leu Gly His Gly Pro Ile Arg Met Val Leu
        35                  40                  45

Ala Ile Leu Ala Phe Leu Arg Phe Thr Ala Ile Lys Pro Ser Leu Gly
    50                  55                  60

Leu Ile Asn Arg Trp Gly Ser Val Gly Lys Lys Glu Ala Met Glu Ile
65                  70                  75                  80

Ile Lys Lys Phe Lys Lys Asp Leu Ala Ala Met Leu Arg Ile Ile Asn
                85                  90                  95

Ala Arg Lys Glu Arg Lys Arg Arg
            100
```

<210> SEQ ID NO 50
<211> LENGTH: 669
<212> TYPE: RNA
<213> ORGANISM: SARS coronavirus
<220> FEATURE:
<223> OTHER INFORMATION: mRNA encoding SARS-CoV-2 Spike (S) RBD protein

<400> SEQUENCE: 50

```
agaguccaac caacagaauc uauuguuaga uuuccuaaua uuacaaacuu gugcccuuuu      60

ggugaaguuu uuaacgccac cagauuugca ucuguuuaug cuuggaacag gaagagaauc     120

agcaacugug uugcugauua uucuguccua uauaauuccg caucauuuuc cacuuuuaag     180

uguuauggag ugucuccuac uaaauuaaau gaucucugcu uuacuaaugu cuaugcagau     240

ucauuuguaa uuagagguga ugaagucaga caaaucgcuc cagggcaaac uggaaagauu     300

gcugauuaua auuauaaauu accagaugau uuuacaggcu gcguuauagc uuggaauucu     360

aacaaucuug auucuaaggu uggugguaau uauaauuacc uguauagauu guuuaggaag     420

ucuaaucuca aaccuuuuga gagagauauu ucaacugaaa ucuaucaggc cgguagcaca     480

ccuuguaaug guguugaagg uuuuaauugu uacuuuccuu uacaaucaua ugguuuccaa     540
```

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cccacuaaug guguugguua ccaaccauac agaguaguag uacuuucuuu ugaacuucua    600

caugcaccag caacuguuug uggaccuaaa aagucuacua auuugguuaa aaacaaaugu    660

gucaauuuc                                                            669
```

What is claimed is:

1. A method for treatment or prophylaxis of a disease in a subject, said method comprising administering to the subject a vaccine composition comprising non-immunomodulating, engineered, plant-derived extracellular vesicles (EVs), wherein said EVs are delimited by a lipid bilayer membrane comprising an outer lipid layer and an inner lipid layer,

wherein said EVs are internally loaded with an exogenous nucleic acid molecule encoding at least one protein antigen;

wherein said EVs have a diameter ranging from 20 to 500 nm;

wherein the membrane potential across the lipid bilayer membrane of said EVs ranges from +5 to −5 mV; and

wherein ≤44% of the EVs in the vaccine composition comprise phosphatidylserine in the outer lipid layer of the lipid bilayer membrane.

2. The method of claim **1**, wherein the loaded exogenous nucleic acid molecule is selected from the group consisting of DNA, cDNA, messenger RNA (mRNA), pre-mRNA, long-chain RNA, coding RNA, single-stranded RNA, double stranded RNA, linear RNA, RNA oligonucleotide, self-replicating RNA (replicon RNA), retroviral RNA, and viral RNA (vRNA).

3. The method of claim **2**, wherein the loaded exogenous nucleic acid molecule is a mRNA molecule comprising a nucleotide sequence selected from the group consisting of SEQ ID NOs: 14, 17, 19 and 50.

4. The method of claim **1**, wherein the encoded at least one protein antigen is selected from the group consisting of tumor antigens, viral antigens, bacterial antigens, fungal antigens and protozoa antigens.

5. The method of claim **4**, wherein the encoded at least one protein antigen is selected from the group consisting of prostate specific antigen (PSA), prostate stem cell antigen (PSCA), prostate specific membrane antigen (PSMA), (six transmembrane epithelial antigen of the prostate 1) (STEAP1), Receptor tyrosine-protein kinase erbB-2, cell surface associated mucin 1 protein (MUC1), Tyrosinase-related protein 2 (TRP-2), Proto-oncogene B-Raf, Proto-oncogene c-Kit, GTPase NRas, melanoma-associated antigen 1, melanoma-associated antigen 1 protein, NY-ESO-1 protein, Spike protein of SARS-COV-2, N protein of SARS-COV-2, M protein of SARS-COV-2, Hemagglutinin protein of influenza A viruses, Hemagglutinin protein of influenza B virus, Neuraminidase protein of influenza A viruses, Neuraminidase protein of influenza B virus, envelope protein of HIV1, envelope protein HIV2, Major Capsid Protein L1 of HPV, Minor Capsid Protein L2 of HPV, glycoprotein of Rabies lyssavirus, glycoprotein of Human Cytomegalovirus, envelope glycoproteins E1E2 of Hepatitis C virus, Fusion protein of RSV, spike glycoprotein of Zaire ebolavirus, Protein prM of Zika virus, Serine protease NS3 of Zika virus, Serine protease subunit NS2B of Zika virus, Envelope protein E of Zika virus, Capsid protein C of Zika virus, Toxoplasma gondii proteins, including dense granule protein 6, rhoptry protein 2A, rhoptry protein 18, surface antigen 1, surface antigen 2A, Toxoplasma gondii apical membrane antigen 1, SARS-CoV-2 Spike(S) RBD protein, and any combination thereof.

6. The method of claim **5**, wherein the encoded at least one protein antigen comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 1-13, 15, 16, 18, and 20-49.

7. The method of claim **1**, wherein the content of the loaded exogenous nucleic acid molecule in the EVs is in the range of from 20 to 200 ng/109 EVs.

8. The method of claim **1**, wherein the EVs are derived from one or more plants selected from the group consisting of: the genus *Citrus*, including lemon and orange; genus *Actinidia*, including kiwifruit; genus *Cucurbita*, including courgette; genus *Brassica*, including cabbage and kale; genus *Punica*, including pomegranate; genus *Vaccinium*, including blueberry, and genus *Apium*, including celery.

9. The method of claim **1**, wherein the vaccine composition further comprises one or more polycationic substances, said one or more polycationic substances being associated with the outer lipid layer of the lipid bilayer membrane of the EVs through electrostatic interactions.

10. The method of claim **9**, wherein the one or more polycationic substances are selected from the group consisting of cationic proteins, including protamine, cationic peptides, polypeptides, polysaccharides, glycerol, polyethylene glycol (PEG), and any combination thereof.

11. The method of claims **1**, wherein the EVs are additionally loaded with one or more sugar molecules, said one or more sugar molecules being associated with the exogenous nucleic acid molecule loaded into the EVs through electrostatic interactions and hydrogen bonding.

12. The method of claim **11**, wherein the one or more sugar molecules are selected from the group consisting of disaccharides, sugar alcohols, polysaccharides, and any combination thereof.

13. The method of claim **1**, wherein the vaccine composition is in a form suitable for oral, intranasal or parenteral administration.

14. A method for preparing a vaccine composition comprising non-immunomodulating, engineered, plant-derived extracellular vesicles (EVs), wherein said EVs are delimited by a lipid bilayer membrane comprising an outer lipid layer and an inner lipid layer,

wherein said EVs are internally loaded with an exogenous nucleic acid molecule encoding at least one protein antigen;

wherein said EVs have a diameter ranging from 20 to 500 nm;

wherein the membrane potential across the lipid bilayer membrane of said EVs ranges from +5 to −5 mV; and wherein ≤44% of the EVs in the vaccine composition comprise phosphatidylserine in the outer lipid layer of the lipid bilayer membrane,

the method comprising the steps of:

(i) contacting and mixing a suspension of the plant-derived EVs with one or more polycationic substances to obtain a first mixture;

(ii) contacting and mixing a preparation of nucleic acid molecules with one or more sugar molecules to obtain a second mixture, said nucleic acid molecules encoding at least one protein antigen;

(iii) admixing said first mixture and said second mixture to obtain a third mixture; and

(iv) adding to said third mixture a pre-determined volume of water, wherein a ratio of said pre-determined volume of water to a volume of the third mixture is from 5:1 to 15:1.

15. The method of claim **14**, further comprising concentrating the vaccine composition obtained in step (iv).

16. The method of claim **14**, wherein the one or more polycationic substances are selected from the group consisting of cationic proteins, including protamine, calcitonin peptides, plectasin, lactoferrin, protamine-like proteins, such as spermine or spermidine, nucleoline, histones, cell penetrating peptides (CPPs);

cationic peptides, including histidine-rich peptides, arginine-rich peptides, lysine-rich peptides, cationic arginine-rich peptides (CARPs); polypeptides, including poly-arginine, poly-lysine, poly-histidine, histidine-rich peptides, arginine-rich peptides, lysine-rich peptides; polysaccharides, including chitosan, glycosaminoglycan such as polysulfated glycosaminoglycan (PSGAG), cationic dextrans; glycerol, polyethylene glycol (PEG), and any combination thereof, and/or wherein the one or more sugar molecules are selected from the group consisting of disaccharides, including trehalose, maltose, lactose, sucrose, cellobiose, chitobiose, kojibiose, nigerose, isomaltose, β,β-trehalose, α,β-trehalose, sophorose, laminaribiose, gentiobiose, trehalulose, turanose, maltulose, leucrose, iso-maltulose, gentiobiulose, mannobiose, melibiose, melibiulose, rutinose, rutinulose, xylobiose; sugar alcohols, including arabitol, erythritol, glycerol, HSHs, isomalt, lactitol, maltitol, mannitol, sorbitol, xylitol; polysaccharides, including starch, glycogen, galactogen, inulin, arabinoxylans, cellulose, chitin and pectin, and any combination thereof.

17. The method of claim **1**, wherein said disease is an infectious disease or cancer.

18. The method of claim **10**, wherein said one or more polycationic substances are present in the vaccine composition in an amount ranging from 0.001 to 2 μg/10^{10} EVs.

19. The method of claim **12**, wherein the content of the one or more sugar molecules in the EVs is in the range of from 0.1 to 10 mg/10^{10} EVs.

20. The method of claim **15**, wherein the vaccine composition obtained in step (iv) is concentrated by filtration.

* * * * *